Recovery-Time Guarantees for Safety Filtering under Sampled Actuation

Technical Notes

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Abstract

These notes supply complete proofs and the reproducible numerical protocol for the accompanying paper. The broad safety formulation permits time-dependent drift and gain and a moving leader. Exact recoverability, spatial minimum entry values, sharp continuous-time fidelity clocks, and the sampled performance theorem concern autonomous dynamics behind a stopped leader. Nominal commands may remain time dependent. The sampled construction certifies constant actuator inputs throughout each hold. A boundary-clearance construction gives a feasible held input and recursive feasibility at an a priori fixed period. For the three optimal-velocity models, an analytic construction certifies .01 and an offline interval calculation certifies .025 throughout the specified recovery episodes, including rest launch. All 192 experiments at these fixed periods apply a certified nominal hold or a stored clearance witness. Verified trajectory enclosures also certify individual longer holds. The energy floor preserves the continuous-time worst-case clock as an upper bound on physical entry time; exact sampled observation adds at most the maximum hold period to detection. An OVM stopping law supplies a separately proved sampled-safe continuation. The numerical section distinguishes certified whole-hold inequalities, ordinary trajectory diagnostics, and empirical performance comparisons. A complete continuous-time comparison is included as an appendix.

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Repository. https://github.com/hosein-matin/Recovery-time-safety

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Reading guide and correspondence with the paper

The conference paper and these notes use the same model, author order, and theorem hypotheses. The correspondence below identifies the detailed arguments supporting its numbered results. Subsections 8.7–8.10 provide the useful fixed-period construction and its complete-episode analytic and interval certificates. Sections 1–5 also develop regular recovery controllers, and Sections 9–12 give extensions and limitations. The continuous-time numerical appendix has a different post-entry continuation from the sampled experiments in Section 13; their cost and motion statistics are not pooled.

Conference result	Detailed statement or proof in these notes
Proposition 1: nonautonomous safety	Proposition 4.0, including all-time face tests and a distance-to-domain proof.
Theorem 2: recoverability criterion	Theorem 2.1; Theorem 3.1 and Proposition 4.1 support its subsequent low-speed entry bound.
Theorem 3: spatial benchmark	Lemmas 6.1–6.2 and Theorem 6.3.
Theorem 4: reflected filter	Theorem 7.1, including the launch construction.
Theorem 5: exact fidelity cost	Theorem 7.2.
Proposition 6: optimal-velocity certificate	Proposition 7.3.
Theorem 7: sampled feasibility and clock	Lemmas 8.1–8.2 and Theorem 8.3.
Proposition 8: a priori clearance certificate	Lemma 8.8 and Theorem 8.9.
Corollary 9: complete-episode periods	Lemma 8.11, Theorem 8.12, and Subsection 8.9.
Computable certificate and continuation	Lemmas 8.4, 8.5, 8.6, and Proposition 8.7.
Sampled numerical protocol	Section 13.

Notation

The variable s denotes forward travel in the spatial analysis; ordinary time is t. A prime on a spatial profile denotes differentiation with respect to s, almost everywhere. Local symbols in examples are defined at their point of use. Table 1 collects the symbols used in the paper and notes.

Table 1: Notation for the paper and technical notes.

Symbol	Meaning
d, v, w	Follower gap, follower speed, and leader speed.
u, k, U	Applied actuator input, nominal input, and actuator magnitude bound.
$a, b, A_{\pm}$	Acceleration drift, positive input gain, and attainable endpoint accelerations $a \pm bU$.
ϵ, γ, M	Minimum standstill gap, headway coefficient, and speed cap.
h, D	Headway margin $d - \gamma v - \epsilon$ and safe domain $\{h \geq 0,\ 0 \leq v \leq M\}$.
$h_g, d_g, \mathcal{T}_g$	Accepted reserve, $d_g = \epsilon + h_g$, and rest target $\{(d, 0) : \epsilon \leq d \leq d_g\}$.
d_η, η	Positive-tolerance entry threshold $d_g + \eta$ and tolerance $\eta > 0$.
λ, h_s	Recovery-reference gain and positive transition scale in the excess-headway factor.
ρ_g, r_λ	Excess-headway factor $(h - h_g)_+/[h_s + (h - h_g)_+]$ and the desired acceleration reference.
N, s_h, s_v	Finite positive blend sharpness and positive headway/speed normalization scales.
β, σ	Speed and boundary blending weights defined in (16) and (18).
u_b, u_{rec}	Actuator-clipped safe backup and the nominal/backup recovery seed in (17).
$\kappa_h, \kappa_v, \kappa_M$	Positive gains for the headway, lower-speed, and upper-speed CBF inequalities.
s, E	Forward travel $d_0 - d$ and kinetic energy per unit mass $v^2/2$.
S, S_0, S_e	Entry travel $d_0 - d_\eta$, maximal geometric travel $d_0 - \epsilon$, and certificate extension beyond S.
$B(s), E^+(s), s_*$	Safety energy ceiling, maximal physically realizable safe profile, and its terminal positive-energy endpoint.
T^*	Minimum feasible first-entry time for the fixed initial state and target; $T^*_\eta(z_0)$, $T^*(z_0, S)$ and $T^*(S)$ display or suppress the same fixed arguments.
θ, q	Performance fraction in $(0, 1)$ and fixed episode floor $q = \theta E^+$.
$f_{\pm}, p_N$	Spatial acceleration endpoints and nominal acceleration evaluated at the physical clock.
E^-_θ, W_θ	Lower-reflected admissible energy profile and its clock; the exact worst-case value for strict interior fidelity.
$\tau(s), \Pi$	Physical elapsed-time clock and an admissible faithful episode filter.
$T_\theta[k], T_\Pi[k]$	First physical entry times under nominal k for the reflected controller and a faithful episode filter, respectively.
τ_η	First time the physical gap reaches $d_g + \eta$; the general strip theorem allows an initial time t_0.
e_u, I_δ	Energy profile for a constant actuator and the interval certified on a fixed spatial prefix of length $M\delta$.
g, P, K	Lower band separation, acceleration bound, and clearance-rate bound used in the fallback certificate.
$\delta_0, \delta_F, \bar{\Delta}$	Analytic existence period, positive fallback duration, and requested maximum physical hold.
$\mathsf{F}, \mathcal{T}_B$	Spatial acceleration field and the ceiling traversal clock.
$J_s, r_\Delta(s), \mathcal{I}_\Delta$	Common ceiling-clock prefix, its endpoint, and its band-feasible held-input interval.
R_q, R_B, u_q, u_B	Tangent boundary input requirements and constant floor/ceiling witnesses.
c_B, c_q, H_B, H_q	Upper opposite-clearance rates and their integrated budgets.
x, x_η, x_e, x_T	Gap clearance $d - \epsilon$, entry clearance, extension clearance, and certified ceiling-contact threshold.
v_{cap}	Current ceiling speed $\min\{M, x/\gamma\}$ in the analytic period construction.
Λ_q, Λ_B	Lipschitz bounds for the two tangent input functions on the headway branch.

Symbol	Meaning
$\Delta_{\mathrm{pre}}, \Delta_{\mathrm{band}}, \Delta_{\mathrm{ext}}$	Analytic pre-tail, tail-band, and extension period bounds; Δ_{band} differs from the post-entry continuation period Δ_{tail}.
$\overline{H}_{B,j}, \overline{H}_{q,j}, \underline{w}_j$	Verified cellwise budget upper bounds and a lower bound for the available band width.
$T_{\mathrm{entry}}, T_{\mathrm{detected}}$	Physical first-entry time and first detection by exact sampled state observation.
$\widehat{z}, \xi_d, \xi_v$	Numerical state path and validated gap/speed error radii, with $\xi = (\xi_d, \xi_v)^\top$.
$Q, \overline{E}_j, \underline{T}_Q$	Verified upper reference, its upper cell endpoint, and a verified lower bound on T^*.
$\mathcal{V}, \alpha$	Desired-velocity law and relaxation coefficient in the optimal-velocity example.
$J_H, I_{1,H}, I_{2,H}, A_H$	Cost, absolute and squared intervention, and acceleration RMS over the common horizon H.

1 Model, safety geometry, and attainable acceleration

Consider the nonlinear longitudinal model

$$\dot{d} = w(t) - v, \qquad \dot{v} = a(t,d,v) + b(t,d,v)u, \qquad |u| \le U, \tag{1}$$

with $U, M, \gamma, \epsilon > 0$, $0 \le w(t) \le M$, and $b \ge b_* > 0$. Assume a, b are continuous and locally Lipschitz in state, uniformly on finite time intervals, w is continuous, and $|a(t,d,v)| + |b(t,d,v)| \le C_T(1 + |d| + |v|)$ on $[0,T]$. Feedbacks are measurable in time and locally Lipschitz in state with the usual finite-interval uniform bounds. An admissible nominal command satisfies $k(t,z) \in [-U, U]$.

Define

$$h = d - \gamma v - \epsilon, \qquad D = \{(d,v) : h \ge 0,\ 0 \le v \le M\}, \qquad A_\pm = a \pm bU. \tag{2}$$

Thus $[A_-, A_+]$ is the attainable acceleration interval. The exact plant viability tests for preserving D, imposed for every $t \ge 0$ on the indicated faces, are

$$A_+ \ge 0 \qquad \text{on } v = 0, \tag{3}$$

$$A_- \le 0 \qquad \text{on } v = M, \tag{4}$$

$$A_- \le (w - v)/\gamma \qquad \text{on } h = 0. \tag{5}$$

All tests include intersections of the faces. At $w = h = v = 0$, the lower speed and headway constraints together require actual acceleration zero.

For recovery, assume from now on that $w \equiv 0$ and a, b are autonomous, unless a result explicitly says otherwise. Let $h_g \ge 0$ specify an accepted standstill gap reserve and put

$$d_g = \epsilon + h_g, \qquad \mathcal{T}_g = \{(d,0) : \epsilon \le d \le d_g\}. \tag{6}$$

Recovery means $\operatorname{dist}(z(t), \mathcal{T}_g) \to 0$. It does not mean finite-time arrival at $\mathcal{T}_g$, preservation of $h \ge h_g$, or convergence to the particular point $(d_g, 0)$.

2 An exact safe-recoverability characterization

Theorem 2.1 (Safe asymptotic gap recovery)**.** *For the autonomous stopped-leader model, the following are equivalent.*

(i) From every $z_0 \in D$ there exists an admissible measurable input whose trajectory remains in D for all future time and satisfies $\operatorname{dist}(z(t), \mathcal{T}_g) \to 0$.

(ii) One autonomous, admissible, locally Lipschitz feedback preserves D and has that convergence property for every initial state in D.

(iii) The three face tests (3)–(5), with $w = 0$, hold, and

$$A_+(d,0) > 0 \qquad \text{for every } d > d_g. \tag{7}$$

Every recovering trajectory converges to a point $(d_\infty, 0)$ with $d_\infty \in [\epsilon, d_g]$.

Proof. The implication (ii)⇒(i) is immediate. For (i)⇒(iii), failure of a face test makes the corresponding constraint derivative strictly outward for every admissible input at a boundary point. Continuity makes this true in a neighborhood, so no safe trajectory can start there.

Suppose (7) fails. The lower-face test implies the existence of $d_* > d_g$ with $G(d_*, 0) = 0$, where $G = A_+$. Consider any safe absolutely continuous trajectory from $(d_*, 0)$ under any admissible measurable input. Define

$$q(t) = d_* - d(t) = \int_0^t v(r)\, dr \ge 0, \qquad Y(t) = q(t) + v(t) \ge 0.$$

Because $b > 0$ and $u \le U$, $\dot{v} \le G(d, v)$. Local Lipschitz continuity of G and $G(d_*, 0) = 0$ give $G(d, v) \le L(q + v)$ while the trajectory remains near $(d_*, 0)$. Consequently

$$\dot{Y} = v + \dot{v} \le (1 + L)Y \quad \text{a.e.}, \qquad Y(0) = 0.$$

Grönwall's inequality gives $Y = 0$. A first-exit argument shows that the trajectory cannot leave the neighborhood; it remains at $(d_*, 0)$ forever. This contradicts recovery. The obstruction applies to every safe measurable input, not just to a particular feedback.

For (iii)⇒(ii), choose $\lambda, h_s > 0$ and define

$$\rho_g(h) = \frac{(h - h_g)_+}{h_s + (h - h_g)_+}, \tag{8}$$

$$r_\lambda(h, v) = \frac{-v + \lambda \rho_g(h)(M - v)}{\gamma}, \tag{9}$$

$$u_b(d, v) = \operatorname{clip}_{[-U,U]} \frac{r_\lambda - a(d, v)}{b(d, v)}. \tag{10}$$

The actual acceleration is

$$p_b = \operatorname{clip}_{[A_-, A_+]} r_\lambda. \tag{11}$$

The feedback is locally Lipschitz and bounded. On $h = 0$, $r_\lambda = -v/\gamma$; on $v = M$, $r_\lambda = -M/\gamma$; on $v = 0$, $r_\lambda \ge 0$. The face tests imply every active tangent inequality for p_b, including at corners. The usual tangent-cone invariance argument gives preservation of D.

Along any safe trajectory, d is nonincreasing and bounded below:

$$d(t) \downarrow d_\infty \ge \epsilon, \qquad \int_0^\infty v(t)\, dt = d_0 - d_\infty < \infty.$$

The trajectory lies in the compact set $D \cap \{d \le d_0\}$. Continuous autonomous coefficients and bounded input bound acceleration there. Hence v is uniformly continuous. A nonnegative, integrable, uniformly continuous function tends to zero, so $v(t) \to 0$.

If $d_\infty > d_g$, the limiting reference and upper attainable acceleration are both strictly positive. Equation (11) then gives $p_b(t) \to p_\infty > 0$, contradicting $v(t) \to 0$. Thus $d_\infty \le d_g$ and the proposed feedback has property (ii). □

No positive lower bound on $A_+(d, 0)$ over the entire unbounded interval $d > d_g$ is required. Compactness supplies a positive lower bound on each fixed excess-gap interval. The necessity proof relies on an autonomous plant and a permanently stopped leader; a time-dependent plant could regain acceleration authority later. Positive authority at every finite time is also insufficient for a time-dependent plant. For example, take $w = 0$, $U = b = \gamma = M = 1$, and

$$a(t, d, v) = e^{-t} - 1 - v.$$

All three face tests hold for every $t \ge 0$, and $A_+(t, d, 0) = e^{-t} > 0$. Nevertheless, a rest start obeys $\dot{v} + v \le e^{-t}$, so $v(t) \le te^{-t}$ and $\int_0^\infty v(t)\, dt \le 1$. Thus an initial gap $d_0 > d_g + 1$ cannot recover to $\mathcal{T}_g$ under any safe admissible input. This is a plant-authority obstruction; a time-uniform progress reserve is needed for the corresponding sufficiency argument.

Corollary 2.2 (Preserving accepted rest)**.** *There is a regular feedback that preserves D, gives the recovery in Theorem 2.1, and keeps every point of $\mathcal{T}_g$ fixed if and only if the conditions in that theorem hold and*

$$A_-(d, 0) \le 0 \qquad (\epsilon \le d \le d_g). \tag{12}$$

Proof. Keeping an accepted rest point fixed requires attainable zero acceleration. The lower-face test already gives $A_+ \ge 0$, so the missing necessary inequality is (12). Conversely, at such a point $r_\lambda = 0$ and $0 \in [A_-, A_+]$, so the feedback (10) has zero actual acceleration. □

For $h_g > 0$, attraction and even invariance of $\mathcal{T}_g$ do not by themselves establish Lyapunov stability of the whole rest segment. The conclusion is asymptotic recovery; stability of this interval is not asserted. For $h_g = 0$, safety gives $v(t) \le (d(t) - \epsilon)/\gamma$ and $d(t) - \epsilon \le d_0 - \epsilon$, so the singleton $(\epsilon, 0)$ is stable relative to D; attraction then supplies the stronger conclusion.

2.1 An explicit plant obstruction

Take $b = U = M = \gamma = 1$, choose $d_* > d_g$, and let

$$a(d, v) = -1 - v + Q(d), \qquad Q(d) = \frac{(d - d_*)^2}{1 + (d - d_*)^2}.$$

Then $A_+ = Q - v$ and $A_- = Q - v - 2$, so the three face tests hold because $0 \le Q < 1$. However, $A_+(d_*, 0) = 0$, and the no-escape argument in Theorem 2.1 applies to every safe measurable input. Replacing Q by $Q + \delta$ for $0 < \delta \le 1$ retains the face tests and gives strictly positive rest authority throughout $d > d_g$. The explicit backup then guarantees recovery. This change alters the attainable plant acceleration, rather than merely changing the filter at the same unrecoverable state.

3 A common progress theorem independent of the filter

Theorem 3.1 (Uniform low-speed progress)**.** *Let $z(t)$ be a forward-complete trajectory that remains in D under a bounded admissible feedback for the autonomous stopped-leader model. Fix its initial state $z(t_0) = (d_0, v_0)$. For $\eta > 0$, put $d_\eta = d_g + \eta$ and $\tau_\eta = \inf\{t \ge t_0 : d(t) \le d_\eta\}$, with $\inf \varnothing = +\infty$. If $d_0 \le d_\eta$, then $\tau_\eta = t_0$. Otherwise, suppose constants $c_\eta > 0$ and $0 < \nu_\eta < M$ satisfy*

$$a(d, v) + b(d, v)u(t, d, v) \ge c_\eta \tag{13}$$

for every $(d, v) \in D$ with $d_\eta \le d \le d_0$ and $0 \le v \le \nu_\eta$, outside one common time-null set. The constants are independent of time on $[t_0, \infty)$. This uniform statewise condition is sufficient; its restriction to the trajectory whenever it is in the strip also suffices. Then

$$\tau_\eta - t_0 \le \frac{(\nu_\eta - v_0)_+}{c_\eta} + \frac{d_0 - d_\eta}{\nu_\eta} < \infty. \tag{14}$$

If this condition holds for every positive η with $d_\eta < d_0$, the trajectory converges to a point of $\mathcal{T}_g$.

Proof. If $v_0 < \nu_\eta$, either the gap threshold is reached first or integration of (13) gives $v = \nu_\eta$ within $(\nu_\eta - v_0)/c_\eta$. Once $v \ge \nu_\eta$, it cannot cross below ν_η before reaching d_η: an interval beginning at $v = \nu_\eta$ and lying below that value afterwards would, by integrating $\dot{v} \ge c_\eta$, give the opposite inequality. Thus, after the possible initial acceleration period, $\dot{d} = -v \le -\nu_\eta$ until the gap threshold is reached. This proves (14).

As in Theorem 2.1, every safe trajectory has $d(t) \to d_\infty$ and $v(t) \to 0$. Finite entry for all positive η forces $d_\infty \le d_g$. □

This condition is stronger than positive acceleration at each individual rest point and time. For example, take $a = 0$, $b = U = M = \gamma = 1$ and

$$u = -v + e^{-t}\rho_g(h)(1 - v).$$

It is admissible and safe on D, and it gives positive acceleration at every excess-gap rest point for every finite t. Yet $\dot{h} = -e^{-t}\rho_g(h)(1 - v) \ge -e^{-t}$. From $h_0 > h_g + 1$, the gap stays above h_g. Pointwise rest positivity alone therefore does not imply recovery. The uniform strip condition is a sufficient certificate, not a necessary condition for every recovering time-dependent controller. Bound (14) concerns any positive gap tolerance, not exact finite-time arrival at $\mathcal{T}_g$.

4 One recovery reference, two safe implementations

For a moving leader, use the reference

$$r_\lambda(t,h,v) = \frac{w(t)/2 - v + \lambda\rho_g(h)(M-v)}{\gamma} \tag{15}$$

in the time-dependent version of (10).

Proposition 4.0 (Nonautonomous safety)**.** *Assume the time-dependent model regularity above and (3)–(5) for every $t \geq 0$. For $h_g \geq 0$, $h_s > 0$, and $\lambda \geq 0$, the feedback*

$$u_b(t,d,v) = \mathrm{clip}_{[-U,U]} \frac{r_\lambda(t,h,v) - a(t,d,v)}{b(t,d,v)}$$

has a unique forward-complete solution from every $z(t_0) \in D$, $t_0 \geq 0$, and that solution remains in D for all $t \geq t_0$.

Proof. Write the closed-loop field as

$$f(t,z) = \big(w(t) - v,\ \mathrm{clip}_{[A_-(t,z),A_+(t,z)]}\, r_\lambda(t,h,v)\big).$$

It is continuous in time and locally Lipschitz in state, uniformly on finite time intervals. Its growth is at most linear because the input is bounded. Hence the physical solution exists uniquely for every finite future time. On $v = 0$ the reference is nonnegative; on $v = M$ it is nonpositive; on $h = 0$ it is at most $(w - v)/\gamma$. Clipping retains each active inequality by the corresponding face test, including when several faces are active. Thus $f(t,y) \in T_D(y)$ for every $t \geq 0$ and $y \in D$.

For completeness, let π_D be Euclidean projection onto the closed convex set D. On a finite time interval choose a common state Lipschitz constant L for f on a compact neighborhood of the trajectory and its projected image. The projection normal inequality and the tangency just proved give

$$\begin{aligned}\frac{d}{dt}\,\mathrm{dist}(z(t),D)^2 &= 2\langle z - \pi_D z, f(t,z)\rangle \\ &\leq 2\langle z - \pi_D z, f(t,z) - f(t,\pi_D z)\rangle \\ &\leq 2L\,\mathrm{dist}(z(t),D)^2 \qquad \text{a.e.}\end{aligned}$$

Grönwall and $z(t_0) \in D$ give zero distance throughout that interval. As the interval was arbitrary, this proves invariance for all future time [1, 2]. □

Safety thus retains the face tests; the asymptotic results below retain the stopped-leader assumptions.

For finite $N > 0$ and positive normalization scales s_h, s_v, define

$$\beta(v) = \tanh(Nv_+/s_v), \tag{16}$$

$$u_{\mathrm{rec}}(t,z) = \beta(v)k(t,z) + (1-\beta(v))u_b(t,z), \tag{17}$$

$$\sigma(h,v) = \tanh(Nh_+/s_h)\tanh(N(M-v)_+/s_v). \tag{18}$$

The same parameters $\lambda, h_s, h_g, N, s_h, s_v$ are used for both implementations. The reference u_{rec} is bounded and regular; it need not itself be safe on the headway and upper-speed faces.

4.1 Boundary-matching blend

Define

$$u_{\mathrm{B}} = \sigma u_{\mathrm{rec}} + (1-\sigma)u_b = \sigma\beta k + (1 - \sigma\beta)u_b. \tag{19}$$

Because $\sigma\beta$ is exactly the three-factor product weight, this is a product blend with a shared recovery reference. It equals the certified backup on all three faces and therefore preserves D. The factorization identifies the reference passed to each safety operation.

4.2 CBF projection

For positive gains $\kappa_h, \kappa_v, \kappa_M$, define

$$L = \max\{A_-, -\kappa_v v\}, \tag{20}$$

$$R = \min\left\{A_+, \frac{w - v + \kappa_h h}{\gamma}, \kappa_M(M - v)\right\}. \tag{21}$$

When $L \le R$, set

$$p_{\mathrm{C}} = \mathrm{clip}_{[L,R]}(a + bu_{\mathrm{rec}}), \qquad u_{\mathrm{C}} = (p_{\mathrm{C}} - a)/b. \tag{22}$$

This is the scalar least-change CBF filter applied to the shared recovery reference. It enforces the actuator bound and three barrier inequalities. Input and acceleration projections are equivalent because $b > 0$ is a positive scalar at each state. The recovery term is incorporated before the safety projection. Adding it afterwards would require a new safety proof.

On its feasible domain the formula is locally Lipschitz. An ambient extension can use the same min/max expression, followed by clipping the input to $[-U, U]$, even where interval endpoints would be reversed outside the certified feasible domain. It agrees with (22) on that domain (D, or $K_{\bar{d}}$ below); no recovery guarantee from unsafe initial states is inferred. Joint feasibility on the certified set and this regular construction suffice for the tangency proof; strict feasibility on an open neighborhood of every corner is not being assumed.

Proposition 4.1 (Recovery inherited by both implementations)**.** *Assume the conditions of Theorem 2.1, a finite N, and any admissible regular nominal command on the infinite horizon. The product blend (19) satisfies Theorem 3.1. The CBF controller (22) also satisfies that theorem when its interval is feasible on a safe invariant domain containing the initial state and the relevant low-speed strips. Both keep every accepted rest point fixed when (12) holds.*

Proof. Fix $d_\eta < d_0$. Strict authority and continuity give $p_b(d, 0) \ge m_\eta > 0$ on $[d_\eta, d_0]$. On a sufficiently thin low-speed strip, $p_b \ge m_\eta/2$. If $b \le B_\eta$ there, then

$$|(a + bu_{\mathrm{rec}}) - p_b| \le 2UB_\eta\beta(v) \le 2UB_\eta Nv/s_v.$$

The corresponding blend perturbation is no larger, since $0 \le \sigma \le 1$. Reducing the strip thickness ensures both seed and blend accelerations are at least $m_\eta/4$, uniformly over time and all admissible nominal commands.

For the CBF projection, at excess-gap rest,

$$R(d, 0) = \min\{A_+(d, 0), \kappa_h(d - \epsilon)/\gamma, \kappa_M M\}.$$

Each term has a positive minimum on $[d_\eta, d_0]$. On a sufficiently thin strip, $R \ge r_\eta > 0$. For a nonempty interval, projecting a value at least $m_\eta/4$ onto an interval with upper endpoint at least r_η gives

$$p_{\mathrm{C}} \ge \min\{m_\eta/4, r_\eta\} > 0.$$

Theorem 3.1 applies to both controllers.

At an accepted rest point, $\beta = \rho_g = 0$ and attainable zero acceleration gives $p_b = a + bu_{\mathrm{rec}} = 0$. The blend also has zero acceleration. The CBF interval has $L = 0$ and $R \ge 0$, so the projection leaves zero unchanged. □

The implementations need not have identical departure acceleration: the CBF upper bounds can reduce the backup's positive acceleration. They share a recovery mechanism and guarantee, not necessarily the same trajectory or recovery time. An ordinary CBF already recovers whenever its nominal reference and feasible interval produce suitable progress. The result provides a guarantee independent of the particular admissible nominal command; it is not a claim that CBFs intrinsically freeze.

5 Constructing CBF feasibility on a compact operating domain

For arbitrary nonlinear coefficients, boundary feasibility does not automatically imply that one fixed set of natural-barrier gains works on the entire unbounded domain D. The following construction makes the shared assumptions explicit.

Proposition 5.1 (Constructive barrier gains)**.** *Assume the autonomous stopped-leader face tests. Choose $\bar{d} \geq \max\{d_0, \epsilon + \gamma M\}$ and put $K_{\bar{d}} = D \cap \{d \leq \bar{d}\}$. Let finite nonnegative constants satisfy*

$$|A_\pm(d, v) - A_\pm(\tilde{d}, \tilde{v})| \leq L_{\pm,d}|d - \tilde{d}| + L_{\pm,v}|v - \tilde{v}| \quad \text{on } K_{\bar{d}}.$$

Positive gains chosen to satisfy

$$\kappa_h \geq \gamma L_{-,d}, \quad \kappa_v \geq \max\{L_{+,v}, 1/\gamma\}, \quad \kappa_M \geq \gamma L_{-,d} + L_{-,v} \tag{23}$$

make (20)–(21) feasible everywhere on $K_{\bar{d}}$. This domain is invariant for the resulting stopped-leader CBF controller.

Proof. Comparing (d, v) with $(d, 0)$ gives $A_+(d, v) \geq -L_{+,v}v$. Comparing with $(\epsilon + \gamma v, v)$ gives $A_-(d, v) \leq -v/\gamma + L_{-,d}h$. Finally, put $\tilde{d} = \max\{d, \epsilon + \gamma M\}$. The point $(\tilde{d}, M)$ belongs to $K_{\bar{d}}$ and $0 \leq \tilde{d} - d \leq \gamma(M - v)$, so

$$A_-(d, v) \leq (\gamma L_{-,d} + L_{-,v})(M - v).$$

Each of the two lower candidates in L is no larger than each of the three upper candidates in R: use $A_- \leq A_+$, the three inequalities just proved, $-\kappa_v v \leq 0 \leq \kappa_M(M - v)$, and $-\kappa_v v \leq -v/\gamma \leq (-v + \kappa_h h)/\gamma$. Thus $L \leq R$. Tangency holds for all three original faces and the additional face $d = \bar{d}$, where $\dot{d} = -v \leq 0$. Applying invariance directly to $K_{\bar{d}}$ proves the assertion. □

Coordinate bounds can be chosen equal to a Lipschitz constant for $A_\pm$ on this compact set. Select strictly positive gains even when a bound in (23) is zero. This supplies a family of fixed-gain CBF realizations for prescribed bounded initial envelopes; it does not assert that one fixed gain triple works on unbounded D. The estimates are sufficient and may be conservative.

For the optimal-velocity specialization $a = \alpha(\mathcal{V}(d) - v)$, $b = 1$, $\alpha > 0$, with a nondecreasing $L_\mathcal{V}$-Lipschitz function $\mathcal{V} \leq M$, the sharper sufficient gains are

$$\kappa_h \geq \alpha\gamma L_\mathcal{V}, \qquad \kappa_v \geq \max\{\alpha, 1/\gamma\}, \qquad \kappa_M \geq \alpha,$$

together with the face certificate. They apply on all of D and for every $w \in [0, M]$. Indeed, $A_+ \geq -\alpha v$ follows from the lower-face test at $d = \epsilon$ and monotonicity of $\mathcal{V}$. Comparing with the headway face gives $A_- \leq -v/\gamma + \alpha L_\mathcal{V} h$, while $A_- \leq \alpha(M - v)$ follows from $\mathcal{V} \leq M$. Pairwise comparison of the two lower and three upper interval candidates then proves feasibility; nonnegative w only enlarges the headway upper bound.

6 Spatial recoverability and exact minimum entry time

Relation to path parameterization. Spatial speed/energy coordinates and minimum-time path tracking are classical tools. Convex formulations appear in [3, 4], while [5] addresses bounded acceleration and velocity along a prescribed path. The results here use scalar order to identify the exact recovery domain, launch obstruction, and physical entry clock for the stated dynamics; the coordinate transformation itself is not claimed as new.

This section treats the autonomous stopped-leader system

$$\dot{d} = -v, \qquad \dot{v} = a(d, v) + b(d, v)u, \qquad |u| \leq U,$$

under the regularity and growth assumptions of the model section. Write $A_\pm = a \pm bU$ and assume the whole-domain face certificate

$$A_+(d,0) \geq 0, \qquad A_-(d,M) \leq 0, \qquad A_-(\epsilon + \gamma v, v) \leq -v/\gamma.$$

The last two inequalities are imposed on the corresponding faces of D, including intersections. We do not assume strict positive rest authority. Admissible inputs in this section are measurable functions taking values in $[-U,U]$ whose trajectories remain in D for all future time. Their existence is supplied by the safe backup constructed earlier.

The phase-plane minimum-time method goes back, in particular, to Bobrow–Dubowsky–Gibson and Shin–McKay [6, 7]. Their path-constrained manipulator problems already include zero endpoint speeds and acceleration/ deceleration switching. Squared-speed and admissible-velocity methods continue this lineage [8, 9]. Neither a distance reparameterization nor departure from rest is claimed as new here. The present result makes the physical branch selection explicit when launch authority vanishes, proves the resulting trapping and infinite-clock alternatives, and identifies an asymptotic recoverability set together with a free-terminal-speed entry value. Whole-domain ceiling viability removes the need for the backward terminal-braking calculation required by prescribed terminal-speed problems. Those classical problems are not asserted to be special cases of the present theorem.

Viability/capture-basin representations of constrained Hamilton–Jacobi problems are also established [10, 11]. The scalar construction below exploits the particular monotone two-state structure within this broader subject. It does not replace the general theory, and a numerical discretization is not identified with the exact continuous optimum.

6.1 A physical maximal energy profile

Fix $z_0 = (d_0, v_0) \in D$ and put $S_0 = d_0 - \epsilon$. When $S_0 > 0$, define, for $0 \leq s < S_0$,

$$B(s) = \frac{1}{2} \min\left\{ \frac{d_0 - s - \epsilon}{\gamma}, M \right\}^2, \tag{24}$$

$$F(s,E) = A_+(d_0 - s, \sqrt{2E}). \tag{25}$$

The ceiling B is positive on $[0, S_0)$, Lipschitz, and piecewise C^1. Its only possible interior kink is where the speed and headway caps coincide. Its almost-everywhere derivative is zero on the speed-cap part and $-(d_0 - s - \epsilon)/\gamma^2$ on the headway-cap part.

Call the initial state *launchable* if $d_0 > \epsilon$ and either $v_0 > 0$ or $A_+(d_0, 0) > 0$. At a nonlaunchable state set $s_*(z_0) = 0$. At a launchable state, set $E^+(0) = v_0^2/2$ and solve

$$(E^+)'(s) = \begin{cases} F(s, E^+(s)), & E^+(s) < B(s), \\ \min\{F(s, E^+(s)), B'(s)\}, & E^+(s) = B(s), \end{cases} \tag{26}$$

up to the endpoint $s_* \leq S_0$ of the first positive-energy interval after departure. More explicitly, $E^+(s) > 0$ for $0 < s < s_*$, and construction stops at the first subsequent zero or at S_0, whichever occurs first. It is not continued through zero by choosing another branch of the spatial equation.

Lemma 6.1 (Construction, clock, and admissible realization)**.** *At every launchable state the profile in (26) exists uniquely, $s_* > 0$, and $E^+(s) \to 0$ as $s \uparrow s_*$. For $0 < q < s_*$, its clock*

$$t^+(q) = \int_0^q \frac{ds}{\sqrt{2E^+(s)}} \tag{27}$$

is finite, strictly increasing, and satisfies $t^+(q) \to \infty$ as $q \uparrow s_*$. Inverting this clock gives a global safe trajectory under an admissible measurable input, with limiting state $(d_0 - s_*, 0)$. A nonlaunchable initial state admits only the stationary safe trajectory.

Proof. Positive-energy construction. Fix a spatial starting point $\sigma < S_0$ with $0 < E_\sigma \le B(\sigma)$. On a sufficiently short interval, restrict energies to a neighborhood bounded away from zero. The mapping F is then continuous and Lipschitz in E, uniformly in s. Indeed, local state-Lipschitz continuity of A_+ and the smoothness of $E \mapsto \sqrt{2E}$ away from zero give this property. For a trial path e, define

$$x_e(s) = E_\sigma + \int_\sigma^s F(r, e(r))\, dr,$$

and the one-sided reflection operator

$$(\mathcal{R}_B x)(s) = x(s) - \max\left\{0,\ \sup_{\sigma \le r \le s} [x(r) - B(r)]\right\}.$$

This operator has Lipschitz constant at most two in the uniform norm. On a sufficiently short interval, the mapping $e \mapsto \mathcal{R}_B x_e$ is therefore a contraction. Bounds on F and the Lipschitz constant of B ensure that the reflected path remains in the chosen positive-energy neighborhood. Its unique fixed point is absolutely continuous. Writing it as $E = x_E - K$, the nondecreasing regulator K increases only at $E = B$ and satisfies

$$K' = \mathbf{1}_{\{E=B\}}(F(s,E) - B')_+ \quad \text{a.e.}$$

This is exactly (26). Conversely, for any absolutely continuous solution of that equation, define

$$K(s) = \int_\sigma^s \mathbf{1}_{\{E(r)=B(r)\}}(F(r, E(r)) - E'(r))\, dr.$$

The differential rule makes K nondecreasing, starting from zero, and increasing only on ceiling contact, with $E = x_E - K \le B$. It is therefore the unique one-sided Skorokhod pair for the driver x_E, so every such solution is a fixed point of the contraction. This proves uniqueness among all absolutely continuous solutions, not only those generated by the construction. The argument also applies across the one ceiling kink, since B is Lipschitz and its derivative is used only almost everywhere.

Admissibility of the selected slope. Below the ceiling the slope is A_+. On the speed-cap part of the ceiling, $B' = 0$ and the face certificate gives $A_- \le B'$. On the headway-cap part, $B' = -v/\gamma$ and the corresponding face certificate again gives $A_- \le B'$. Thus

$$A_-(d_0 - s, \sqrt{2E^+}) \le (E^+)'(s) \le A_+(d_0 - s, \sqrt{2E^+}) \quad \text{a.e.}$$

When $A_+ < B'$, the selected slope is A_+ and the path leaves the ceiling downward; the argument does not assume that sliding along the ceiling is always possible.

A positive-authority rest start. If $v_0 = 0$ and $g_0 = A_+(d_0, 0) > 0$, then $B(0) > 0$. The unique local full-throttle time trajectory, with $u = U$, satisfies

$$v(t) = g_0 t + o(t), \quad s(t) = \frac{g_0 t^2}{2} + o(t^2), \quad E(t) = \frac{g_0^2 t^2}{2} + o(t^2).$$

For small positive time it is safe and has strictly increasing s, so it defines a spatial solution satisfying

$$E^+(s) = g_0 s + o(s). \tag{28}$$

Any continuous nonnegative solution of (26) with this initial value is below the ceiling near zero and, by continuity of F and $g_0 > 0$, satisfies $cs \le E(s) \le Cs$ for suitable $c, C > 0$. For two such solutions,

$$|F(s, E_1) - F(s, E_2)| \le L\frac{2|E_1 - E_2|}{\sqrt{2E_1} + \sqrt{2E_2}} \le \frac{C_1}{\sqrt{s}}|E_1 - E_2|.$$

The coefficient is integrable at zero. Grönwall's inequality therefore proves uniqueness through departure; the positive-energy construction continues that solution afterward. Equation (28) also makes the initial singularity in the clock integrable.

Continuation and the terminal zero. All spatial states lie in the compact set $\{\epsilon \le d \le d_0,\ 0 \le v \le M\}$. The acceleration endpoints and B' are bounded there, so the profile has bounded slope. It can cease to continue at an interior point $s_* < S_0$ only if its limit is zero. At S_0, the ceiling itself tends to zero. Hence $E^+(s) \to 0$ at either endpoint. If $s_* < S_0$, then $B(s_*) > 0$ and the profile is strictly below the ceiling near its endpoint. The lower-speed face gives $A_+(d_0 - s_*, 0) \ge 0$. If this value were positive, then $(E^+)' \ge c > 0$ near s_*, contradicting approach to zero from positive values. Therefore

$$A_+(d_0 - s_*, 0) = 0 \qquad (s_* < S_0). \tag{29}$$

Physical realization and divergence of the clock. For every $q < s_*$, continuity and positivity of E^+ away from zero, together with (28) when needed, make $t^+(q)$ finite. Invert the clock on its range and set

$$d(t) = d_0 - s(t), \qquad v(t) = \sqrt{2E^+(s(t))}.$$

The chain rule gives $\dot{d} = -v$ and $\dot{v} = (E^+)'(s(t))$ almost everywhere. The startup portion is the full-throttle trajectory just constructed. At other times the energy is positive, so the inverse and the chain rule are ordinary local changes of variable. The selected acceleration is attainable, and

$$u(t) = \frac{(E^+)'(s(t)) - a(d(t), v(t))}{b(d(t), v(t))}$$

is measurable and belongs to $[-U, U]$ almost everywhere.

Suppose first that $s_* < S_0$ and that the clock had a finite endpoint T_*. Near that endpoint the physical trajectory obeys the locally Lipschitz full-throttle dynamics and tends to $(d_*, 0)$, where $d_* = d_0 - s_*$ and $A_+(d_*, 0) = 0$. Set $Y = d - d_* + v > 0$ there. Local Lipschitz continuity gives $|A_+(d, v)| \le LY$, and hence

$$\dot{Y} = -v + A_+(d, v) \ge -(1 + L)Y.$$

It follows that $Y(t) \ge Y(t_1)e^{-(1+L)(t-t_1)}$ on a terminal interval. This contradicts $Y \to 0$ at a finite T_*. If $s_* = S_0$, then

$$E^+(s) \le \frac{(S_0 - s)^2}{2\gamma^2}, \qquad t^+(q) \ge \gamma \int_{s_1}^{q} \frac{ds}{S_0 - s} \longrightarrow \infty.$$

Thus the inverse clock is defined for every future time, and the limiting state is $(d_0 - s_*, 0)$.

Nonlaunchable states. If $d_0 = \epsilon$, safety and $\dot{d} = -v \le 0$ force $d = \epsilon$ and $v = 0$ forever. The face certificate makes zero acceleration attainable at this corner. Otherwise nonlaunchability means $v_0 = 0$ and $A_+(d_0, 0) = 0$. For any safe input, put $q = d_0 - d = \int_0^t v \ge 0$ and $Y = q + v \ge 0$. Local Lipschitz continuity yields

$$\dot{Y} = v + \dot{v} \le (1 + L)Y, \qquad Y(0) = 0,$$

until a possible exit from a small neighborhood. Grönwall and a first-exit argument prevent that exit and force $Y \equiv 0$. The stationary trajectory is attainable with $u = U$ because $A_+(d_0, 0) = 0$. This proves the final assertion. □

6.2 Comparison with every safe measurable control

Lemma 6.2 (Spatial domination and elapsed time)**.** *Let z_0 be launchable, and let an arbitrary safe measurable input produce traveled distance $s_u(t) = d_0 - d_u(t)$ and total travel $S_u = \lim_{t\to\infty} s_u(t)$.*

On every spatial interval actually traversed by that trajectory, its energy $e_u(s)$ is well defined and Lipschitz. For $0 \le s < \min\{S_u, s_*\}$,

$$e_u(s) \le E^+(s).$$

For every positive level $q < \min\{S_u, s_*\}$, the time of first arrival at $s = q$ is at least $t^+(q)$. Moreover, $S_u \le s_*$.

Proof. Bounded input and the compact safe state envelope bound actual acceleration by a constant C. If $s_u(t_1) = s_u(t_2)$ with $t_1 < t_2$, then nonnegative speed and $\int_{t_1}^{t_2} v = 0$ give $v = 0$ throughout that interval. Thus energy has the same value at all times corresponding to one traveled-distance value. The spatial function e_u is consequently well defined. Furthermore,

$$\left|\frac{v(t_2)^2 - v(t_1)^2}{2}\right| = \left|\int_{t_1}^{t_2} v(t)p(t)\,dt\right| \le C\big(s_u(t_2) - s_u(t_1)\big),$$

so it is Lipschitz in s. Where $e_u > 0$, the chain rule yields $e_u' = p \le F(s, e_u)$. On its zero set the derivative of the nonnegative absolutely continuous function e_u is zero almost everywhere; $F(s, 0) \ge 0$ by the face certificate. Hence $e_u' \le F(s, e_u)$ almost everywhere on its whole spatial domain. Safety also gives $e_u \le B$.

Fix $q < \min\{S_u, s_*\}$. On the set where $e_u > E^+$, the latter profile is strictly below B and follows F. For $W = (e_u - E^+)_+$, the positive-part chain rule and local Lipschitz continuity in velocity give

$$W'(s) \le \frac{C_2}{\sqrt{2E^+(s)}} W(s) \quad \text{a.e. on } (0, q).$$

This coefficient is integrable by Lemma 6.1. Since $W(0) = 0$, Grönwall gives $e_u \le E^+$ up to q. No monotonicity of A_+ in velocity is required.

Let $T_u(q)$ be the finite first time at distance $q < S_u$. The change-of-variable formula for the nondecreasing absolutely continuous map s_u gives, by first using truncated nonnegative integrands and then monotone convergence,

$$\int_0^q \frac{ds}{\sqrt{2e_u(s)}} = \int_0^{T_u(q)} \mathbf{1}_{\{v_u(t)>0\}}\,dt \le T_u(q).$$

The convention $1/0 = +\infty$ is used inside the spatial integral; the substitution is justified with the truncations even if the trajectory contains rests. Spatial domination now yields

$$T_u(q) \ge \int_0^q \frac{ds}{\sqrt{2E^+(s)}} = t^+(q).$$

If $S_u > s_*$, choose one level $\bar{q}$ with $s_* < \bar{q} < S_u$. The finite time $T_u(\bar{q})$ bounds $T_u(q)$ for all $q < s_*$, contradicting $t^+(q) \to \infty$. Thus $S_u \le s_*$. The case $S_u = 0$ is immediate, and the nonlaunchable case has already been resolved by the no-escape argument. □

6.3 The exact set and the exact entry value

Define the asymptotic recoverability set by

$$\mathcal{C}_{\text{asy}} = \{z_0 \in D : \text{some safe admissible input gives } \operatorname{dist}(z(t), \mathcal{T}_g) \to 0\}.$$

This asymptotic notion is kept distinct from finite-time arrival at the rest target. For $\eta > 0$, define

$$T_\eta^*(z_0) = \inf_{u \text{ safe}} \inf\{t \ge 0 : d_u(t) \le d_g + \eta\},$$

with $\inf \varnothing = +\infty$.

Theorem 6.3 (Exact asymptotic recoverability and minimum gap-entry time)**.** *Under the assumptions of this section,*

$$\mathcal{C}_{\text{asy}} = \{(d_0, v_0) \in D : d_0 - s_*(d_0, v_0) \le d_g\}. \tag{30}$$

For $S_\eta = d_0 - d_g - \eta$,

$$T^*_\eta(z_0) = \begin{cases} 0, & S_\eta \le 0, \\ \displaystyle\int_0^{S_\eta} \frac{ds}{\sqrt{2E^+(s)}}, & 0 < S_\eta < s_*, \\ +\infty, & S_\eta > 0,\ S_\eta \ge s_*. \end{cases} \tag{31}$$

The minimum limiting gap among all safe trajectories is $d_0 - s_*$*. The maximal-profile trajectory attains that limiting gap and every finite positive entry-time value in* (31)*.*

Proof. Every safe trajectory has $d(t) \downarrow d_\infty \ge \epsilon$, and $\int_0^\infty v = d_0 - d_\infty < \infty$. Its acceleration is bounded on the compact state envelope, so v is uniformly continuous and tends to zero. Recovery therefore holds precisely when $d_\infty \le d_g$. Lemma 6.2 bounds total travel by s_*, whereas Lemma 6.1 gives one safe global trajectory with total travel exactly s_*. This proves (30) and the minimum limiting-gap assertion.

If $S_\eta \le 0$, the entry condition is initially satisfied. For $0 < S_\eta < s_*$, comparison gives the displayed integral as a lower bound, and the maximal-profile realization attains it. If $S_\eta > s_*$, total-travel domination prevents entry. If $S_\eta = s_* > 0$, any purported finite entry time would dominate $t^+(q)$ for every $q < s_*$, contradicting its divergence. For a nonlaunchable state, $s_* = 0$ and the stationary trajectory resolves the remaining case. □

For $d_0 > d_g$, an equivalent form of (30) is physical launchability together with $E^+(s) > 0$ for every $0 < s < d_0 - d_g$. The upper endpoint $s = d_0 - d_g$ is excluded: zero energy there can describe asymptotic recovery to $(d_g, 0)$. Every state with $d_0 \le d_g$ belongs to the set, including positive-speed states, because every safe continuation has limiting gap at most its initial gap.

The whole-domain rest-authority criterion is also transparent from this construction. If $A_+(d, 0) > 0$ at every $d > d_g$, an interior terminal zero cannot have $d_0 - s_* > d_g$, by (29); hence all initial states recover. If that condition fails, a nonlaunchable excessive rest state does not recover. The earlier backup construction additionally provides one common regular feedback in the whole-domain case.

6.4 Extension to a viable velocity ceiling

The same reduction applies to a moderately more general domain. Let $V : [\epsilon, \infty) \to [0, \infty)$ be locally Lipschitz, positive on (ϵ, ∞), with $V(\epsilon) = 0$. Assume V is piecewise C^1 with finitely many pieces on each compact interval and finite one-sided derivatives at their endpoints. Define

$$D_V = \{(d, v) : d \ge \epsilon,\ 0 \le v \le V(d)\}, \qquad \mathcal{T}_{g,V} = \{(d, 0) : \epsilon \le d \le d_g\}.$$

Retain the autonomous input-affine model and its local state-Lipschitz regularity, and impose

$$\begin{aligned} A_+(d, 0) &\ge 0 \quad (d \ge \epsilon), \\ A_-(d, V(d)) &\le -V(d)V'(d) \quad (d > \epsilon), \\ A_-(\epsilon, 0) &\le 0. \end{aligned} \tag{32}$$

The ceiling inequality is imposed on every smooth piece and for each of the one-sided derivatives at a kink. These simultaneous inequalities are sufficient face conditions; a weaker nonsmooth tangent-cone characterization is not needed here.

Proposition 6.4 (Spatial reduction for a viable ceiling)**.** *Under these assumptions, Lemmas 6.1 and 6.2 and Theorem 6.3 remain valid with D_V and $\mathcal{T}_{g,V}$ in place of D and $\mathcal{T}_g$, respectively, and with*

$$B_V(s) = \tfrac{1}{2}V(d_0 - s)^2 \tag{33}$$

in place of B. The launchability convention and the instruction to stop at the first zero after departure are unchanged.

Proof. For fixed d_0, the state envelope with $\epsilon \le d \le d_0$ is compact, because V is bounded on that interval. Its spatial ceiling is positive before S_0, Lipschitz, and piecewise C^1. Consequently the reflection contraction, positive-authority startup, and continuation arguments in Lemma 6.1 apply without change. On the ceiling,

$$B_V'(s) = -V(d_0 - s)V'(d_0 - s),$$

so (32) makes the selected slope attainable almost everywhere. Finitely many ceiling kinks on the compact interval do not affect this conclusion. At the corner, the first and last conditions in (32) make zero acceleration attainable.

An interior terminal zero still satisfies $A_+(d_*, 0) = 0$, and the same local Lipschitz no-escape argument makes its clock infinite. At the corner, local Lipschitz continuity and $V(\epsilon) = 0$ give $V(d) \le L(d - \epsilon)$ near ϵ. Hence

$$E^+(s) \le \tfrac{1}{2}L^2(S_0 - s)^2,$$

which again makes the clock diverge. Spatial domination uses only the common ceiling, local Lipschitz continuity of A_+, and the integrable departure singularity, all of which hold here. Finally every safe trajectory has bounded acceleration in its compact envelope, $d \downarrow d_\infty$, and $\int_0^\infty v = d_0 - d_\infty < \infty$, so $v \to 0$. The proof of Theorem 6.3 now applies verbatim. □

This extension concerns the spatial optimal-control result. The particular backup, blend, and interval-CBF formulas for D do not automatically transfer to D_V; their feasibility and regularity would have to be checked for the new ceiling. Time dependence, a moving leader, arbitrary state constraints, terminal speed requirements, or a jerk state likewise require separate arguments. The original headway/speed ceiling is recovered with $V(d) = \min\{(d - \epsilon)/\gamma, M\}$.

6.5 Interior HJB interpretation and its boundary limits

Fix a gap-entry threshold $d_\eta = d_g + \eta$ and let T denote the value just constructed, either on D or on D_V. On any open subset of

$$\{(d, v) : d > d_\eta,\ 0 < v < V(d)\}$$

on which T is finite and continuous, the dynamic programming principle gives the interior viscosity equation

$$vT_d + \max_{p \in [A_-(d,v), A_+(d,v)]} (-pT_v) = 1. \tag{34}$$

Here V is the corresponding ceiling for the domain being used. For sufficiently short times, every control from an interior point stays in the domain, so the Hamiltonian uses the full attainable acceleration interval. The usual smooth-test-function argument for the dynamic programming principle yields (34); at differentiability points it is the classical equation [12]. Thus the reflected ODE and quadrature provide a direct verification and pointwise value evaluation for this monotone class. One integration is required per initial state; this is not one scalar integration for the entire two-dimensional value field.

A global boundary characterization requires additional care. State-constrained minimum-time and constrained Hamilton–Jacobi theories distinguish boundary notions and impose hypotheses for

their comparison principles [13, 14, 10]. We do not claim those hypotheses or a global continuous-viscosity uniqueness theorem here. In fact, continuity at the rest target can fail in an example already covered by the theorem. For $a = -1 - v$ and $b = U = M = \gamma = 1$, write $S = d - d_\eta$. Then

$$T(d, v) = \begin{cases} 0, & S \le 0, \\ -\log(1 - S/v), & 0 < S < v, \\ +\infty, & S > 0,\ S \ge v. \end{cases}$$

At $(d_\eta, 0)$ the value is zero, whereas along $d = d_\eta + v/2$ with $v \downarrow 0$ it is $\log 2$. This does not obstruct the exact trajectory/value theorem or the local interior interpretation above; it rules out an unsupported global continuity assertion.

6.6 Why zero-energy branches and terminal speed matter

The spatial equation must not be solved through an arbitrary zero-energy branch. For example, set

$$a = -1, \qquad b(v) = 1 + v_+, \qquad U = M = \gamma = 1.$$

The attainable endpoints on D are $A_+ = v$ and $A_- = -2 - v$, and the face tests hold. Every safe time-domain trajectory from rest satisfies $\dot{v} \le v$ and remains fixed. Yet

$$E' = \sqrt{2E}, \quad E(0) = 0$$

has the branch $E = s^2/2$. Its clock $\int_0^q ds/s$ diverges at the initial endpoint, so it is not a finite-time departure. The launchability convention explicitly excludes this branch.

The entry target has free terminal speed. To see the distinction from arrival at $\mathcal{T}_g$, take

$$a = 0, \quad b = U = M = \gamma = 1, \quad \epsilon = 1, \quad h_g = 1, \quad (d_0, v_0) = (9/4, 0).$$

Full acceleration reaches $d = d_g = 2$ safely at $T = 1/\sqrt{2}$ with $v = 1/\sqrt{2}$. A rest-to-rest trajectory of duration T under $|\dot{v}| \le 1$ travels at most $T^2/4$. At least $1/4$ distance is needed to enter $\mathcal{T}_g$, so stopped arrival requires $T \ge 1$. Acceleration $+1$ for $1/2$ followed by -1 for $1/2$ attains this time safely. Thus the stopped-target minimum is 1, although the minimum gap-entry time to d_g is $1/\sqrt{2}$. Terminal braking requires a backward controllability condition, as in the classical phase-plane and later path parameterization methods [6, 7, 8].

When $h_g = 0$, no non-target safe state reaches $\mathcal{T}_g$ in finite time. Indeed,

$$\dot{d} \ge -\frac{d - \epsilon}{\gamma} \quad \Longrightarrow \quad d(t) - \epsilon \ge (d_0 - \epsilon)e^{-t/\gamma} > 0.$$

Positive-tolerance entry is consequently the appropriate finite benchmark for that target.

The maximizing policy takes maximum attainable acceleration in the interior and uses the maximal admissible ceiling slope on an active ceiling. It may be discontinuous at contact; attainment above is asserted over measurable admissible inputs, not over the paper's narrower class of regular feedbacks. No jerk or actuator-rate constraint is included. Adding such a constraint would change the optimal-control problem and its state dimension.

6.7 An analytic partial set and a filter limitation

Take

$$a(d, v) = -1 - v, \qquad b = U = M = \gamma = 1.$$

Then $A_+ = -v$, $A_- = -2 - v$, and all face tests hold, while $A_+(d, 0) = 0$ everywhere. For every admissible control,

$$\dot{h} = 1 - u \ge 0.$$

Consequently recovery requires $h_0 \le h_g$. Conversely, $u = 1$ gives

$$v(t) = v_0 e^{-t}, \quad d(t) = d_0 - v_0(1 - e^{-t}), \quad h(t) = h_0.$$

The exact result is therefore

$$\mathcal{C}_{\text{asy}} = D \cap \{h \le h_g\}, \qquad s_* = v_0, \qquad E^+(s) = \frac{1}{2}(v_0 - s)^2 \quad (0 \le s < v_0).$$

For $0 < S_\eta < v_0$,

$$T_\eta^* = -\log\left(1 - \frac{S_\eta}{v_0}\right).$$

It is zero when the threshold is initially satisfied and infinite when positive required travel is at least v_0. The example separates inability to restart from the ability of a moving state to recover using its existing velocity.

The exact set is a plant-level existence statement. It does not automatically become the domain of recovery of either implemented filter. To see this, fix $h_g > 0$ and a boundary point $h = h_g$, $0 < v < 1$ of this example's recoverability set. The current backup has $\rho_g = 0$, $r_\lambda = -v$, and $u_b = 1$. For an admissible nominal $k < 1$, the recovery blend gives

$$u_B = 1 - \sigma\beta(1 - k) < 1, \qquad \dot{h} = \sigma\beta(1 - k) > 0.$$

It immediately leaves $h \le h_g$, after which recovery is impossible because h cannot decrease.

For the recovery CBF at the same point,

$$R = -v, \qquad L = \max\{-2 - v, -\kappa_v v\}, \qquad p_{\text{rec}} = -v - \beta(1 - k) < -v.$$

If $\kappa_v > 1$, then $L < -v$ and projection still gives $p_C < -v$, hence $\dot{h} = -v - p_C > 0$. These gains are feasible for this plant: $R = -v$ throughout D and $L \le -v$ when $\kappa_v \ge 1$. In the special case $\kappa_v = 1$, $L = R = -v$ and the filter forces $u = 1$, preserving the full set.

The original filter inheritance theorem remains valid because it assumes strict rest authority outside the target. Extending that guarantee to a partial recoverability set requires a separate condition that preserves that set; safety of D alone does not provide it. The numerical study therefore separates the exact plant-level set, the optimal-entry benchmark, and the observed success sets of the implemented controllers.

6.8 Relation to the progress estimate and numerical evaluation

The exact value supplies a lower bound for every safe controller. The uniform low-speed progress theorem supplies an upper bound for a specified controller when its hypotheses hold. For such a controller and a common tolerance,

$$T_\eta^* \le T_\eta^{\text{controller}} \le \frac{(\nu_\eta - v_0)_+}{c_\eta} + \frac{d_0 - d_g - \eta}{\nu_\eta}.$$

The upper bound proves progress of the implementation and cannot be replaced by an optimal lower bound. Ratios against T_η^* are meaningful only when $0 < T_\eta^* < \infty$; initially satisfied thresholds and unreachable events are reported separately. For general nonlinear coefficients, evaluation entails one scalar constrained ODE and a quadrature per initial state, not a universal closed form or a single integration producing the entire value field. A finite numerical approximation is not claimed to be a rigorous lower bound unless its error is enclosed. The minimum-time value also has a different objective from the finite-horizon quadratic HJB nominal used in the matched filter comparison.

7 A feasible performance floor and its exact worst-case clock

The preceding energy profile describes what the dynamics permit. This section constructs a filter whose entry time is controlled relative to that profile, uniformly over its nominal command. The construction is an episode-based, reflected controller. Its acceleration can jump when a performance boundary becomes active; it is not claimed to be a locally Lipschitz feedback, to have bounded jerk, or to retain its guarantee under an arbitrary sampled implementation.

Fix a launchable initial state $z_0 = (d_0, v_0)$ and a physical-gap entry level $d_\eta < d_0$ such that

$$S = d_0 - d_\eta \in (0, s_*(z_0)).$$

The maximal profile E^+ and ceiling B are those of Section 6; they are computed once for this episode. A target already satisfied has entry time zero and does not activate the construction. Set $E_0 = v_0^2/2$ and

$$f_\pm(s, E) = A_\pm(d_0 - s, \sqrt{2E}), \tag{35}$$

$$q(s) = \theta E^+(s), \qquad 0 < \theta < 1. \tag{36}$$

Thus $q < B$ throughout $[0, S]$. At a rest start $q(0) = E^+(0) = 0$, while the upper ceiling remains strictly positive. On every compact subinterval of $(0, S]$, q is positive. At a rest start,

$$q(s) = \theta g_0 s + o(s), \qquad g_0 = A_+(d_0, 0) > 0. \tag{37}$$

All derivatives of the profiles and ceiling below are understood almost everywhere. Values at an isolated ceiling kink do not affect a trajectory or a clock.

The central design test is the scalar inequality

$$q'(s) \le f_+(s, q(s)) \qquad \text{a.e. on } [0, S]. \tag{F}$$

It tests acceleration at the floor speed $\sqrt{\theta}\, v_+(s)$, not at the optimal-profile speed $v_+(s)$. Therefore it is an additional condition, rather than a consequence of the original face tests. An analytic, whole-domain certificate for the running model is provided below. The construction must not be used with an uncertified floor.

For an explicit separation, take $a = -0.9 - v$ and $b = U = M = \gamma = 1$. Then $A_+ = 0.1 - v$ and $A_- = -1.9 - v$; all original face tests hold, and rest authority is strictly positive. Start in the interior with $v_0 = 0.8$ and $d_0 > \epsilon + 0.8$. For $\theta = 1/4$, the maximal profile initially has $(E^+)' = -0.7$, so $q' = -0.175$, whereas $f_+(s, q) = 0.1 - 0.4 = -0.3$ at departure. By continuity this viability failure persists on a positive spatial interval. Original domain viability and strict rest authority therefore do not certify every scaled performance floor.

7.1 The reflected controller and physical initialization

For this section the nominal $k(t, d, v) \in [-U, U]$ is jointly locally Lipschitz in (t, d, v) on finite time/state sets. This includes constant commands and the continuous piecewise-interpolated nominal tables used in the experiments. It is a more specific regularity class than a command that is merely measurable in time. There is no common prescribed Lipschitz constant across the nominal class, and the time guarantees below do not depend on that constant. Write

$$p_N(s, E, \tau) = a(d_0 - s, \sqrt{2E}) + b(d_0 - s, \sqrt{2E})k(\tau, d_0 - s, \sqrt{2E}).$$

Define the spatial trajectory and its physical clock together by

$$E' = \begin{cases} p_N(s, E, \tau), & q < E < B, \\ \max\{p_N(s, q, \tau), q'\}, & E = q, \\ \min\{p_N(s, B, \tau), B'\}, & E = B, \end{cases} \tag{38}$$

$$\tau(s) = \int_0^s \frac{dr}{\sqrt{2E(r)}}, \qquad E(0) = E_0. \tag{39}$$

The two boundaries are separated, so their rules never conflict. The physical trajectory is obtained by inverting τ and setting

$$d(t) = d_0 - s(t), \qquad v(t) = \sqrt{2E(s(t))}, \qquad u(t) = \frac{E'(s(t)) - a(d(t), v(t))}{b(d(t), v(t))}.$$

This is a causal controller: its fixed profile uses the plant and the initial state, while p_N uses the current physical time and current state. It needs no future values of the nominal command.

At a rest start, its initialization is the no-waiting physical branch. In particular the right initial acceleration is

$$p_0 = \max\{a(d_0, 0) + b(d_0, 0)k(0, d_0, 0), \theta g_0\} > 0. \tag{40}$$

If the controller is evaluated at $s = E = 0$ at any physical time, the same rule uses $k(t, d_0, 0)$ and the fixed positive threshold θg_0. This explicitly excludes an arbitrary waiting interval. Denote the first physical entry time of this reflected controller under nominal k by

$$T_\theta[k](z_0, S) = \inf\{t \geq 0 : d(t) \leq d_\eta\};$$

when finite it equals $\tau(S)$. Merely requiring $E \geq q(s)$ would not do so: the stationary initial state satisfies that inequality, and $\frac{d}{dt}(E - q(s)) = v(p - q')$ vanishes at rest.

Theorem 7.1 (Well-posed, rate-certified reflected filter)**.** *Assume the original face certificate and the regularity assumptions above, and suppose* (F) *holds. Equations* (38)–(39)*, with the no-waiting initialization, have a unique spatial trajectory and physical realization up to entry. The input is measurable, belongs to* $[-U, U]$ *almost everywhere, and the physical state remains in* D*. For every admissible nominal,*

$$T^*(z_0, S) \leq T_\theta[k](z_0, S) \leq \frac{T^*(z_0, S)}{\sqrt{\theta}}. \tag{41}$$

The filter applies the nominal whenever it is compatible with the active safety and performance tangent conditions. At an active boundary it is the nearest admissible acceleration to the nominal acceleration, and equivalently the nearest admissible input to k*.*

Proof. Construction away from zero speed. On a positive-energy neighborhood, p_N is locally Lipschitz in (E, τ), and $(2E)^{-1/2}$ is locally Lipschitz in E. The boundaries q, B are Lipschitz and strictly separated. On a sufficiently short interval only one boundary can be encountered: this follows from their positive separation and bounds on the nominal slope and boundary slopes. At a lower boundary the reflection operator is

$$(\mathcal{R}_q x)(s) = x(s) + \max\left\{0, \sup_{\sigma \leq r \leq s} [q(r) - x(r)]\right\};$$

at an upper boundary it is

$$(\mathcal{R}_B x)(s) = x(s) - \max\left\{0, \sup_{\sigma \leq r \leq s} [x(r) - B(r)]\right\}.$$

Both have uniform-norm Lipschitz constant at most two on each prefix of the interval. Applying the relevant reflection to

$$x_e(s) = E_\sigma + \int_\sigma^s p_N(r, e(r), \tau_e(r))\, dr, \quad \tau_e(s) = \tau_\sigma + \int_\sigma^s (2e(r))^{-1/2}\, dr$$

is a contraction on a sufficiently short interval. If neither boundary is near, use the identity operator. The fixed point is absolutely continuous and satisfies (38) almost everywhere: the regulator acts only at the relevant boundary. To verify uniqueness among all absolutely continuous solutions of the displayed differential rule, take any such solution and set

$$K_q(s) = \int_\sigma^s \mathbf{1}_{\{E(r)=q(r)\}}(E'(r) - p_N(r, E(r), \tau(r)))\, dr,$$
$$K_B(s) = \int_\sigma^s \mathbf{1}_{\{E(r)=B(r)\}}(p_N(r, E(r), \tau(r)) - E'(r))\, dr.$$

Both regulators are nondecreasing, start from zero, and grow only on their respective contacts. Also $E = x_E + K_q - K_B \in [q, B]$. Thus (E, K_q, K_B) is the unique two-sided Skorokhod triple for its driver x_E. Every solution is consequently a fixed point of the same local contraction. This proves local existence and uniqueness, including through a Lipschitz ceiling kink.

Construction through a rest start. Here $q(s) \geq cs$ and $q(s) \leq Cs$ near zero, for positive constants c, C, by (37). The upper ceiling is separated from zero, so only the lower reflection is relevant near departure. Consider continuous trial profiles with $e(0) = 0$ and

$$q(s) \leq e(s) \leq Hs$$

on $[0, \delta]$, where H is sufficiently large. Nominal acceleration is bounded on the corresponding compact neighborhood, and the lower reflection maps this class into itself after increasing H if needed. Equip differences with the weighted norm

$$\|e_1 - e_2\|_w = \sup_{0<s\leq\delta} \frac{|e_1(s) - e_2(s)|}{s}.$$

This is a complete metric on the closed trial class. The clock is well defined and satisfies $\tau_e(s) \leq C_0\sqrt{s}$. Moreover,

$$\begin{aligned}|\tau_{e_1}(s) - \tau_{e_2}(s)| &\leq C_1 \int_0^s r^{-3/2}|e_1(r) - e_2(r)|\, dr \\ &\leq C_2\sqrt{s}\, \|e_1 - e_2\|_w.\end{aligned}$$

Local Lipschitz continuity of a, b, k gives

$$|p_N(s, e_1, \tau_{e_1}) - p_N(s, e_2, \tau_{e_2})| \leq C_3\sqrt{s}\, \|e_1 - e_2\|_w.$$

Indeed, the speed difference is bounded by $C|e_1 - e_2|/\sqrt{s}$ and the clock difference was just bounded. After integration and lower reflection, the fixed-point map therefore has weighted Lipschitz constant at most $C_4\sqrt{\delta}$. Taking δ small proves existence and uniqueness through launch.

Continuity of the nominal acceleration and the first-order expansion of q imply, by the reflection formula,

$$E(s) = p_0 s + o(s), \quad \tau(s) = \sqrt{2s/p_0} + o(\sqrt{s}).$$

Consequently $v(t) = p_0 t + o(t)$ and $s(t) = p_0 t^2/2 + o(t^2)$. This proves the explicit physical launch and excludes a delayed branch. Any other absolutely continuous solution has bounded slope by the differential rule and obeys $q \leq E \leq Hs$ after adjusting H. Its contact regulators again identify it with the lower Skorokhod reflection of its driver, so the weighted uniqueness argument applies to it.

Admissibility and continuation. The nominal acceleration lies in $[f_-, f_+]$. At the lower boundary, the maximum in (38) stays below f_+ by (F) and stays above f_-. At the upper boundary, the original face certificate gives $f_-(s, B) \le B'$, so the minimum also stays in $[f_-, f_+]$. Thus the selected slope is attainable everywhere, almost everywhere in s. The reflection keeps $q \le E \le B$. Away from launch, q has a positive minimum on each closed spatial subinterval. Bounded slopes and the finite clock bound therefore allow continuation to S; no additional zero-speed state or waiting interval can occur before entry.

The inverse-clock chain rule gives $\dot{d} = -v$ and $\dot{v} = E'$. The startup expansion handles the only zero-speed endpoint. Hence the physical input is admissible and the trajectory is safe. It can be extended after entry with the existing safe backup, so the maximal-profile comparison applies and yields $E \le E^+$. Since $q = \theta E^+ \le E$, integration of the clocks gives (41). The active rules are scalar interval projections. Because $b > 0$, minimizing the instantaneous acceleration change and minimizing the instantaneous input change have the same projection. This proves the fidelity assertion; no claim of minimum *integrated* intervention is made. □

7.2 The exact worst-case value under strict interior fidelity

Fixed-time CLF/CBF design already treats safety, input constraints, and feasibility-dependent attraction domains [15, 16]. Robust fixed-time CBF–CLF design also handles disturbances and state-estimation error under input limits [17]; time-varying CBFs encode temporal-logic tasks [18]. Those prescribed requirements differ from the nominal-class extremal clock studied here.

Timed reach–avoid filtering constructs contractive implicit sets from backup trajectories [19]; recurrence filtering uses backup reachability and timed activation to enforce returns [20]. Faithful finite-time return with a prescribed nominal region is also established [21]. Path control barrier functions use adjustable path progression to address input-constrained feasibility, liveness, and completion-time bounds [22]. For the fixed episode floor $q = \theta E^+$ and the strict fidelity class defined below, we identify a common exact worst clock over admissible nominals and relate it to the same plant's minimum first-entry value. The result is neither a general prescribed-time synthesis theorem nor an optimality claim over all safety/liveness filters.

The active floor arc is a constrained sliding motion, understood through the absolutely continuous reflected energy solution, not an unproved identification with a Filippov extension. The reflection construction uses the classical Skorokhod map [23].

We now make the fidelity class explicit. An *admissible faithful episode filter* assigns a physical trajectory to every nominal in the class above, preserves D and $E \ge q$, makes no initial wait, and applies the nominal strictly above the floor whenever it is tangent to the original safety domain. In spatial terms, at $E > q$ the applied acceleration is p_N if $E < B$, and also if $E = B$ and $p_N \le B'$. At the floor it may modify the nominal as needed for feasibility. Here no initial wait means $s(t) > 0$ for every $t > 0$ before entry, including a rest start. Since $q(s) > 0$ for $s > 0$, the floor also excludes any subsequent rest before entry. The class requires actual trajectories with admissible bounded inputs, not impulses. For a filter Π and nominal k, define

$$T_\Pi[k](z_0, S) = \inf\{t \ge 0 : d_{\Pi[k]}(t) \le d_\eta\}, \qquad \inf \varnothing = +\infty.$$

The controller just constructed shows that the class is nonempty. Anticipatory controllers that modify an otherwise admissible nominal strictly above the floor are outside this strict fidelity class.

Define the lower-reflected profile by

$$(E_\theta^-)' = \begin{cases} f_-(s, E_\theta^-), & E_\theta^- > q(s), \\ \max\{f_-(s, q(s)), q'(s)\}, & E_\theta^- = q(s), \end{cases} \qquad E_\theta^-(0) = E_0. \tag{42}$$

It is precisely (38) for $k = -U$: on the upper ceiling, the original face certificate already makes f_- admissible, so the upper reflection is inactive. The preceding theorem therefore establishes its unique physical realization.

Theorem 7.2 (Sharp nominal-uniform clock)**.** *Under the hypotheses of Theorem 7.1, define*

$$W_\theta(z_0, S) = \int_0^S \frac{ds}{\sqrt{2E_\theta^-(s)}}. \tag{43}$$

Every admissible faithful episode filter Π *satisfies*

$$\sup_k T_\Pi[k](z_0, S) = W_\theta(z_0, S) \le \frac{T^*(z_0, S)}{\sqrt{\theta}}. \tag{44}$$

The constant braking nominal $k \equiv -U$ attains the supremum. Consequently the infimum over this nonempty filter class of its worst-nominal entry time is also W_θ.

If $v_0 = 0$ and the floor is fully admissible,

$$f_-(s, q(s)) \le q'(s) \le f_+(s, q(s)) \quad a.e., \tag{45}$$

then $E_\theta^- = q$ and

$$W_\theta(z_0, S) = T^*(z_0, S)/\sqrt{\theta}. \tag{46}$$

For a positive-speed start and $0 < \theta < 1$, the latter bound is strict, although (44) remains exact.

Proof. First, every no-wait physical trajectory preserving the floor reaches S in finite time. Otherwise monotone travel would converge to $s_\infty \in [0, S]$. No initial wait excludes $s_\infty = 0$. Continuity and positivity of q on $(0, S]$ then bound $v(t) \ge \sqrt{2q(s(t))}$ away from zero for all sufficiently large times, contradicting bounded travel. Thus its spatial clock extends to the full interval $[0, S]$ before the comparisons below are integrated.

Let e be the spatial energy of any admissible floor-preserving trajectory. Its acceleration satisfies $e' \ge f_-(s, e)$ almost everywhere. Where $E_\theta^- > e$, the lower profile is strictly above q and obeys $(E_\theta^-)' = f_-(s, E_\theta^-)$. The positive-part comparison therefore gives

$$((E_\theta^- - e)_+)' \le L(s)(E_\theta^- - e)_+, \qquad L(s) \le C/\sqrt{q(s)}.$$

This coefficient is integrable through a rest start by (37); elsewhere it is bounded locally. Grönwall yields $e \ge E_\theta^-$. With no initial wait and $e > 0$ afterward, the physical elapsed time is its spatial clock, so $T \le W_\theta$.

Now set $k = -U$ and let e be the trajectory of any faithful filter. Where $e > E_\theta^-$, one has $e > q$. The nominal acceleration f_- is tangent to the original ceiling whenever that ceiling is active, so fidelity gives $e' = f_-(s, e)$. Since $(E_\theta^-)' \ge f_-(s, E_\theta^-)$, the opposite positive-part comparison yields $e \le E_\theta^-$. Thus $e = E_\theta^-$ and the braking nominal attains W_θ for every filter in the class. The comparison $E_\theta^- \ge q = \theta E^+$ supplies the simple upper bound.

If $E_0 = 0$ and (45) holds, q itself solves (42) with the required initial value. Uniqueness proves $E_\theta^- = q$, and its clock is the scaled optimal clock. If $E_0 > 0$, then $E_\theta^-(0) = E_0 > \theta E_0 = q(0)$. Continuity makes this inequality strict on a spatial interval of positive length before any positive target S. Its clock is therefore strictly smaller than the clock of q, proving the last statement. □

The theorem concerns strict interior fidelity, not all safe controllers. An anticipatory controller may obtain a smaller worst-case time by altering the nominal before the floor is reached. It therefore changes the fidelity requirement. Likewise a CBF condition that restricts inputs throughout the interior is not automatically a member of this class; its effective acceleration interval must be used if an analogous exact worst-case characterization is desired.

7.3 An analytic certificate for the optimal-velocity class

For $a(d,v) = \alpha(V(d) - v)$, $\alpha > 0$, $b = 1$, put

$$C(d) = \alpha V(d) + U, \qquad \bar{v}(d) = \min\{(d-\epsilon)/\gamma, M\}.$$

Then $A_+ = C - \alpha v$ and $A_- = C - 2U - \alpha v$.

Proposition 7.3 (Fully admissible scaled profiles)**.** *Suppose the original face certificate holds and, for every $d \geq \epsilon$,*

$$\alpha\bar{v}(d) \leq 2C(d), \qquad C(d) \leq 2U. \tag{47}$$

Then every $q = \theta E^+$, $0 < \theta < 1$, satisfies (45). Thus every launchable rest-start episode has the exact worst-case ratio $1/\sqrt{\theta}$ for the faithful filter class, attained by $k = -U$. For a moving start, (43) is the exact worst-case value and the $1/\sqrt{\theta}$ bound is strict.

Proof. Write $r = \sqrt{\theta}$ and $v = \sqrt{2E^+}$. Since $(E^+)' \leq C - \alpha v$, the upper-floor margin obeys

$$\begin{aligned} f_+(s,q) - q' &\geq C - \alpha r v - r^2(C - \alpha v) \\ &= (1-r)[(1+r)C - \alpha r v] \geq 0, \end{aligned}$$

where $\alpha v \leq \alpha\bar{v} \leq 2C$ gives $(1+r)C - \alpha r v \geq (1-r)C \geq 0$.

When the maximal profile uses the full-throttle slope,

$$\begin{aligned} q' - f_-(s,q) &= r^2(C - \alpha v) - (C - 2U - \alpha r v) \\ &= 2U - (1-r^2)C + \alpha(r - r^2)v \geq 0. \end{aligned}$$

When it slides along the speed cap, $q' = 0$ and $f_-(s,q) = C - 2U - \alpha r M \leq 0$. When it slides along the headway ceiling, $q' = -r^2 v/\gamma$, so

$$q' - f_-(s,q) = 2U - C + v(\alpha r - r^2/\gamma).$$

The expression on the right is concave in $r \in [0,1]$. At $r = 0$ it is $2U - C \geq 0$; at $r = 1$ it is $-A_-(d,v) - v/\gamma \geq 0$ by the headway face certificate. It is therefore nonnegative throughout the interval. If the maximal profile leaves a ceiling under full throttle, the first calculation applies. These branches exhaust the construction almost everywhere. □

This certificate holds analytically for all three numerical models on the whole gap range $d \geq \epsilon$. For C2, $C(d) \geq 5 - \tanh 2 > 4$, while $\alpha M = 1.96$. For C3, $C(d) \geq 5$, while $\alpha M = 5.096$. For C1, put $d_c = \epsilon + \gamma M = 0.412$. On $[\epsilon, d_c]$ the derivative of $2C(d) - \alpha\bar{v}(d)$ is $2\alpha\,\mathrm{sech}^2 d - \alpha/\gamma < 0$; on $[d_c, \infty)$ it is nonnegative. Its global minimum is

$$2C(d_c) - \alpha M = 1.919938\ldots > 0.$$

Finally the upper bounds on C are respectively 5.093529, 5.035973, and 8.9, all below $2U = 10$. These are analytic certificates using the fixture formulas, rather than sign checks on a sampled grid.

7.4 Interpretation, scope, and implementation

The parameter θ gives an operationally meaningful tradeoff. For a certified rest start in the preceding optimal-velocity class, its worst-case entry-time factor is exactly $1/\sqrt{\theta}$. This remains a design choice: the theorem does not choose the acceptable intervention cost or guarantee a favorable value of the nominal objective. For moving starts, the initial energy buffer makes the exact factor W_θ/T^* smaller. A short double-integrator segment illustrates why universal equality

would be false: if $E_+(s) = E_0 + s$ and the braking profile $E_0 - s$ stays above $\theta(E_0 + s)$ before the target, the faithful filter follows that braking profile and never reaches its performance floor.

The endpoint $\theta = 1$ is implemented separately by the original maximal profile. It achieves T^* and permits no trajectory-level energy slack relative to that profile. At $\theta = 0$, the positive floor and startup argument disappear; no conclusion is obtained by substituting zero into the rate theorem. The blend and recovery CBF of Section 4 are not asserted to be limits of this reflected family. Their entry-time bounds are finite uniformly over admissible nominals for fixed certified data, even when their competitive ratios are large.

Profiles remain fixed for the duration of each episode. Arbitrarily recomputing E^+ from the current state resets the performance floor and is not covered by the proof. At entry the controller may hand over to the existing safe recovery law. The theorem guarantees the first entry into a physical-gap tolerance with free terminal speed; it does not impose terminal braking or exact positive-gap tracking. If the floor certificate holds for the entire interval $[0, s_*)$ and the episode is continued, every positive distance below s_* is traversed in finite time, while the clock diverges at s_*. The limiting gap is then $d_0 - s_*$, so the controller realizes the maximal available travel for that certified episode. This last statement is conditional on the whole-profile certificate and does not assert preservation of every partial recoverability set for every plant in the original class.

The reflected formulation is also a direct numerical specification: integrate the nominal dynamics until an active floor or ceiling is met; follow the corresponding feasible boundary while the nominal points outward; release it when the nominal points inward. The clock is integrated on the same spatial trajectory, with the rest singularity removed by a substitution such as $s = r^2$. A state projection applied after an unconstrained numerical step is not this control law. Acceleration changes at contact can be discontinuous. Total acceleration variation and jump sizes are therefore reported alongside intervention and cost; finite-difference jerk, if shown at a specified sampling step, is not an intrinsic bounded-jerk guarantee.

8 Sampled safety and recovery: construction and complete proofs

This section treats the autonomous dynamics with a stopped leader. The input held on a sampling interval is the actuator u; in general the net acceleration $a(d, v) + b(d, v)u$ still varies during that interval. The sampled nominal values can be arbitrary elements of $[-U, U]$. No regularity or prediction of future nominal values is required.

Fixed-period CBF margins [24], self-triggered CLF–CBF control [25], and flow-based zero-order CBFs [26] already provide intersample safety certificates. The construction here couples a fixed episode energy floor to a safety ceiling, uses scalar ordering to prove nonemptiness of a spatial-prefix input interval through rest launch, and compares the resulting recovery clock with W_θ and T^*. Scalar clipping itself is not a new general alternative to CBF optimization; the larger physical-hold test below need not define an interval. Sampled CBFs with state uncertainty and input delays are treated in [27]; mechanical constraints and constructive actuator-limited simultaneous feasibility are developed in [28, 29]. Our exact-model certificate exploits scalar ordering and the specified floor/ceiling geometry. It does not extend those methods to a broader mechanical system or disturbance class.

8.1 Fixed episode and positive separation

Fix a launchable state (d_0, v_0) and a positive-tolerance entry travel $S \in (0, s_*)$. Choose an extension S_e such that

$$0 < S < S_e < s_*, \qquad \min_{[0,S_e]} B > 0. \tag{48}$$

The maximal profile E^+ is computed for this same initial state and is not reset at subsequent sampling instants. Fix $0 < \theta < 1$, set $q = \theta E^+$, and let W_θ denote the lower-profile clock in (43). Assume on the extended interval that

$$q'(s) \le f_+(s, q(s)) \quad \text{a.e.}, \qquad f_-(s, B(s)) \le B'(s) \quad \text{a.e.} \tag{49}$$

The second inequality follows from the original safety faces. Both q and B are Lipschitz; $q < B$, and $q > 0$ except possibly at the initial rest point. At such a point,

$$E^+(s) = g_0 s + o(s), \qquad q(s) = \theta g_0 s + o(s), \qquad g_0 = A_+(d_0, 0) > 0. \tag{50}$$

In particular, the band has positive energy width even though its lower boundary meets zero at launch. The extended interval permits a hold to continue safely after it crosses the entry level and before entry is detected at the next sample.

Let

$$\begin{aligned} \mathcal{B}_e &= \{(s, E) : 0 \le s \le S_e,\ q(s) \le E \le B(s)\}, \\ F(s, E, u) &= a(d_0 - s, \sqrt{2E}) + b(d_0 - s, \sqrt{2E})u. \end{aligned} \tag{51}$$

Choose verified constants

$$\begin{aligned} 0 < g &\le \min_{[0, S_e]} (B - q), \\ P &\ge \sup_{(s,E) \in \mathcal{B}_e,\ |u| \le U} |F(s, E, u)|, \\ K &= P + \max\{L_q, L_B\}, \qquad L_q \ge \mathrm{Lip}(q), \quad L_B \ge \mathrm{Lip}(B), \end{aligned} \tag{52}$$

$$\delta_0 = \frac{1}{M} \min\{S_e - S, g/(4K)\}. \tag{53}$$

If $K = 0$, interpret its contribution as $+\infty$. In the launchable episodes considered here $K > 0$. These are ordinary bounds on a compact set. For the headway/speed ceiling one may use $L_B = M/\gamma$ and

$$g = (1 - \theta) \min B, \qquad L_q = \theta \max\{P, L_B\}. \tag{54}$$

Indeed $E^+ \le B$, and the reflected maximal profile has slope equal to either f_+ or the active ceiling slope. These choices do not require an exact numerical minimum of $B - q$.

8.2 A computable spatial certificate

At a pre-entry state $(s, E) \in \mathcal{B}_e$, $s < S$, and for a constant $u \in [-U, U]$, solve the scalar initial-value problem

$$e_u'(r) = F(r, e_u(r), u), \qquad e_u(s) = E. \tag{55}$$

For positive initial energy this is the ordinary locally unique solution until its first zero. At $(s, E) = (0, 0)$, restrict the candidate to

$$p_0(u) := a(d_0, 0) + b(d_0, 0)u \ge \theta g_0 > 0 \tag{56}$$

and take the positive spatial profile induced by the unique physical constant-input trajectory. It satisfies $e_u(r) = p_0(u)r + o(r)$. Consequently its clock is finite near zero and has no waiting interval. Candidates with nonpositive subsequent energy are rejected.

For $0 < \delta \le \delta_0$, set $\ell = M\delta$ and define

$$\begin{aligned} I_\delta(s, E) = \{u \in [-U, U] : {} & e_u \text{ exists on } [s, s + \ell], \\ & q(r) \le e_u(r) \le B(r) \quad (s \le r \le s + \ell), \\ & E > 0 \text{ or } p_0(u) \ge \theta g_0 > 0\}. \end{aligned} \tag{57}$$

Thus (56) is included explicitly at rest. A certificate checks the entire spatial interval, not only selected nodes. For explicitly evaluable model coefficients with known local bounds, validated scalar integration or analytic comparison bounds implement this test. The extreme-input witness below is itself an analytic certificate and avoids deciding an equality by floating-point sampling.

Lemma 8.1 (Closed interval and an explicit feasible witness)**.** *For every pre-entry band state and $0 < \delta \le \delta_0$, $I_\delta(s, E)$ is a nonempty closed interval. A feasible witness is*

$$u_F(s, E) = \begin{cases} U, & B(s) - E \ge g/2, \\ -U, & B(s) - E < g/2. \end{cases} \tag{58}$$

Thus the fallback does not require evaluating the exact midpoint of the numerically represented floor.

Proof. Order. Since $b > 0$, $F(r, e, u)$ is increasing in u. Scalar comparison gives $e_{u_1} \le e_{u_2}$ for $u_1 \le u_2$ on every common positive-energy interval. At a rest start, the initial slopes have the same order. For profiles bounded below by q, the coefficient in the positive-part comparison can be bounded by $C/\sqrt{q(r)}$. By (50), this is integrable at zero. The usual integral comparison therefore extends through launch. If two inputs are feasible, every intermediate input is between their profiles and hence exists on the full interval and is feasible. This proves the interval property.

Closedness. For a sequence of feasible inputs $u_j \to u$, their profiles are bounded in the compact band and have slope bounded by P. Arzelà–Ascoli supplies a uniformly convergent subsequence. The integral equation, continuity of F, and uniform convergence show that its limit solves (55) with input u and satisfies the closed band constraints. At rest, the closed inequality (56) gives a uniformly positive initial acceleration. The limiting spatial solution is the unique physical positive-launch branch: its comparison coefficient is integrable as above, and its clock is finite. Thus u is feasible. Boundedness by $[-U, U]$ makes the interval compact.

Nonemptiness. Suppose first that $B(s) - E \ge g/2$ and select $u = U$. The first inequality in (49) makes q a subsolution for the full-throttle scalar equation. Hence $e_U \ge q$, including at a rest start, where $g_0 > \theta g_0$. Until any first upper exit,

$$B(r) - e_U(r) \ge B(s) - E - (P + L_B)(r - s) \ge g/2 - K\ell \ge g/4 > 0.$$

Thus an upper exit is impossible. Boundedness and positive energy away from the initial rest point provide continuation over the full interval.

If instead $B(s) - E < g/2$, then

$$E - q(s) = (B(s) - q(s)) - (B(s) - E) > g/2.$$

Select $u = -U$. The ceiling is a supersolution by the second inequality in (49), so $e_{-U} \le B$. Until any first lower exit,

$$e_{-U}(r) - q(r) > g/2 - (P + L_q)(r - s) \ge g/4 > 0.$$

This excludes a lower exit and again gives continuation. In both cases $s + \ell \le S + M\delta_0 \le S_e$, so no profile beyond the specified extension is used. □

Lemma 8.2 (From spatial certification to an actual held input)**.** *If $u \in I_\delta(s, E)$, the unique physical constant-input solution starting at the corresponding band state exists and stays in D and above the performance floor for a physical duration of at least δ. Its travel during that interval is at most $M\delta$.*

Proof. The clock of the certified profile is

$$\tau(r) = \int_s^r \frac{d\xi}{\sqrt{2e_u(\xi)}}.$$

It is finite and strictly increasing on the certified interval, with the integrable positive-launch singularity allowed at the left endpoint. Since $e_u \le B \le M^2/2$,

$$\tau(s + M\delta) \ge \frac{M\delta}{M} = \delta.$$

Invert the clock on $[0, \delta]$. The chain rule gives the original physical equations with the same constant u, including the launch expansion at rest. Local uniqueness of the original plant identifies this trajectory with the physical held-input solution. Its membership in the band implies every original state constraint. □

8.3 Fixed and adaptive sampling theorems

A hold certificate may be conservative: acceptance must imply the stated inequalities, whereas rejection need not imply true infeasibility. For a selected duration δ it must certify all of the following:

1. the actual held-input trajectory stays in D;
2. its travel remains in $[0, S_e]$ and its energy is at least $q(s(t))$ throughout the hold;
3. if the initial episode state is at rest, it follows a positive, no-waiting physical launch.

The spatial test above is one such certificate. The trajectory-tube test below can certify larger physical holds. Its accepted set need not be an interval, since different inputs can travel different distances in the same physical time.

For sample times $0 = t_0 < t_1 < \cdots$, define the physical first-entry and exact-observation detection times by

$$T_{\text{entry}} = \inf\{t \ge 0 : d(t) \le d_\eta\}, \qquad T_{\text{detected}} = \inf\{t_n : d(t_n) \le d_\eta\}, \qquad \inf \varnothing = +\infty.$$

The floor-preserving rule below proves these times finite.

Theorem 8.3 (Recursive feasibility, intersample safety, and recovery clock)**.** *Assume (48)–(49) and the original autonomous regularity and whole-domain face conditions. Sampled state observation is exact.*

For a fixed $0 < \delta \le \delta_0$, the controller

$$u_n = \operatorname{proj}_{I_\delta(s_n, E_n)} k_n, \qquad |k_n| \le U, \tag{59}$$

held until the next sample, is well defined and recursively feasible until detected entry. The constant δ_0 is a uniform existence bound and need not be a practical hold period. A fixed period exceeding δ_0 requires a separate certificate at every hold; its feasibility does not follow from the uniform bound alone.

More generally, fix $\bar\Delta > 0$ and $\delta_F = \min\{\bar\Delta, \delta_0\}$. At every pre-entry update allow any certified pair (u_n, δ_n) with $|u_n| \le U$ and $\delta_F \le \delta_n \le \bar\Delta$. If a longer hold is not certified, use $(u_F(s_n, E_n), \delta_F)$, or another member of $I_{\delta_F}(s_n, E_n)$. The resulting physical trajectory is safe through detected entry, has no accumulating update times, and satisfies

$$T^* \le T_{\text{entry}} \le W_\theta \le T^*/\sqrt{\theta}, \qquad T_{\text{detected}} \le W_\theta + \bar\Delta. \tag{60}$$

The nominal may be tested first and is then left unchanged whenever its hold is accepted by the specified certificate. This assertion is preservation under that sufficient certificate, rather than maximal preservation over all physically feasible held inputs.

Proof. The interval lemma and hold lemma prove the first-step feasibility and safety of (59), including at zero speed. At a later pre-entry sample, the new state remains in the same fixed episode band. The interval lemma applies to that state with the same constants and extension, proving induction and recursive feasibility.

For the more general rule, every accepted pair certifies the entire hold by definition. If a test fails, the interval lemma supplies the explicit fallback. Hence the algorithm always has a certified pair. Every duration is at least $\delta_F > 0$, so no finite time contains infinitely many updates. The piecewise-constant admissible input and local plant regularity give a unique physical solution across the updates. Its speed is strictly positive after the no-waiting launch until entry: at every positive travel $s \le S$, the floor is positive.

The trajectory therefore has an absolutely continuous spatial energy profile E_Δ. The first comparison in the proof of Theorem 7.2 uses only an admissible input, the floor inequality, and the absence of initial waiting. It does not use strict fidelity. Consequently $E_\Delta \ge E_\theta^-$, and

$$T_{\text{entry}} = \int_0^S \frac{ds}{\sqrt{2E_\Delta(s)}} \le \int_0^S \frac{ds}{\sqrt{2E_\theta^-(s)}} = W_\theta \le \frac{T^*}{\sqrt{\theta}}.$$

In particular entry cannot fail to occur: otherwise the same identity on every attained prefix, together with the positive floor on compact positive-travel intervals, would extend the trajectory to S within this finite clock. The physical trajectory is safe and can be extended from any band state by the original continuous-time safe backup; hence the plant-optimal lower bound $T^* \le T_{\text{entry}}$ applies.

The hold containing first entry ends at most $\bar{\Delta}$ later. Its certificate uses the extended band and supplies safety over this remaining portion. Thus detection has the asserted bound. The nominal-preservation assertion follows by testing and accepting that candidate before searching for alternatives. □

Scope of the sampling claim. The fixed-period theorem guarantees feasibility only for its explicit δ_0. The adaptive theorem guarantees feasibility for any requested maximum period by retaining a positive minimum fallback duration. A numerical run that accepts the requested period on every hold is a fixed-period run; its fallback count and actual hold durations must be reported. The fallback bound can be very small, especially as $\theta \uparrow 1$ or as the target approaches the stopped corner. For $h_g = 0$, the smallest ceiling scales as $\eta^2/(2\gamma^2)$. No uniform useful sampling period over those limits is claimed.

The bound by W_θ shows that certified sampling preserves the continuous-time worst-case upper clock. This is a consequence of the same lower-profile comparison, not a new comparison method. It does not order the sampled and continuous trajectories for a given nominal, nor assert that the worst-case clock over sampled controllers is exactly W_θ. The sampled fidelity rule permits anticipation, whereas the sharp equality of Theorem 7.2 concerns strict interior fidelity.

The one-period detection allowance in (60) uses exact sampled target observation. A numerical observer that declares entry only when a verified upper gap bound satisfies $d_{\text{upper}} \le d_\eta$ can conservatively delay detection by more than one period after physical entry. We do not transfer the universal one-period statement to that observer without an additional observation-error argument. Instead, the reported numerical runs certify their observed time directly using the lower optimal-clock bound derived below.

Theorem 7.2 concerns a different, strict continuous-time fidelity class. Anticipatory sampled interventions need not belong to that class, so its exact adversarial value cannot be transferred without a separate proof. Likewise a finite candidate search is not claimed to minimize instantaneous or integrated intervention over all safe holds. A compulsory full-throttle launch step, when used by an implementation, is an explicitly specified initialization rather than a nominal-preservation claim at rest.

8.4 A posteriori certification of a physical hold

The following certificate avoids assuming that dense numerical samples prove an intersample inequality. It also separates plant-integration error from error in the precomputed benchmark.

For the running model $\dot{d} = -v$, $\dot{v} = \alpha(\mathcal{V}(d) - v) + u$, let $\mathcal{V}$ be nondecreasing. Fix a gap interval K_d and a verified Lipschitz bound $L_{\mathcal{V}}$ for $\mathcal{V}$ on K_d. Let $\widehat{z} = (\widehat{d}, \widehat{v})$ be a continuous piecewise differentiable approximation of $z = (d, v)$ for a fixed u. Define its defects

$$\begin{aligned} r_d &= \dot{\widehat{d}} + \widehat{v}, \\ r_v &= \dot{\widehat{v}} - \alpha(\mathcal{V}(\widehat{d}) - \widehat{v}) - u. \end{aligned} \tag{61}$$

Suppose verified nonnegative bounds $\rho_d(t), \rho_v(t)$ dominate $|r_d|, |r_v|$ almost everywhere. Solve the cooperative linear error equation for $\xi = (\xi_d, \xi_v)^\top$ with a componentwise initial enclosure

$$\dot{\xi} = H\xi + \rho, \quad H = \begin{pmatrix} 0 & 1 \\ \alpha L_{\mathcal{V}} & -\alpha \end{pmatrix}, \quad \xi(0) = \xi_0 \geq |z(0) - \widehat{z}(0)|. \tag{62}$$

Require the complete gap tube $[\widehat{d}(t) - \xi_d(t), \widehat{d}(t) + \xi_d(t)]$ to stay in $\operatorname{int} K_d$ throughout the hold. Verify this inclusion after computing ξ with the already chosen K_d and $L_{\mathcal{V}}$. The bounds and this equation may be applied successively on smaller cells. Error radii are carried across cells and, when validating a complete numerical rollout, across sample instants; they are not reset to zero without a new exact state observation.

Lemma 8.4 (Validated hold tube)**.** *The exact trajectory for the same constant actuator satisfies*

$$|d(t) - \widehat{d}(t)| \leq \xi_d(t), \qquad |v(t) - \widehat{v}(t)| \leq \xi_v(t). \tag{63}$$

Let Q be a verified upper enclosure of E^+ on the extended spatial interval. Put

$$J(t) = [d_0 - \widehat{d}(t) - \xi_d(t),\, d_0 - \widehat{d}(t) + \xi_d(t)].$$

The following inequalities, certified for every time in the hold, are sufficient for its acceptance:

$$\begin{aligned} \widehat{v} - \xi_v &\geq 0, & \widehat{v} + \xi_v &\leq M, \\ \widehat{d} - \gamma\widehat{v} - \epsilon - \xi_d - \gamma\xi_v &\geq 0, & J(t) &\subset [0, S_e], \\ \tfrac{1}{2}(\widehat{v} - \xi_v)^2 &\geq \theta \sup_{r \in J(t)} Q(r). \end{aligned} \tag{64}$$

At an initial rest state a positive-launch certificate is additionally required. An analytic certified prefix can be combined with the tube certificate on the rest of the hold. The quantity $\sup_{r\in J(t)} Q(r)$ may be replaced by any directly verified upper bound on $\sup_{r\in J(t)} E^+(r)$, such as the range enclosure constructed in Section 8.5.

Proof. Write $e_d = d - \widehat{d}$ and $e_v = v - \widehat{v}$. Until a first exit of d from K_d, their equations and the Lipschitz bound give the upper right derivative inequalities

$$\begin{aligned} D^+|e_d| &\leq |e_v| + \rho_d, \\ D^+|e_v| &\leq \alpha L_{\mathcal{V}}|e_d| - \alpha|e_v| + \rho_v. \end{aligned}$$

The matrix H is Metzler, so its exponential is entrywise nonnegative. The comparison formula for this cooperative linear system yields (63). The verified tube inclusion in $\operatorname{int} K_d$ precludes that exit, so the comparison holds throughout the hold. The first lines of (64) then imply the three

physical safety inequalities and the stated travel range. Since the certified speed lower bound is nonnegative, the last line gives

$$v(t)^2/2 \geq \tfrac{1}{2}(\widehat{v} - \xi_v)^2 \geq \theta \sup_{r \in J(t)} Q(r) \geq \theta E^+(d_0 - d(t)).$$

Thus every accepted hold satisfies the performance floor as well. The explicit launch condition excludes the stationary branch that would otherwise satisfy a zero-energy inequality at zero travel. □

Rounded polynomial joins. The same lemma applies cell by cell when stored polynomial pieces have small, explicitly bounded jumps. If the approximation changes from $\widehat{z}^-$ to $\widehat{z}^+$ at a cell or sample boundary, the exact physical state remains continuous. The triangle inequality therefore permits the update

$$\xi^+ \geq \xi^- + |\widehat{z}^+ - \widehat{z}^-|$$

componentwise, with an outward enclosure of the jump. The discrepancy between a stored polynomial endpoint and the next state center is handled in the same way. No claim that rounded Hermite pieces join exactly is needed, and no uncertainty is discarded at those joins.

Finite computation and roundoff. For a piecewise polynomial $\widehat{z}$, its derivatives are explicit. Polynomial extrema or interval bounds give whole-cell bounds on r_d, on the safety expressions, and on the polynomial components of r_v; an interval enclosure of $\mathcal{V}$ bounds its remaining component. On a cell where ρ is constant,

$$\xi(t+h) = e^{Hh}\xi(t) + \int_0^h e^{H(h-r)}\rho\, dr.$$

Verified enclosures of this expression and of the entire cell, rather than endpoint values alone, implement (64). Interval arithmetic, analytic envelopes, or explicit error bounds must account for numerical integration and roundoff. A small ODE solver tolerance, by itself, is not such a bound. The same requirement applies to the upper profile enclosure Q. Conservatism of this sufficient test is allowed: an uncertain candidate is rejected or subdivided, and an uncertified rejection is never called physical infeasibility. The fallback in Theorem 8.3 supplies the existence guarantee. In particular, the analytic fallback is accepted by its comparison proof and the constants (54); it is *not* required to pass the conservative Q-based numerical test. An upper enclosure Q need not be continuous or satisfy a floor viability condition. It is used only to imply the true floor inequality. Consequently a positive enclosure error at a state on the exact floor may reject a numerical candidate without obstructing the analytic fallback.

Once speed nonnegativity has been separately certified throughout a hold, actual travel is nondecreasing. The spatial enclosure used for the last inequality in (64) may therefore be intersected with an already verified lower travel bound, such as zero at the episode launch. Its upper endpoint must still be certified to lie at or below S_e. This removes artificial negative travel created solely by a symmetric numerical error enclosure.

Launch. For a rest start, constant full throttle produces the physical maximal profile until its first ceiling contact. A verified prefix remaining strictly below the ceiling therefore obeys $E = E^+ \geq \theta E^+$ and launches with acceleration $g_0 > 0$. This provides an analytic startup certificate; once strictly positive floor and safety margins are established, the trajectory-tube test can continue. One must certify the absence of an earlier ceiling contact on the prefix, rather than assume it from a plot. The fallback interval proof gives a general analytic launch duration when a larger one is not certified.

8.5 Certified upper enclosure of the reference profile

The trajectory certificate uses a verified upper enclosure of E^+, rather than treating the nominal profile integrator as an exact safety oracle. The implementation in `profile_bounds.py` uses the following construction for the autonomous optimal-velocity models. Write

$$C(s) = \alpha \mathcal{V}(d_0 - s) + U. \tag{65}$$

Since $\mathcal{V}$ is nondecreasing, C is nonincreasing in s. Partition $[0, S_e]$ into stored spatial nodes $s_0 < \cdots < s_N$. On each cell choose a positive verified upper bound

$$\overline{C}_j \geq C(s_j) \geq C(s) \quad (s_j \leq s \leq s_{j+1}),$$

and choose $0 < \underline{\alpha} \leq \alpha$. These bounds include the specified parameter and arithmetic enclosures. The models have positive rest authority throughout this interval, so the required positive constant bounds are available directly. Let $\overline{B}(s)$ be a verified upper evaluation of the explicit safety ceiling.

Define $\Phi_{c,\beta}(h; e_0)$ as the energy at spatial distance h for the constant-coefficient equation

$$e' = c - \beta\sqrt{2e}, \qquad e(0) = e_0 \geq 0, \quad c, \beta > 0. \tag{66}$$

This flow has an elementary physical-time representation. With $v_0 = \sqrt{2e_0}$ and $v_\infty = c/\beta$,

$$\begin{aligned} v_c(t) &= v_\infty + (v_0 - v_\infty)e^{-\beta t}, \\ X_c(t) &= v_\infty t + (v_0 - v_\infty)\frac{1 - e^{-\beta t}}{\beta}. \end{aligned} \tag{67}$$

For $h > 0$, $X_c(t_h) = h$ has a unique positive solution, and

$$\Phi_{c,\beta}(h; e_0) = \tfrac{1}{2} v_c(t_h)^2. \tag{68}$$

The formula also covers launch from rest: $v_c(t) > 0$ for $t > 0$ and X_c is strictly increasing away from its zero-speed endpoint.

Starting from $\overline{E}_0 \geq v_0^2/2$, form the endpoint bounds

$$\overline{E}_{j+1} = \min\left\{\overline{B}(s_{j+1}),\, \overline{\Phi}_{\overline{C}_j, \underline{\alpha}}(s_{j+1} - s_j; \overline{E}_j)\right\}, \tag{69}$$

where the overline on Φ denotes a verified upper evaluation, including the spatial-length enclosure. The online point enclosure is

$$Q(s) = \min\{\overline{B}(s),\, \overline{E}_j + \overline{C}_j(s - s_j)\}, \quad s_j \leq s < s_{j+1}, \tag{70}$$

with outward evaluation of the addition and product and with the endpoint bound at $s = s_N$.

Lemma 8.5 (Upper reference recursion and whole-cell query)**.** *Every endpoint satisfies $E^+(s_j) \leq \overline{E}_j$, and $E^+(s) \leq Q(s)$ on the complete extended interval. For an interval $[a, b] \subset [0, S_e]$, a verified upper bound on $\sup_{[a,b]} E^+$ is obtained by taking the maximum over intersected cells of*

$$\min\{\overline{B}(a_j),\, \overline{E}_j + \overline{C}_j(b_j - s_j)\}, \quad a_j = \max\{a, s_j\}, \quad b_j = \min\{b, s_{j+1}\}. \tag{71}$$

An explicit upper bound on the ceiling over $[a_j, b_j]$ can replace $\overline{B}(a_j)$ if a different ceiling is used. No assumption that the maximal profile remains on the ceiling after first contact is required.

Proof. The upper-reflected maximal profile obeys, almost everywhere,

$$(E^+)' \leq C(s) - \alpha\sqrt{2E^+} \leq \overline{C}_j - \underline{\alpha}\sqrt{2E^+} \quad \text{on cell } j.$$

This is true both on free arcs and on ceiling arcs, since reflection can only decrease the unrestricted energy slope. Starting from $E^+(s_j) \le \overline{E}_j$, scalar comparison with (66) gives

$$E^+(s) \le \Phi_{\overline{C}_j,\underline{\alpha}}(s - s_j; \overline{E}_j).$$

The comparison also holds at zero energy: the scalar right-hand side is continuous and nonincreasing in energy, so the positive part of a putative excess cannot increase. A positive constant c supplies the unique positive-launch flow in (66). Taking the minimum with the independent upper bound $E^+(s_{j+1}) \le B(s_{j+1}) \le \overline{B}(s_{j+1})$ proves the endpoint induction.

On the same cell, $(E^+)' \le \overline{C}_j$ implies $E^+(s) \le \overline{E}_j + \overline{C}_j(s - s_j)$, which proves (70) after capping by the ceiling. The affine upper bound is nondecreasing, while the headway/speed ceiling is nonincreasing. Their values at b_j and a_j respectively therefore bound both functions throughout the intersected cell. Taking their minimum and then the maximum over cells proves (71). □

Verified evaluation of the constant-authority flow. The numerical root solver supplies only a proposed clock value. The implementation enlarges a bracket until interval evaluation proves

$$\sup \mathbf{X}_c(t_-) \le \inf \mathbf{h}, \qquad \inf \mathbf{X}_c(t_+) \ge \sup \mathbf{h}, \tag{72}$$

where bold symbols denote interval enclosures. Thus the true clock lies in $[t_-, t_+]$ independently of the root solver's error estimate. If $v_0 < v_\infty$, speed and energy increase with time, so evaluation at t_+ gives the upper energy. If $v_0 > v_\infty$, evaluation at t_- gives it instead. The appropriate endpoint of an enclosed spatial length is chosen by the same monotonicity. If the verified intervals for v_0 and v_∞ overlap, the implementation uses the safe all-time bound $\frac{1}{2}\max\{\overline{v}_0, \overline{v}_\infty\}^2$. The actual speed is a convex combination of those two endpoint speeds in (67), which justifies this branch.

The code evaluates the transcendental clock expressions with interval arithmetic and enlarges interval-to-binary64 conversions outward. Online additions, products, and ceiling evaluations are also rounded outward. Consequently the certificate does not use the plotting profile, an unverified ODE tolerance, or an unverified scalar-root tolerance as an error bound. The enclosure can have downward jumps at cell boundaries, and is not itself asserted to be a viable performance floor. The implemented test $E \ge \theta Q$ is only a sufficient test for $E \ge \theta E^+$. The analytic fallback continues to use the original floor certificate and bypasses any uncertain numerical query.

Lemma 8.6 (A verified lower bound on the optimal clock)**.** *For $0 < S \le S_e$, let $h_j = \max\{0, \min\{S, s_{j+1}\} - s_j\}$ and omit cells with $h_j = 0$. The endpoint enclosures above give*

$$\ell_j = \frac{2h_j}{\sqrt{2(\overline{E}_j + \overline{C}_j h_j)} + \sqrt{2\overline{E}_j}},$$
$$T^*(S) \ge \underline{T}_Q(S) := \sum_{j:h_j>0} \max\left\{\ell_j, \frac{h_j}{M}, \gamma \log \frac{d_0 - s_j - \epsilon}{d_0 - s_j - h_j - \epsilon}\right\}. \tag{73}$$

The terms are finite and positive, including a rest-start cell with $\overline{E}_0 = 0$.

Proof. On cell j, the proof of Lemma 8.5 gives $E^+(s_j + r) \le \overline{E}_j + \overline{C}_j r$. Consequently

$$\int_0^{h_j} \frac{dr}{\sqrt{2E^+(s_j + r)}} \ge \int_0^{h_j} \frac{dr}{\sqrt{2(\overline{E}_j + \overline{C}_j r)}} = \frac{2h_j}{\sqrt{2(\overline{E}_j + \overline{C}_j h_j)} + \sqrt{2\overline{E}_j}}.$$

The rationalized expression evaluates the integrable launch singularity without subtracting close square roots. Summing gives the affine contribution. Independently $v \le M$ and $v \le (d - \epsilon)/\gamma$

give the other two cellwise lower clock integrals. The maximum of three lower bounds is still a lower bound on that cell's clock, including a cell straddling the ceiling switch. Summing proves (73). □

The implementation `clock_lower(S)` encloses spatial lengths, rounds denominators upward, encloses the logarithm with interval arithmetic, and rounds each contribution and accumulated sum downward. Outward coefficient bounds weaken the corresponding ceiling terms as necessary. Thus its returned value is a numerical lower enclosure of the true optimal clock, not merely a quadrature estimate. For an interval-certified target observer, let $\overline{T}_{\mathrm{obs}}$ be a verified upper bound on its actual declaration time. The separate per-run check

$$\overline{T}_{\mathrm{obs}} \le \frac{\underline{T}_Q(S)}{\sqrt{\theta}} + \bar{\Delta} \tag{74}$$

with the right side evaluated downward proves $T_{\mathrm{obs}} \le T^*/\sqrt{\theta} + \bar{\Delta}$ for that run. This numerical check verifies the $T^*/\sqrt{\theta}$ bound; it does not claim verification of the stronger W_θ bound for the conservative interval observer. This stronger sufficient numerical check avoids claiming a universal one-period delay for a conservative interval observer. The whole-hold certificates must remain valid through its declaration, including any holds after true physical entry; checking only the clock inequality would not establish intersample safety.

8.6 An analytic sampled-safe continuation for the running model

The recovery theorem ends at detected tolerance entry. This subsection provides an all-time sampled continuation for the numerical plant class. Its terminal rest position remains within the tolerance-expanded target; it does not assert exact arrival at $\mathcal{T}_g$.

Proposition 8.7 (Admissible headway-tangent held continuation)**.** *Suppose $b = 1$, $a = \alpha(\mathcal{V}(d) - v)$, $\alpha > 0$, and $\alpha\gamma \le 1$. Let $\mathcal{V}$ be nondecreasing and let $L_{\mathcal{V}}$ be a verified Lipschitz bound on $[\epsilon, \epsilon + \gamma M]$. Assume the original stopped-leader face tests. Start from any state in*

$$D_{\mathrm{tail}} = D \cap \{d \le \epsilon + \gamma M\}.$$

At each sample set

$$u_n = -\alpha\mathcal{V}(d_n) + (\alpha - 1/\gamma)v_n \tag{75}$$

and hold this value for a fixed period

$$0 < \Delta < \Delta_{\mathrm{tail}} := \int_{1/\gamma}^{\infty} \frac{dr}{r^2 + \alpha L_{\mathcal{V}}} = \begin{cases} \arctan(\gamma\sqrt{\alpha L_{\mathcal{V}}})/\sqrt{\alpha L_{\mathcal{V}}}, & L_{\mathcal{V}} > 0, \\ \gamma, & L_{\mathcal{V}} = 0. \end{cases} \tag{76}$$

Every command belongs to $[-U, U]$. The physical trajectory remains in D_{tail} for all time, its speed tends to zero, and its gap tends to a value in $[\epsilon, d_{\mathrm{start}}]$. In particular, if the continuation starts at detected entry with $d_{\mathrm{start}} \le d_\eta$, it preserves the tolerance target for all future time.

Proof. Actuator bounds. In D_{tail}, $0 \le v \le \bar{v}(d) = (d - \epsilon)/\gamma \le M$. Since $\alpha - 1/\gamma \le 0$, the held command is nonincreasing in v for fixed d. At $v = 0$, the rest face $\alpha\mathcal{V}(d) + U \ge 0$ gives $u \le U$. At $v = \bar{v}(d)$, the headway face gives

$$\alpha\mathcal{V}(d) - \alpha\bar{v}(d) - U \le -\bar{v}(d)/\gamma,$$

or equivalently $u \ge -U$. These endpoint bounds control every intermediate speed.

One hold. If $v_n = 0$, the command cancels the drift exactly and uniqueness gives the stationary trajectory. Suppose $v_n > 0$ and write $p = \dot{v}$. At the start of the hold, $p(0) = -v_n/\gamma$. Since u_n is

constant, the chain rule for the Lipschitz nondecreasing function $\mathcal{V}$ gives, almost everywhere while $v > 0$ and $d \in [\epsilon, \epsilon + \gamma M]$,

$$\dot{p} = -\alpha p - \alpha m(t) v, \qquad 0 \le m(t) \le L_{\mathcal{V}}. \tag{77}$$

The derivative exists almost everywhere; the asserted chain-rule coefficient follows from monotonicity and the Lipschitz bound. An integrating factor shows $p(t) \le p(0)e^{-\alpha t} < 0$ as long as $v > 0$. Thus speed is decreasing on that interval.

Set $r = -p/v$. Then $r(0) = 1/\gamma$ and

$$\dot{r} = r^2 - \alpha r + \alpha m(t) \le r^2 + \alpha L_{\mathcal{V}}.$$

The solution of $R' = R^2 + \alpha L_{\mathcal{V}}$, $R(0) = 1/\gamma$, is finite on $[0, \Delta_{\text{tail}})$. Scalar comparison gives $r \le R$. Because $v(t) = v_n \exp(-\int_0^t r(\tau)d\tau)$, speed cannot reach zero before Δ_{tail} while the gap stays in the stated interval.

The initial headway derivative is zero. Furthermore,

$$\ddot{h} = -p - \gamma\dot{p} = (\alpha\gamma - 1)p + \alpha\gamma m(t) v \ge 0.$$

Hence $\dot{h} \ge 0$ and $h(t) \ge h_n \ge 0$ on the hold. Speed decreases, so the cap is respected; gap also decreases and remains above ϵ by headway safety. These inequalities exclude a first exit from D_{tail} and extend the estimates over the complete hold.

All-time continuation and limit. Reapply the argument at each sample. Integration of $p(t) \le -(v_n/\gamma)e^{-\alpha t}$ gives

$$v_{n+1} \le \rho_\Delta v_n, \qquad \rho_\Delta = 1 - \frac{1 - e^{-\alpha\Delta}}{\alpha\gamma} \in (0, 1). \tag{78}$$

To check positivity, observe $\Delta_{\text{tail}} \le \gamma$ and $1 - e^{-\alpha\Delta} < \alpha\Delta < \alpha\gamma$. Therefore the sample speeds converge geometrically to zero, and monotonicity within each hold gives $v(t) \to 0$. Gap is decreasing and bounded below by ϵ, hence has a limit in the stated interval. No equality with d_g is inferred. □

This continuation has a fully analytic period certificate and avoids transferring a continuous-time backup guarantee to a sampled controller without proof. It assumes exact sampled state feedback; a state estimation or measurement-error claim would need the corresponding robust margin. During a numerical validation, nonzero simulation error must still be propagated or enclosed rather than silently discarded.

8.7 An a priori certificate for a useful held-input period

8.7.1 Setting and the ceiling clock

Consider the autonomous stopped-leader dynamics

$$\dot{d} = -v, \qquad \dot{v} = a(d, v) + b(d, v)u, \qquad |u| \le U,$$

under the regularity and face assumptions of Theorem 8.3. In particular, a, b are locally Lipschitz in the physical state, $b \ge b_* > 0$, and the original safety domain is $D = \{d - \gamma v - \epsilon \ge 0,\ 0 \le v \le M\}$. Fix one launchable episode, its original initial state (d_0, v_0), an entry travel $S > 0$, and an extension $S_e > S$ below the first terminal zero of its maximal profile. The episode profile is not reset at sampling instants. Write

$$s = d_0 - d, \quad E = v^2/2, \quad B(s) = \tfrac{1}{2}\min\{M, (d_0 - s - \epsilon)/\gamma\}^2, \quad q(s) = \theta E^+(s), \quad 0 < \theta < 1,$$

and

$$\mathsf{F}(r, e, u) = a(d_0 - r, \sqrt{2e}) + b(d_0 - r, \sqrt{2e})u.$$

Here F denotes the spatial field F in (51). On $[0, S_e]$, assume that B, q are Lipschitz, $0 \le q < B$, $\min B > 0$, and

$$q'(r) \le \mathsf{F}(r, q(r), U), \qquad \mathsf{F}(r, B(r), -U) \le B'(r) \quad \text{for almost every } r. \tag{79}$$

The floor is positive except possibly at an initial rest state. In that case, $v_0 = 0$, $q(0) = 0$, and

$$q(r) = \kappa r + o(r), \qquad \kappa = \theta g_0 > 0, \qquad g_0 = a(d_0, 0) + b(d_0, 0)U. \tag{80}$$

All spatial derivatives below are interpreted almost everywhere; finitely or infinitely many Lipschitz-profile kinks are permitted.

Define the fastest ceiling clock on a prefix by

$$\mathcal{T}_B(s, r) = \int_s^r \frac{d\xi}{\sqrt{2B(\xi)}}. \tag{81}$$

For a proposed fixed period $\Delta > 0$, require

$$\Delta \le \mathcal{T}_B(S, S_e). \tag{82}$$

For every $s \in [0, S]$, let $r_\Delta(s) \le S_e$ be the unique endpoint with $\mathcal{T}_B(s, r_\Delta(s)) = \Delta$, and set $J_s = [s, r_\Delta(s)]$, $\ell_s = r_\Delta(s) - s$. The endpoint exists because the integrand is positive and $\mathcal{T}_B(s, S_e) \ge \mathcal{T}_B(S, S_e)$. Every profile below B takes at least Δ physical time to traverse J_s. Thus the interval is a common spatial prefix for all candidate held inputs, despite their different physical travel.

The prefix is explicit for the headway/speed ceiling. Put $x = d_0 - s - \epsilon$ and define its endpoint gap $x_\Delta = x - \ell_s$. Then

$$x_\Delta = \begin{cases} xe^{-\Delta/\gamma}, & x \le \gamma M, \\ x - M\Delta, & x > \gamma M, \quad \Delta \le (x - \gamma M)/M, \\ \gamma M \exp\left[-\dfrac{\Delta - (x - \gamma M)/M}{\gamma}\right], & x > \gamma M, \quad \Delta > (x - \gamma M)/M. \end{cases} \tag{83}$$

In particular, $\ell_s = x(1 - e^{-\Delta/\gamma})$ on the headway branch, and always $\ell_s \le M\Delta$.

8.7.2 Two constant boundary witnesses

On a prefix $J = J_s$, define the required boundary inputs

$$R_q(r) = \frac{q'(r) - a(d_0 - r, \sqrt{2q(r)})}{b(d_0 - r, \sqrt{2q(r)})},$$
$$R_B(r) = \frac{B'(r) - a(d_0 - r, \sqrt{2B(r)})}{b(d_0 - r, \sqrt{2B(r)})}. \tag{84}$$

Select constants $u_q, u_B \in [-U, U]$ such that

$$u_q \ge \operatorname*{ess\,sup}_J R_q, \qquad u_B \le \operatorname*{ess\,inf}_J R_B. \tag{85}$$

At $s = 0$, $q(0) = 0$, also impose the explicit launch guard

$$a(d_0, 0) + b(d_0, 0)u_q \ge \kappa. \tag{86}$$

These requirements are feasible by (79) and $\kappa = \theta g_0 < g_0$: one can always take $u_q = U$, $u_B = -U$. The least conservative exact choices within this construction are $u_q = \max\{-U, \operatorname{ess\,sup}_J R_q\}$, enlarged by the launch guard when needed, and $u_B = \min\{U, \operatorname{ess\,inf}_J R_B\}$. Verified upper and lower enclosures may replace the exact extrema.

Choose nonnegative integrable functions c_B, c_q satisfying, for almost every $r \in J$ and every $e \in [q(r), B(r)]$,

$$\mathsf{F}(r, e, u_q) - B'(r) \le c_B(r), \qquad q'(r) - \mathsf{F}(r, e, u_B) \le c_q(r). \tag{87}$$

For example, the positive parts of the corresponding extrema over the compact energy interval give such bounds. Set

$$H_B(J) = \int_J c_B(r)\, dr, \qquad H_q(J) = \int_J c_q(r)\, dr. \tag{88}$$

In implementation, verified upper bounds on these two integrals are sufficient. The a priori condition is

$$\boxed{u_q \le u_B \quad \text{or} \quad H_B(J_s) + H_q(J_s) \le B(s) - q(s).} \tag{89}$$

It is a test on model/profile bounds and a spatial prefix; it does not require integrating a trajectory for each candidate input.

Lemma 8.8 (A feasible constant actuator from boundary clearances)**.** *Fix $s \in [0, S]$ and suppose (89) holds. For every $E \in [q(s), B(s)]$ there is a held input whose physical spatial profile remains in the band on J_s. If $u_q \le u_B$, every $u \in [u_q, u_B]$ has this property. Otherwise one feasible choice is*

$$u_F(s, E) = \begin{cases} u_q, & B(s) - E \ge H_B(J_s), \\ u_B, & B(s) - E < H_B(J_s). \end{cases} \tag{90}$$

A rest start uses a positive, no-waiting physical launch with initial acceleration at least κ.

Proof. By (85), q is a subsolution of $e' = \mathsf{F}(r, e, u_q)$ and B is a supersolution of $e' = \mathsf{F}(r, e, u_B)$. For positive energy the usual scalar comparison theorem applies. At the initial rest point the launch guard and physical local uniqueness give $e(r) = p_0 r + o(r)$ with $p_0 \ge \kappa > 0$. In a comparison with q, local Lipschitz continuity in speed gives an energy difference coefficient bounded by $C/\sqrt{q(r)}$. This coefficient is integrable at zero by (80), so the positive-part Grönwall argument also compares through launch. Consequently u_q preserves the floor and u_B preserves the ceiling wherever their positive profiles exist. Monotonicity in u gives both properties for every $u \in [u_q, u_B]$ when this interval is nonempty.

Consider the budget alternative. If $B(s) - E \ge H_B(J_s)$, use u_q. Until an upper contact, the floor is already preserved and

$$B(r) - e(r) \ge B(s) - E - \int_s^r c_B(\xi)\, d\xi \ge \int_r^{r_\Delta(s)} c_B(\xi)\, d\xi \ge 0.$$

This excludes an upper exit. To make the equality case explicit, an interior first contact can occur only when the remaining integral is zero. Then $c_B = 0$ almost everywhere on the remaining prefix, so B is a supersolution there as well. Comparison from that contact excludes a subsequent exit. Contact at the right endpoint is allowed.

Otherwise $B(s) - E < H_B(J_s)$, and the budget condition implies

$$E - q(s) = B(s) - q(s) - (B(s) - E) > H_q(J_s).$$

Use u_B. Its ceiling is preserved, and until any lower contact,

$$e(r) - q(r) \ge E - q(s) - \int_s^r c_q(\xi)\, d\xi > 0.$$

Thus a lower exit is impossible. If the initial state is at rest, $B(0) - E = B(0) - q(0) \geq H_B(J_0)$, so the first branch, with its launch guard, is necessarily chosen. Bounded slopes, the positive floor away from launch, and $\min B > 0$ give continuation over the full prefix in each case. □

Theorem 8.9 (A priori fixed-period feasibility and recovery)**.** *Suppose* (82) *and* (89) *hold for every $s \in [0, S]$, with the episode assumptions above. For each pre-entry state $q(s) \leq E \leq B(s)$, the set*

$$\begin{aligned}\mathcal{I}_\Delta(s, E) = \{u \in [-U, U] : q \leq e_u \leq B \text{ throughout } J_s,\\ E > 0 \text{ or } a(d_0, 0) + b(d_0, 0)u \geq \kappa\}\end{aligned} \tag{91}$$

is a nonempty closed interval. Here $e_u(s) = E$, $e'_u = \mathsf{F}(r, e_u, u)$, and at rest e_u is the positive branch induced by the physical constant-input solution.

Selecting any member of this interval at each update, and holding it for the fixed physical period Δ, is recursively feasible through detected target entry. The trajectory remains in the original safety domain and above q throughout every hold. For the initial episode state and exact sampled state observation,

$$T^* \leq T_{\text{entry}} \leq W_\theta \leq T^*/\sqrt{\theta}, \qquad T_{\text{detected}} \leq W_\theta + \Delta. \tag{92}$$

In particular, projection of an arbitrary sampled nominal $k_n \in [-U, U]$ onto $\mathcal{I}_\Delta(s_n, E_n)$ is well defined. The explicit choice in Lemma 8.8 supplies a fallback using the a priori certificate alone. Exact computation of the interval endpoints is not required: any sound inner approximation $\widehat{\mathcal{I}}_\Delta(s, E) \subseteq \mathcal{I}_\Delta(s, E)$ that contains a certified witness from that lemma is nonempty, and selecting any of its members retains the guarantees. A singleton witness is sufficient. If projection is used, a closed inner approximation makes a nearest point exist; the projection need not be unique unless that approximation is convex.

Proof. Nonemptiness follows from Lemma 8.8. For two feasible inputs $u_1 \leq u_2$, scalar comparison orders their profiles because $b > 0$. The same comparison bounds every intermediate-input profile between them. At rest the closed positive-acceleration guard is inherited by intermediate inputs, and the integrable $C/\sqrt{q}$ comparison extends the ordering through zero energy. Thus every intermediate input is feasible.

For closedness, let feasible u_j tend to u. Their profiles have uniformly bounded slopes on a common compact band. Arzelà–Ascoli and the integral equation give a limit profile with input u in the same closed band. The launch guard passes to the limit. Positive-energy uniqueness, or the integrable comparison at the rest start, identifies this profile with the physical branch. Therefore the input set is closed; it is compact as a subset of $[-U, U]$.

For a feasible profile, its clock satisfies

$$\tau_u(r_\Delta(s)) = \int_s^{r_\Delta(s)} \frac{dr}{\sqrt{2e_u(r)}} \geq \mathcal{T}_B(s, r_\Delta(s)) = \Delta.$$

The clock is finite and strictly increasing, including its integrable $r^{-1/2}$ launch singularity. Inverting it on $[0, \Delta]$ gives the unique physical held-input trajectory. It stays in the certified prefix and band. If the endpoint is pre-entry, the same certificate applies again, proving recursion. The uniform extension covers the last hold crossing S.

There is no waiting interval at launch, and the positive floor prevents later waiting before entry. Finite entry is not assumed in this argument: if travel converged to some $s_\infty \leq S$ without entry, positive launch would give $s_\infty > 0$. Continuity and $q(s_\infty) > 0$ would then bound v away from zero, contradicting convergence of travel. Thus entry occurs in finite time, and the physical trajectory has an absolutely continuous spatial energy profile on the entire entry interval. Comparing it with the lower reflected profile defining W_θ uses only admissibility and $e \geq q$: where $E_\theta^- > e \geq q$, the lower profile follows f_- while $e' \geq f_-(r, e)$. The positive-part comparison, again with its integrable

launch coefficient, gives $e \geq E_\theta^-$. Integrating clocks yields $T_{\text{entry}} \leq W_\theta \leq T^*/\sqrt{\theta}$. The original whole-domain face assumptions give a safe backup continuation from the state at detected entry. Concatenating it with the certified finite trajectory produces a globally safe admissible input, so the definition of T^* gives $T^* \leq T_{\text{entry}}$. Exact sampled detection occurs at most one period after physical entry. Fixed positive Δ also rules out accumulation of updates. □

Combining a spatial fallback with physical-hold acceptance. The nominal may alternatively be accepted whenever an independent sound certificate proves its actual physical hold of duration Δ safe, above q, inside the extension, and compatible with the positive launch guard of Theorem 8.3. Such a command need not belong to $\mathcal{I}_\Delta$: it need not preserve the band over the entire common spatial prefix if it travels less far during that physical hold. If this test rejects, use the spatial witness from Lemma 8.8. Each accepted hold ends in the same band, and the witness exists at every subsequent pre-entry state; induction therefore gives fixed-period feasibility. The admissibility/floor clock comparison in the proof above and exact-observation detection bound apply unchanged. The physical acceptance set need not be an interval. This combined rule is the one implemented in Section 13.10.

8.7.3 Preservation of the general existence result

Write $\mathcal{B}_e = \{(s, E) : 0 \leq s \leq S_e,\ q(s) \leq E \leq B(s)\}$.

Proposition 8.10 (The local certificate contains the coarse bound)**.** *Let $g \leq \min(B - q)$, $P \geq \sup_{\mathcal{B}_e, |u| \leq U} |\mathsf{F}|$, $L_B \geq \mathrm{Lip}(B)$, and $L_q \geq \mathrm{Lip}(q)$ be the global constants of (52). The certificate is satisfied if*

$$0 < \Delta \leq \min\left\{\mathcal{T}_B(S, S_e), \frac{g}{M(2P + L_B + L_q)}\right\}. \tag{93}$$

A zero denominator is interpreted as $+\infty$. In particular, every period allowed by the original $\delta_0 = M^{-1}\min\{S_e - S, g/(4K)\}$, $K = P + \max\{L_B, L_q\}$, satisfies this refined sufficient condition.

Proof. Take the admissible extreme witnesses $u_q = U$, $u_B = -U$ and the constant closing-rate bounds $c_B = P + L_B$, $c_q = P + L_q$. Since $\ell_s \leq M\Delta$, their sum of budgets is at most $M\Delta(2P + L_B + L_q) \leq g \leq B(s) - q(s)$. Also $\mathcal{T}_B(S, S_e) \geq (S_e - S)/M$ and $2P + L_B + L_q \leq 2K$. Thus the coarse period is included. □

The refined certificate therefore retains positive-period existence for the general model class. A useful numerical value does require verified local model/profile bounds satisfying (89) over the entire specified episode. Pointwise floating-point evaluations, or successful holds along selected trajectories, do not establish this uniform statement. The theorem does not assert that the fixed physical-time feasible input set is an interval: interval structure is proved on the common spatial prefix J_s.

8.7.4 Simplification for the running optimal-velocity model

For $a = \alpha(\mathcal{V}(d) - v)$, $b = 1$, $\alpha > 0$, F is nonincreasing in energy. Consequently the exact closing-rate choices reduce to two scalar boundary evaluations:

$$\begin{aligned} c_B(r) &= [\mathsf{F}(r, q(r), u_q) - B'(r)]_+, \\ c_q(r) &= [q'(r) - \mathsf{F}(r, B(r), u_B)]_+. \end{aligned} \tag{94}$$

No energy-variable optimization is needed. A verified partition of the travel coordinate can enclose the required boundary inputs and the two nonnegative integrals. To certify a fixed period, the partition must cover every possible start travel $s \in [0, S]$, including prefixes crossing launch/ceiling transitions; a grid without interval control between its nodes is insufficient.

Scope. The result strengthens the feasibility part of Theorem 8.3 while preserving its recovery clock. It does not assert strict continuous-time fidelity for the held law, sharpness of the sampled worst-case clock, a real-time computation deadline, or improved cost or motion metrics. The explicit witness and the candidate-search controller in Section 13 are different input-selection rules; performance comparisons apply to the rule actually simulated.

8.8 An analytic period for complete optimal-velocity recovery episodes

Consider

$$\dot{d} = -v, \qquad \dot{v} = \alpha(\mathcal{V}(d) - v) + u, \qquad |u| \le U, \tag{95}$$

where $\alpha, \gamma, M, U > 0$ and $\mathcal{V}$ is nondecreasing and $L_{\mathcal{V}}$-Lipschitz on the extended episode. Write

$$x = d - \epsilon, \qquad \bar{v}(x) = \min\{M, x/\gamma\}, \qquad B(s) = \tfrac{1}{2}\bar{v}(d_0 - s - \epsilon)^2, \qquad C(d) = \alpha\mathcal{V}(d) + U.$$

Let E^+ be the maximal physical energy profile for the fixed initial condition (d_0, v_0), and put $q = \theta E^+$, $0 < \theta < 1$. The profile is fixed throughout the episode. Assume the original face tests and the fully admissible floor condition

$$\mathsf{F}(s, q(s), -U) \le q'(s) \le \mathsf{F}(s, q(s), U), \qquad \mathsf{F}(s, B(s), -U) \le B'(s) \quad \text{a.e.}, \tag{96}$$

where $\mathsf{F}(s, E, u) = C(d_0 - s) - U - \alpha\sqrt{2E} + u$. In addition assume

$$\alpha\gamma \le 1, \qquad 0 < C_{\min} \le C(d) \le C_{\max} \quad (\epsilon + x_e \le d \le d_0), \tag{97}$$

where $0 < x_e < x_\eta = d_\eta - \epsilon$ is the profile-extension clearance. These hypotheses hold for the three reported models and the two reported values of θ; Proposition 7.3 and the bounds below establish (96).

8.8.1 A closed-form ceiling-contact certificate

Choose $x_T \in (0, \gamma M]$, set $d_T = \epsilon + x_T$, $v_T = x_T/\gamma$, and suppose $d_0 > d_T$ and $v_0 \le v_T$. Let c_T be a lower bound for $C(d)$ on $[d_T, d_0]$, with $c_T > \alpha v_T$. Define

$$D_T(v_0) = \frac{c_T}{\alpha^2} \log \frac{c_T - \alpha v_0}{c_T - \alpha v_T} - \frac{v_T - v_0}{\alpha}. \tag{98}$$

Lemma 8.11 (Certified onset of the headway-ceiling profile)**.** *If $D_T(v_0) \le d_0 - d_T$, then $E^+(s) = B(s)$ whenever $x = d_0 - s - \epsilon \le x_T$ within the extended episode.*

Proof. The constant-authority physical trajectory $\dot{v} = c_T - \alpha v$, $\dot{s} = v$, launched from v_0, reaches v_T after traveling the distance in (98). Indeed $v(t) = c_T/\alpha + (v_0 - c_T/\alpha)e^{-\alpha t}$ and integration of v gives that formula. Its free speed increases because $v_0 \le v_T < c_T/\alpha$.

Let E_c be its spatial energy reflected at the same nonincreasing ceiling B. Before contact it equals the free energy. After contact it equals B: the ceiling speed is then at most the free equilibrium c_T/α, so $c_T - \alpha\sqrt{2B} \ge 0 \ge B'$. Thus E_c reaches B no later than $s_T = d_0 - d_T$. On $[0, s_T]$, $C(d_0 - s) \ge c_T$, and scalar comparison for the upper-constrained equations gives $E_c \le E^+ \le B$. For example, on $E_c > E^+$ the latter is below its ceiling, while $E_c' \le c_T - \alpha\sqrt{2E_c}$; the positive difference cannot grow. This comparison also covers physical launch from rest. Consequently $E^+(s_T) = B(s_T)$.

For $x \le x_T \le \gamma M$, the ceiling slope is $B' = -\bar{v}/\gamma$, and

$$\mathsf{F}(s, B, U) - B' = C(d_0 - s) + (1/\gamma - \alpha)\bar{v} \ge C_{\min} > 0.$$

The maximal profile therefore remains on the ceiling throughout the rest of the extended episode. □

8.8.2 Explicit bounds for the two regions

Define

$$\begin{aligned} D_0 &= C_{\max} + [2U - (1-\theta)C_{\min}]_+, \\ \Delta_{\text{pre}} &= \frac{(1-\theta)v_T}{2\{D_0 + (\alpha + 1/\gamma)v_T\}}, \\ \kappa_q &= \frac{\alpha\sqrt{\theta}}{\gamma} - \frac{\theta}{\gamma^2}, \qquad \kappa_B = \frac{\alpha}{\gamma} - \frac{1}{\gamma^2}, \\ \Lambda_q &= \alpha L_{\mathcal{V}} + |\kappa_q|, \qquad \Lambda_B = \alpha L_{\mathcal{V}} + |\kappa_B|, \\ A_T &= \frac{\gamma^2(\Lambda_q + \Lambda_B)}{1-\theta}, \qquad z_T = \frac{1}{2 + \sqrt{4 + 2A_T}}, \end{aligned} \tag{99}$$

$$\Delta_{\text{band}} = -\gamma \log(1 - z_T), \qquad \Delta_{\text{ext}} = \gamma(1 - x_e/x_\eta). \tag{100}$$

The quantity D_0 is positive, since $C_{\max} > 0$. Here Δ_{band} concerns the recovery band; it is distinct from the post-entry continuation period in (76). All constants are prior coefficient or geometry bounds; they require no integration of candidate held trajectories.

Theorem 8.12 (Useful analytic fixed period for a complete OVM episode)**.** *Under the original autonomous regularity and whole-domain face assumptions, (96)–(97), and the ceiling-contact test of Lemma 8.11, every fixed period*

$$0 < \Delta \le \min\{\Delta_{\text{pre}}, \Delta_{\text{band}}, \Delta_{\text{ext}}\} \tag{101}$$

admits a constant actuator value preserving $q \le E \le B$ throughout the hold from every pre-entry band state. An explicit feasible value is given below. At the episode's initial rest state it makes a positive physical launch. At the initial rest hold, every accepted alternative must also satisfy $C(d_0) - U + u \ge \theta C(d_0) > 0$; alternatively, use the displayed full-throttle launch witness. Applying a band-certified nominal input satisfying this launch guard when available, and the witness otherwise, is recursively feasible at the fixed period. With exact state observations,

$$T^* \le T_{\text{entry}} \le W_\theta \le T^*/\sqrt{\theta}, \qquad T_{\text{detected}} \le W_\theta + \Delta. \tag{102}$$

These claims apply through detected entry. The post-entry controller and any numerical observation error require their separate certificates.

Explicit pre-tail witness. At a state with $x \ge x_T$, put $v_{\text{cap}} = \bar{v}(x)$ and

$$\ell = v_{\text{cap}}\Delta, \qquad c_U = C_{\max} + v_{\text{cap}}/\gamma, \qquad c_L = [2U - (1-\theta)C_{\min}]_+ + \alpha v_{\text{cap}}.$$

Select $u = U$ if $B(s) - E \ge \ell c_U$, and select $u = -U$ otherwise. The proof below shows that in the second case $E - q(s) > \ell c_L$.

Explicit tail witness. At a state with $x < x_T$, use the prefix $\ell = x(1 - e^{-\Delta/\gamma})$. On this region define

$$\begin{aligned} R_q(x) &= -\alpha\mathcal{V}(\epsilon + x) + \kappa_q x, \\ R_B(x) &= -\alpha\mathcal{V}(\epsilon + x) + \kappa_B x, \\ u_q &= \min\{U, R_q(x) + (\alpha L_{\mathcal{V}} - \kappa_q)_+\ell\}, \qquad u_B = R_B(x), \\ D_x &= (1-\theta)x/\gamma^2, \qquad c_U = D_x + \Lambda_q\ell, \qquad c_L = D_x + \Lambda_B\ell. \end{aligned}$$

Select $u = u_q$ if $B(s) - E \ge \ell c_U$, and $u = u_B$ otherwise. Each formula uses only the observed state, the fixed reference value $q(s)$ when evaluating the band membership, and known model constants. The choice between the witnesses uses only $B(s) - E$ and the displayed bounds.

Proof of Theorem 8.12. Pre-tail region. On a prefix beginning at $x \geq x_T$, the ceiling speed is at most $v_{\mathrm{cap}} = \bar{v}(x)$. The full-throttle field preserves the floor by (96). Its rate of loss of upper clearance is bounded by

$$\mathsf{F}(r, q(r), U) - B'(r) \leq C_{\max} + v_{\mathrm{cap}}/\gamma = c_U.$$

This also bounds $\mathsf{F}(r, e, U) - B'(r)$ for every $e \geq q(r)$, because F decreases with energy. The full-braking field preserves the ceiling. The reflected maximal profile always satisfies $(E^+)' \leq C - \alpha\sqrt{2E^+}$, and hence

$$\begin{aligned} q' - \mathsf{F}(r, B, -U) &\leq 2U - (1-\theta)C + \alpha\bar{v} - \theta\alpha\sqrt{2E^+} \\ &\leq [2U - (1-\theta)C_{\min}]_+ + \alpha v_{\mathrm{cap}} = c_L. \end{aligned}$$

The same upper bound applies to $q' - \mathsf{F}(r, e, -U)$ for $e \leq B$. Thus an initial upper clearance of ℓc_U makes U feasible, and an initial lower clearance of ℓc_L makes $-U$ feasible. The sum of the available clearances is $B - q \geq (1-\theta)v_{\mathrm{cap}}^2/2$.

Then

$$\ell(c_U + c_L) \leq \tfrac{1}{2}(1-\theta)v_{\mathrm{cap}}^2$$

follows from (99), because $v_{\mathrm{cap}} \geq v_T$ and $v_{\mathrm{cap}}/[D_0 + (\alpha + 1/\gamma)v_{\mathrm{cap}}]$ is increasing. At least one witness is therefore feasible by the first-exit clearance argument. The estimate covers a prefix that crosses the splitting threshold or a ceiling/profile kink, since its derivative bounds hold almost everywhere throughout the extended band.

Tail region. Lemma 8.11 gives $q = \theta B$. The tangent actuator values for the floor and ceiling are R_q and R_B. They belong to $[-U, U]$: this is (96) for R_q, and the original headway-face test together with $\mathsf{F}(B, U) > B'$ for R_B. For $y \in [x - \ell, x]$, monotonicity and the Lipschitz bound of $\mathcal{V}$ imply

$$R_q(y) \leq R_q(x) + (\alpha L_{\mathcal{V}} - \kappa_q)_+ \ell \quad \text{and} \quad R_B(y) \geq R_B(x),$$

where the second inequality uses $\kappa_B \leq 0$. The cap at U in u_q preserves the first inequality because $R_q(y) \leq U$. Therefore u_q preserves the floor and u_B preserves the ceiling by scalar comparison.

Moreover

$$\begin{aligned} 0 \leq u_q - R_q(y) &\leq [(\alpha L_{\mathcal{V}} - \kappa_q)_+ + (\kappa_q)_+]\ell \leq \Lambda_q \ell, \\ 0 \leq R_B(y) - u_B &\leq \Lambda_B \ell. \end{aligned}$$

Since $q' - B' = (1-\theta)y/\gamma^2 \leq D_x$, these inequalities give the displayed opposite-clearance rates c_U, c_L. Writing $z = \ell/x$, the sufficient width condition becomes

$$\ell(c_U + c_L) \leq B - q \quad \Longleftarrow \quad 2z + A_T z^2 \leq \tfrac{1}{2}.$$

The positive root is z_T, so (100) gives the required condition and proves existence of the stated witness.

Clock, extension, and recursion. In the pre-tail region, $\int_s^{s+\ell} (2B)^{-1/2}\, dr \geq \ell/v_{\mathrm{cap}} = \Delta$. In the tail it equals $\gamma \log[x/(x-\ell)] = \Delta$. Thus each feasible spatial prefix covers the requested physical hold. The pre-tail prefix satisfies $x - \ell \geq x(1 - \Delta/\gamma)$; the tail prefix satisfies the stronger inequality $x - \ell = xe^{-\Delta/\gamma} \geq x(1 - \Delta/\gamma)$. For every pre-entry state $x \geq x_\eta$, the extension bound in (100) therefore gives $x - \ell \geq x_e$.

The only possible zero-energy point in the band is its initial rest point. At that point the pre-tail rule selects U, since its upper clearance is at least the required budget; the launch acceleration is $C(d_0) > 0$. The prefix is the unique physical no-waiting branch. All later holds start at positive energy. Reapplying the construction at each sample proves recursive feasibility and excludes accumulation of update times. Finite entry follows from the positive floor as in Theorem 8.9. The same floor comparison gives $E_\Delta \geq E_\theta^-$, hence the upper entry bounds in (102); the final hold adds at most Δ to detection. The original whole-domain face certificate allows a safe backup continuation after detected entry. Concatenation therefore places the certified finite trajectory in the admissible input class defining T^*, and gives $T^* \leq T_{\mathrm{entry}}$. □

8.8.3 Certified constants for the reported models

The conservative decimal bounds in Table 2 apply simultaneously to the rest and moving starts $h_0 = .8$, $v_0 \in \{0, .4\}$, $d_0 = \epsilon + h_0 + \gamma v_0$: The available rest-start distances to the thresholds

Table 2: Analytic coefficient and ceiling-contact bounds.

	$C_{\min}$	$C_{\max}$	c_T	x_T	upper bound on $D_T(0)$
C1	2.55	4.36	3.05	.20	.479
C2	4.05	4.78	4.43	.40	.253
C3	5.00	6.79	5.76	.45	.246

are $.60, .40, .35$, respectively. The moving start increases the available distance and decreases $D_T(v_0)$. These inequalities verify Lemma 8.11 without evaluating a numerical maximal profile. The conditions $\alpha\gamma \le 1$ and $2U \ge (1-\theta)C_{\min}$ are immediate. Moreover, these same bounds give

$$C_{\max} \le 2U, \qquad \alpha M \le 2C_{\min}, \qquad C_{\max} - 2U + (1/\gamma - \alpha)M \le 0. \tag{103}$$

Together with $C_{\min} > 0$, they certify all original safety faces on this extended domain. They also certify full floor admissibility: putting $r = \sqrt{\theta}$ and $v = \sqrt{2E^+}$, the upper margin on a free arc is $(1-r)[(1+r)C - \alpha r v] \ge 0$; reflection can only reduce $(E^+)'$. The lower margin on a free arc is $2U - (1-r^2)C + \alpha(r-r^2)v \ge 0$; on the speed cap it is $2U - C + \alpha r M \ge 0$. On the headway ceiling it is $2U - C + v(\alpha r - r^2/\gamma)$, a concave function of r whose endpoint values are nonnegative by (103). This proves (96) for the specified models without requiring a numerically estimated floor.

With $x_e = .002$ and the smaller target clearance $x_\eta = .01$, Table 3 gives the period bounds as rounded decimal diagnostics: The comparisons proving a common period $\Delta = .01$ have been

Table 3: Components of the analytic sufficient period.

Model	θ	Δ_{pre}	Δ_{band}	Δ_{ext}
C1	.25	.0187056	.0512863	.16
C1	.50	.0120861	.0478647	.16
C2	.25	.0285384	.0740541	.24
C2	.50	.0179862	.0677825	.24
C3	.25	.0240394	.0884285	.28
C3	.50	.0150861	.0849453	.28

verified in 51 outward interval checks at 60 decimal digits by `ovm_analytic_constants.py` in `sampling_certificate/`; its JSON output records every comparison and interval. No candidate trajectory or approximate reference profile enters that calculation. The tracking-cost target $x_\eta = .21$ is a subset of the same pre-entry episode and has a larger extension allowance, so the same period applies. Thus this is a full-episode all-band-state certificate for all 24 combinations of model, target, start, and θ, uniformly over the nominal.

Implementation and scope. The period is certified separately for each fixed initial-state/target episode, using its own E^+ throughout execution. It is uniform over that episode's band states and sampled nominals; the finite collection uses six initial-state maximal profiles and two values of θ, giving twelve profile/band choices. Each is used for both targets, hence twenty-four model/start/fraction/target combinations. The analytic construction explicitly uses $\alpha\gamma \le 1$ and $C_{\min} > 0$, particularly to retain ceiling contact and to make R_B monotone on the headway branch.

For fixed plant, episode, and contact constants, provided the floor conditions hold as $\theta \uparrow 1$, the combined sufficient bound in (101) is of order $1-\theta$. Indeed

$$\frac{\Delta_{\mathrm{pre}}}{1-\theta} \longrightarrow \frac{v_T}{2\{C_{\max} + 2U + (\alpha + 1/\gamma)v_T\}} > 0.$$

Since $\Lambda_q + \Lambda_B$ stays bounded, $\Delta_{\mathrm{band}} \geq c\sqrt{1-\theta}$ for some $c > 0$ when θ is sufficiently close to one, while $\Delta_{\mathrm{ext}} > 0$ stays fixed. Thus Δ_{pre} eventually limits the combined bound. The tail bound itself need not have linear scaling.

The constructive witness is an analytic fallback. It does not assert that the finite candidate search of Section 13 will find a feasible input without this fallback, nor that this witness improves the reported cost or motion comparisons. Exact-state feasibility is separate from the propagated numerical enclosures used by the offline experiments in Section 13. The theorem certifies a useful specified period; it does not certify an implementation runtime deadline.

8.9 An offline certificate for the common period 0.025

The analytic construction already gives a useful period without a numerical reference profile. A sharper value can be certified before running a controller by verifying (89) over the complete pre-entry band. This calculation covers every state in that band; it does not infer feasibility from a collection of trajectories. All parameters below are exact decimal real data, and periods use normalized model time.

The models have $\epsilon = .02$, $M = 1.96$, $U = 5$, and $\dot{v} = \alpha(\mathcal{V}(d) - v) + u$:

Model	α	γ	$\mathcal{V}(d)$
C1	2.6	.20	$\tanh d - \tanh 2$
C2	1.0	.30	$\tanh d - \tanh 2$
C3	2.6	.35	$\mathrm{clip}_{[0,1.5]}(.8(d-.1))$

The initial states satisfy $h_0 = .8$, $v_0 \in \{0, .4\}$, and $d_0 = \epsilon + h_0 + \gamma v_0$. Both $\theta = .25, .5$ are tested. The extension is $d - \epsilon = .002$, and certification continues to the gap-entry target $d - \epsilon = .01$. The tracking-cost entry target $d - \epsilon = .21$ is covered by the same calculation because its pre-entry interval is a subset. Thus twelve profile/fraction certificates cover all twenty-four model, target, start, and fraction combinations, uniformly over the sampled nominal commands.

8.9.1 A finite certificate for a continuum of sample positions

Partition the travel coordinate into cells $A_j = [s_j, s_{j+1}]$ of exact rational length 10^{-4}. For every $s \in A_j$, enclose its ceiling-clock prefix J_s by a single larger prefix $\widehat{J}_j = [s_j, s_{k(j)}]$. The terminal position of the ceiling flow is monotone in its starting position, so it is sufficient to evaluate that flow from s_{j+1}, enclose the endpoint from above, and round it forward to a grid node. Reject a cell if this enlarged prefix leaves the specified extension.

On $\widehat{J}_j$, obtain constant inputs $u_{q,j}, u_{B,j}$ satisfying (85), with the explicit launch guard in the first rest-start cell. Let $\overline{H}_{B,j}, \overline{H}_{q,j}$ enclose the two nonnegative budgets from above, and let

$$0 < \underline{w}_j \leq \inf_{s \in A_j} \{B(s) - q(s)\}.$$

Then the finite set of inequalities

$$\overline{H}_{B,j} + \overline{H}_{q,j} \leq \underline{w}_j \quad \text{for every pre-entry cell } j \tag{104}$$

implies the certificate in Theorem 8.9 at every start position. Indeed the enlarged prefix bounds all actual prefixes, and its nonnegative integrals bound every corresponding subinterval. The width bound holds throughout the entire starting cell. No discretization of the energy coordinate is needed, since the OVM field is decreasing in energy and (94) places the two worst cases on the boundaries.

8.9.2 Verified profile and derivative enclosures

Write $C(s) = \alpha\mathcal{V}(d_0 - s) + U$, which is positive and nonincreasing. On a spatial cell, endpoint interval evaluations give $c_- \le C(s) \le c_+$. Constant-authority equations

$$y' = c - \alpha\sqrt{2y}, \qquad y(0) = e \ge 0,$$

bound the free scalar flow from below and above. An elementary enclosure is sufficient. Put $f_e = c - \alpha\sqrt{2e}$ and, for cell length $h > 0$, define

$$a_h = \max\{0, e + h\min(f_e, 0)\}, \qquad b_h = e + h\max(f_e, 0).$$

The solution moves monotonically toward $c^2/(2\alpha^2)$ and its derivative magnitude cannot increase. Hence $y([0,h]) \subset [a_h, b_h]$, and integration of the derivative bounds gives

$$e + h\big(c - \alpha\sqrt{2b_h}\big) \le y(h) \le e + h\big(c - \alpha\sqrt{2a_h}\big). \tag{105}$$

The lower endpoint can be intersected with $[0, \infty)$. Every expression is evaluated with outward interval arithmetic.

For the lower comparison, reflection is obtained by taking $\min\{y, B\}$. On the headway branch, $c - \alpha\sqrt{2B} - B' = c + (1/\gamma - \alpha)\sqrt{2B} > 0$. On a flat speed cap a free flow starting below the cap either cannot reach it, or remains constrained there after contact. These observations justify the reflected constant-authority lower comparison for $\alpha\gamma \le 1$. For the upper comparison, E^+ is a subsolution of the free c_+ equation and is separately at most B, so $\min\{y_+, B\}$ bounds it from above. This argument remains valid when a rounded upper initial value slightly exceeds the exact ceiling.

The enclosures are propagated cell by cell. A cell whose energy upper bound is below its lowest ceiling is entirely on the free branch. If a lower free-flow endpoint exceeds the ceiling upper bound, actual ceiling contact is proved by that node. Permanent headway contact follows from $C > 0$ and $\alpha\gamma \le 1$. The verifier also checks the cap geometry: C1 cannot reach the speed cap, whereas C2 and C3 satisfy $C > \alpha M$ on the cap. Full throttle points outward there, so upper reflection enforces sticking. Cells intersecting a cap kink receive explicit coefficient and energy checks.

On free cells, with $r = \sqrt{\theta}$ and $v^+ = \sqrt{2E^+}$, the correlated boundary formulas are

$$\begin{aligned} R_q &= U - (1-\theta)C + \alpha(r - \theta)v^+, \\ q' - a(d_0 - s, \sqrt{2B}) &= U - (1-\theta)C - \alpha\theta v^+ + \alpha\sqrt{2B}. \end{aligned}$$

On ceiling cells, replace these by

$$\begin{aligned} R_q &= \theta B' - C + U + \alpha r\sqrt{2B}, \\ q' - a(d_0 - s, \sqrt{2B}) &= \theta B' - C + U + \alpha\sqrt{2B}. \end{aligned}$$

Cells containing an unresolved contact use the hull of both branches. One-sided ceiling slopes are both included at a kink. This bounds the derivative almost everywhere without estimating an exact contact location. Finally, outward nonnegative Riemann sums enclose the budgets on each enlarged prefix.

The implementation uses interval arithmetic with 45 decimal digits for model functions, profile comparisons, and ceiling flows. Conversion to binary64 is enlarged in the appropriate direction. Nonnegative sums use a conservative forward-error bound, and products, budget sums, and widths are rounded outward. These are applications of standard validated-numerics principles [30]; the enclosure equations above, rather than an ODE solver's tolerance estimate, supply the mathematical error bounds.

8.9.3 Results and operational meaning

All twelve certificates pass (104) at $\Delta = .025$. Together they cover 101,600 sample-location cells, each including all its intermediate positions and all band energies. Table 4 reports maxima over both initial velocities. Ratios are rounded upward to three decimal places. The largest unrounded certified ratio is 0.7477265063542601, so every displayed budget lies below the available width.

Model	θ	analytic Δ	verified Δ	budget/width
C1	0.25	.01	.025	0.567
C1	0.50	.01	.025	0.748
C2	0.25	.01	.025	0.441
C2	0.50	.01	.025	0.646
C3	0.25	.01	.025	0.383
C3	0.50	.01	.025	0.568

Table 4: Uniform a priori periods for complete episodes. The analytic column uses the explicit construction of Theorem 8.12; the larger value is an offline interval certificate for Theorem 8.9. The last column is $\max_j(\overline{H}_{B,j} + \overline{H}_{q,j})/\underline{w}_j$. It is a certificate diagnostic, not a trajectory-performance metric.

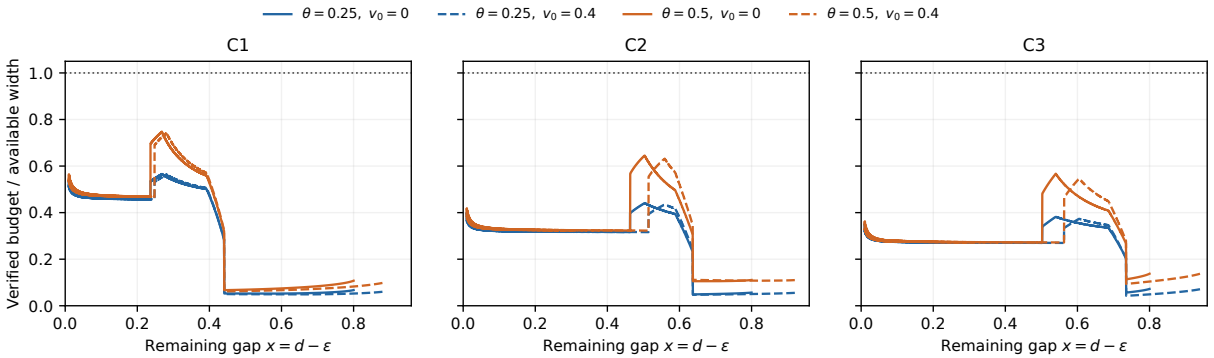


Figure 1: Certified upper budget/width bounds on complete sample-location cells for $\Delta = .025$. The dotted line is the sufficient-condition threshold. Travel proceeds from larger to smaller remaining gap x. The largest values occur near the transition of the maximal reference to the safety ceiling. These are offline certificate bounds, not simulated controller trajectories.

Figure 1 locates the largest clearance ratios near profile contact. The stored data provide the inputs, budget upper bounds, and width lower bounds for every cell. With exact state information, a table lookup and the comparison $B(s) - E \geq \overline{H}_{B,j}$ select $u_{q,j}$; otherwise $u_{B,j}$ is feasible. The launch test is checked explicitly. No future nominal values are needed. A nominal hold may still be accepted by a separate sound certificate; the table supplies a guaranteed alternative.

The certificate is sufficient. It does not identify the largest feasible period, and rejection of a larger period would not prove physical infeasibility. In particular, this calculation makes no

uniform .05-period claim. The .05 runs in Section 13 use the whole-hold certificates of (64); their trajectory-specific scope is distinct from this uniform certificate.

Reproduction files are in `sampling_certificate/`. The main verifier is `certify_balanced.py`; its report is `certified_results_0025.json`. Twelve detailed certificate arrays and a separate analytic constants checker are also supplied. The package README and `CERTIFICATE_METHOD.txt` specify dependencies, exact inputs, enclosure steps, and reproduction commands. This calculation uses model and profile enclosures directly; the experiment trajectories are not used as a safety oracle.

8.10 Scope of the a priori period guarantees

The boundary-clearance certificate strengthens the feasibility statement in Theorem 8.3 while retaining its recovery clock. The exact spatial benchmark, the strict-fidelity theorem, and the distinction between nonautonomous safety and autonomous performance retain their stated hypotheses. The analytic optimal-velocity specialization uses additional model structure; the general certificate in Theorem 8.9 applies to the broader autonomous class.

The periods refer to normalized model time. A positive sampling period is not a certificate that an implementation finishes its calculations before the next update. The numerical study in Section 13 reports decision-time overruns for its candidate search. Section 13.10 separately implements nominal-first acceptance with the stored clearance witness and measures its decision times in a serialized replay. Neither measured timing distribution is a worst-case execution-time bound.

The theorems assume exact sampled state observations. Applying a stored clearance threshold to an uncertain state requires bounding the actual clearance and retaining the corresponding margin. The whole-hold state-error certificate in Section 8 provides one way to account for such numerical uncertainty. A proposed input must pass a sound band certificate before acceptance; a pointwise nominal request alone does not certify the full hold.

The stored witnesses establish feasibility and allow a separately certified nominal hold to be accepted. They do not assert minimum intervention over all safe held inputs, an acceleration-variation bound, or superiority in common-horizon costs. The .05 comparisons in Section 13 concern the specified candidate-search controller. Section 13.10 distinguishes its 192 fixed-period witness-policy runs from those preserved records. The validated .025 period is sufficient for the specified episodes, not a maximum allowable period or a uniform .05-period claim.

9 Leaders with finite total travel

The stopped-leader recovery theorem extends to a leader whose total forward travel is finite. Its speed need not become identically zero after a finite time, or even converge to zero. The autonomous plant assumption is retained: compactness then supplies a time-independent bound on the acceleration under every bounded command.

Theorem 9.1 (Recovery behind a finite-travel leader)**.** *Assume autonomous a, b with the stated regularity and growth bounds, the three face conditions (3)–(5) evaluated at $w = 0$, and strict rest authority (7). Fix $\lambda, h_s > 0$ and use the moving-leader reference (15). Let the continuous leader speed satisfy*

$$0 \le w(t) \le M, \qquad S = \int_0^\infty w(t)\,dt < \infty. \tag{106}$$

From every initial state $(d_0, v_0) \in D$, the backup (10) and the recovery blend (19) preserve D and satisfy

$$v(t) \longrightarrow 0, \qquad d(t) \longrightarrow d_\infty \in [\epsilon, d_g]. \tag{107}$$

The same conclusion holds for the recovery CBF (22) if its interval at $w = 0$ is feasible throughout $K_{\bar{d}} = D \cap \{d \le \bar{d}\}$ for some $\bar{d} > d_0 + S$. The nominal may be any admissible regular command on $[0, \infty)$; no additional nominal sign or convergence assumption is required.

Proof. The stopped-leader face certificate implies the moving-leader face certificate, because $w \ge 0$ only relaxes the headway upper bound. The backup and blend therefore preserve D. For the CBF, write $R(w, z)$ for the upper endpoint (21). The lower endpoint is independent of w, and

$$R(w, z) \ge R(0, z), \qquad w \ge 0. \tag{108}$$

Thus its interval stays feasible throughout $K_{\bar{d}}$, and the controller satisfies the original boundary tangent inequalities there. Use the regular, bounded ambient extension described after (22) to obtain a forward-complete solution. Until a putative first crossing of $d = \bar{d}$, local boundary invariance preserves the original three constraints.

On any such safe interval, integration of the spacing equation gives

$$d(t) = d_0 + \int_0^t w(r)\, dr - \int_0^t v(r)\, dr. \tag{109}$$

Since $v \ge 0$, $d(t) \le d_0 + S < \bar{d}$; hence the artificial upper boundary cannot be reached. This closes the CBF safety argument. For all three controllers, $d \ge \epsilon$ and (109) consequently imply

$$\epsilon \le d(t) \le d_0 + S, \qquad 0 \le \int_0^t v(r)\, dr \le d_0 - \epsilon + S. \tag{110}$$

Thus v is integrable. The state lies in the compact set $D \cap \{d \le d_0 + S\}$, and bounded input and continuous autonomous coefficients bound $|\dot{v}|$. Therefore v is uniformly continuous. A nonnegative, uniformly continuous, integrable function tends to zero, giving $v(t) \to 0$. Both travel integrals in (109) have finite limits, so $d(t) \to d_\infty$ for some $d_\infty \ge \epsilon$.

Suppose $d_\infty > d_g$. Let

$$P_0(z) = \mathrm{clip}_{[A_-(z), A_+(z)]}\, r_\lambda(0, h(z), v),$$

where the first argument denotes leader speed. At $z_\infty = (d_\infty, 0)$, both the reference and A_+ are strictly positive. Hence $P_0(z_\infty) > 0$. Continuity gives a neighborhood of z_∞ and a constant $c > 0$ such that $P_0 \ge 2c$ there. Since

$$r_\lambda(w, h, v) \ge r_\lambda(0, h, v),$$

monotonicity of interval projection gives $p_b(w, z) \ge P_0(z)$. Thus the backup has $\dot{v} \ge 2c$ eventually. Let B be a bound for b on the compact state envelope. The actual seed acceleration $p_{\mathrm{rec}} = a + b u_{\mathrm{rec}}$ satisfies

$$|p_{\mathrm{rec}} - p_b| \le 2UB\beta(v), \qquad |a + b u_{\mathrm{B}} - p_b| \le 2UB\beta(v). \tag{111}$$

Because $\beta(v) \to 0$, both seed and blend accelerations are at least c eventually, independently of the admissible nominal. For the CBF,

$$R(0, z_\infty) = \min\left\{A_+(d_\infty, 0), \frac{\kappa_h(d_\infty - \epsilon)}{\gamma}, \kappa_M M\right\} > 0.$$

By continuity and (108), $R(w(t), z(t)) \ge r > 0$ eventually. Projection onto a nonempty interval then gives $p_{\mathrm{C}} \ge \min\{c, r\} > 0$. In every case, a uniform positive lower acceleration bound for all sufficiently large times contradicts $v(t) \to 0$. Consequently $d_\infty \le d_g$. □

9.1 Regularity, feasibility, and interpretation

No conclusion $w(t) \to 0$ was used. Continuous, bounded, nonnegative integrable functions can have fixed-height pulses whose widths are summable and need not tend to zero. The monotonicity in w of the backup reference and CBF upper endpoint avoids that obstruction. For a physical leader with uniformly continuous speed, integrability would additionally imply $w \to 0$, but this extra assumption is not needed for the follower theorem.

For the general coefficient class, an available upper bound $\bar{S} \geq S$ permits the choice

$$\bar{d} > \max\{d_0 + \bar{S}, \epsilon + \gamma M\}.$$

The gain construction (23) on $K_{\bar{d}}$ then ensures the required stopped-leader feasibility; nonnegative w preserves it by (108). This is a guarantee for the specified initial envelope and travel budget. It does not assert that every moving-leader trajectory beginning anywhere in $K_{\bar{d}}$ remains there: on its artificial upper face, $\dot{d} = w - v$ can be positive. The bound (110) is the correct reason the particular trajectory does not reach that face. For the optimal-velocity specialization with globally feasible gains, no compact-domain gain redesign is needed.

The theorem is a sufficiency extension. Strict stopped-leader rest authority need not be necessary for a specified moving leader: positive leader travel can move an excessive rest state to a different gap where forward acceleration is attainable. Likewise, the theorem does not assert that an accepted rest state stays fixed while its leader is moving.

Proposition 9.2 (A first-entry bound with finite leader travel)**.** *Under the safety and finite-travel assumptions above, fix $d_\eta = d_g + \eta < d_0$. If constants $c_\eta > 0$ and $0 < \nu_\eta < M$ satisfy $\dot{v} \geq c_\eta$ almost everywhere whenever $d_\eta \leq d \leq d_0 + S$ and $v \leq \nu_\eta$, then the first entry time $\tau_\eta = \inf\{t \geq 0 : d(t) \leq d_\eta\}$ obeys*

$$\tau_\eta \leq \frac{(\nu_\eta - v_0)_+}{c_\eta} + \frac{d_0 - d_\eta + S}{\nu_\eta}. \tag{112}$$

Proof. Before entry, speed reaches ν_η within the first term, unless entry occurs sooner. The low-speed condition prevents a subsequent downcrossing before entry, by the same interval integration argument as in Theorem 3.1. If t_a is that acceleration time, then for $t \geq t_a$ before entry,

$$d(t) \leq d_0 + S - \nu_\eta(t - t_a).$$

This yields (112). □

The bound concerns first entry. A moving leader can reopen a gap; the convergence conclusion (107) separately gives eventual approach to the accepted interval.

10 A finite-platoon consequence

Corollary 10.1 (Recovery through a finite platoon)**.** *Consider n followers with*

$$\dot{d}_i = v_{i-1} - v_i, \qquad \dot{v}_i = a_i(d_i, v_i) + b_i(d_i, v_i)u_i, \qquad i = 1, \ldots, n.$$

Use a common speed cap M, individually admissible command bounds and safety domains D_i, and assume each autonomous plant satisfies its stopped-leader face certificate and strict rest authority outside $d_{g,i} = \epsilon_i + h_{g,i}$. Each follower uses the backup, recovery blend, or recovery CBF of Theorem 9.1, with the required feasibility on its operating envelope. Suppose $(d_i(0), v_i(0)) \in D_i$ and

the head speed is continuous with $0 \le v_0 \le M$ *and* $S_0 = \int_0^\infty v_0(t)\,dt < \infty$*. Then every follower remains safe and*

$$v_i(t) \to 0, \qquad d_i(t) \to d_{i,\infty} \in [\epsilon_i, d_{g,i}], \tag{113}$$

$$S_i := \int_0^\infty v_i(t)\,dt \le S_0 + \sum_{j=1}^{i}(d_j(0) - \epsilon_j). \tag{114}$$

Proof. The first follower satisfies Theorem 9.1, giving its convergence and $S_1 \le S_0 + d_1(0) - \epsilon_1$. Its speed is continuous, lies in $[0, M]$, and is integrable, so it is an admissible finite-travel leader for follower 2. Inductively, the same theorem gives the convergence of follower i and $S_i \le S_{i-1} + d_i(0) - \epsilon_i$. Iteration proves (114). □

A known upper bound on S_0 supplies all downstream travel bounds and therefore all compact CBF design envelopes before simulation. Heterogeneous speed caps can also be used when the upstream bound fits the downstream controller's assumed leader-speed range. If the head is permanently stopped and all followers are already at accepted rest states, the cancellation condition (12) for each plant preserves that configuration. The corollary is a finite-chain recovery result. It establishes neither string stability nor a convergence rate independent of n, and it does not select one prescribed positive terminal spacing.

11 Exact excess-rest equilibria of the ordinary CBF filter

The following proposition diagnoses the ordinary safety filter, which projects the nominal command directly, without the shared recovery reference. It specializes the equilibrium mechanisms studied in [31, 32]; no claim of a new general CBF-equilibrium phenomenon is made.

Proposition 11.1 (Nominal sign and excess-rest equilibria)**.** *Assume* $w = 0$*, autonomous* a, b*, strict rest authority* (7)*, a feasible CBF interval on the operating domain, and positive gains. Let* $k(z) \in [-U, U]$ *be an autonomous regular nominal, and put*

$$p_{\mathrm{nom}} = a + bk, \qquad p_{\mathrm{C}}^0 = \mathrm{clip}_{[L,R]}\, p_{\mathrm{nom}}.$$

Define $\mathcal{E}_{\mathrm{C}}^0 = \{(d, 0) \in D : p_{\mathrm{C}}^0(d, 0) = 0\}$ *on the certified operating domain. Then the excess-rest equilibria of the resulting closed loop are exactly*

$$\mathcal{E}_{\mathrm{C}}^0 \cap \{(d, 0) : d > d_g\} = \{(d, 0) : d > d_g,\ p_{\mathrm{nom}}(d, 0) \le 0\}, \tag{115}$$

with both sets restricted to the certified operating domain.

Proof. At an excess-rest point,

$$L(d, 0) = \max\{A_-(d, 0), 0\} \ge 0,$$

whereas

$$R(d, 0) = \min\left\{A_+(d, 0), \frac{\kappa_h(d - \epsilon)}{\gamma}, \kappa_M M\right\} > 0.$$

If $p_{\mathrm{nom}}(d, 0) \le 0$, nominal admissibility implies $A_-(d, 0) \le p_{\mathrm{nom}}(d, 0) \le 0$. Therefore $L(d, 0) = 0$ and the projection returns zero. Conversely, a projection onto a feasible interval with $L \ge 0$ and $R > 0$ can equal zero only when $L = 0$ and the projected value is nonpositive. Since $\dot{d} = 0$ at $v = 0$, this condition is exactly the equilibrium condition. Uniqueness of the regular closed-loop solutions identifies the constant trajectory from each such point. □

The accepted-rest cancellation assumption (12) is only imposed for $d \le d_g$; it is not used to assert $L = 0$ at every excess-rest state. In the proof, $L = 0$ follows at a candidate equilibrium from admissibility and the nominal sign itself.

Three cases must be distinguished. If $p_{\mathrm{nom}} = 0$, the nominal rest equilibrium is inherited. If $p_{\mathrm{nom}} < 0$, the unfiltered vector field is $(0, p_{\mathrm{nom}}) \neq 0$ and points outside the nonnegative-speed constraint; projection to zero creates a rest equilibrium. If $p_{\mathrm{nom}} > 0$, the positive feasible upper endpoint ensures that departure is retained. Thus safety alone does not guarantee departure, while strict rest authority ensures that this filter does not extinguish a positive nominal departure acceleration. The proposition characterizes equilibrium locations, not their local stability.

For example, $a = 0$, $b = U = M = \gamma = 1$, and $k \equiv -1$ satisfy the face tests. With $\kappa_v = 1$ and any positive headway and upper-speed gains, the interval is feasible. At every excess-rest point, $L = 0 < R$ and the projection of -1 is zero, although the unfiltered system has nonzero acceleration there.

Setting $k \equiv 0$ tests the sign of the drift $a(d, 0)$, not of a zero net acceleration. Consequently, a zero-input ablation freezes only where the drift is nonpositive. For a time-dependent nominal $k(t, z)$, the same algebra gives the instantaneous zero-acceleration condition. A constant rest trajectory from time t_0 exists exactly when

$$a(d, 0) + b(d, 0)k(t, d, 0) \le 0 \quad \text{for almost every } t \ge t_0,$$

under the same feasible-interval and strict-authority assumptions. A sign at one instant is insufficient to infer permanent freezing for a finite-horizon HJB nominal or its continuation.

12 An obstruction to unknown additive-disturbance robustness

Proposition 12.1 (Forced-corner obstruction)**.** *For the stopped-leader model*

$$\dot{d} = -v, \qquad \dot{v} = a(d, v) + b(d, v)u + \delta(t), \qquad |\delta(t)| \le \bar{\delta},$$

no single controller independent of the unmeasured disturbance can preserve all of D for every such disturbance when $\bar{\delta} > 0$. This includes deterministic causal controllers using only the state history and a fixed initial controller state.

Proof. At $z_* = (\epsilon, 0)$, the lower-speed and headway tangent inequalities require the actual acceleration to be both nonnegative and nonpositive, hence zero. If $p = a + bu$, a common command for all disturbance values would require $p - \bar{\delta} \ge 0$ and $p + \bar{\delta} \le 0$, which are incompatible.

The same obstruction holds for state-history controllers without relying on a static-feedback interpretation. Every safe trajectory from z_* satisfies

$$0 \le \int_0^t v(r)\, dr = \epsilon - d(t) \le 0.$$

It must therefore have $d(t) = \epsilon$ and $v(t) = 0$ for all t. For the constant disturbances $\delta = \bar{\delta}$ and $\delta = -\bar{\delta}$, the controller observes identical constant state histories but would need the two different commands

$$u(t) = -\frac{a(\epsilon, 0) + \bar{\delta}}{b(\epsilon, 0)}, \qquad u(t) = -\frac{a(\epsilon, 0) - \bar{\delta}}{b(\epsilon, 0)}$$

almost everywhere. One deterministic state-history controller cannot produce both from the same history and initial controller state. □

Separate tightened face tests can conceal this incompatibility if different commands are allowed on the two intersecting faces. Robust invariance requires one command to satisfy all active

constraints for every possible disturbance. Consequently the rest-authority infimum is not a positive whole-domain robustness radius for this uncertainty model. Instantaneous disturbance measurement, a different safety domain, or uncertainty that vanishes at the forced corner changes the problem and requires a separately stated theorem. No positive unknown-additive-disturbance margin is asserted for this model.

13 Sampled-actuation experiments and reproducibility

This section specifies the sampled experiments. The separate continuous-time comparison in the following appendix uses different post-entry dynamics and is not pooled with these data.

13.1 Models and comparison matrix

The controlled dynamics are $\dot{d} = -v$, $\dot{v} = \alpha(\mathcal{V}(d) - v) + u$ with $\epsilon = .02$, $M = 1.96$, $U = 5$. Table 5 specifies the three models used in both numerical studies. All quantities are normalized. The three

Table 5: Normalized optimal-velocity models.

Model	α	γ	$\mathcal{V}(d)$
C1	2.6	.20	$\tanh d - \tanh 2$
C2	1.0	.30	$\tanh d - \tanh 2$
C3	2.6	.35	$\mathrm{clip}_{[0,1.5]}(.8(d - .1))$

models are transparent stress tests, not a calibrated vehicle fleet. The gap cost penalizes physical gap magnitude; it does not vanish at $d = \epsilon$. It is a specified quadratic benchmark objective, not a calibrated fuel or comfort cost.

Initial states satisfy $h_0 = .8$, $v_0 = 0$ or .4, and $d_0 = \epsilon + h_0 + \gamma v_0$. Gap recovery has $h_g = 0$ and cost $.2u^2 + 50d^2$; the tracking-cost task has $h_g = .2$ and cost $.2u^2 + v^2 + 5(h - .2)^2$. Physical-gap entry uses $d_\eta = \epsilon + h_g + .01$ and does not constrain the entry speed. The four nominal commands are the stored HJB policy, $k = 0$, $k = \mathrm{clip}_{[-U,U]}(-a)$, and $k = -U$. The HJB policy is the same ten-unit finite-horizon policy as in the continuous-time comparison; no controller-specific retraining is performed.

The main matrix contains 48 model/task/start/nominal combinations and uses $\theta = .25, .5$, with a requested hold duration .05. Each setting is run with the certified band controller, the naive reflected-law ZOH implementation, and the progress-aware controller with an intersample safety test. The four stress cases are C3/gap/moving/HJB, C3/gap/rest/braking, C1/gap/rest/braking, and C2/tracking/moving/HJB. Each also uses requested periods $.0125, .025, .1$ and both floor fractions, adding 24 cases per method to the 96 main cases. They are fixed before the production run. Outcomes are not used to select initial conditions or omit unsuccessful cases.

13.1.1 HJB nominal and policy identity

The HJB nominal minimizes the same task-specific quadratic running cost used for the corresponding experiment over $0 \le t \le 10$, with zero terminal cost, stopped leader, and $u \in [-5, 5]$. It is a finite-horizon quadratic-cost nominal, not the minimum-entry controller defining T^*. The supplied action tables have shape $201 \times 301 \times 129$ in (t, d, v). Their time spacing is .05; the retained gap grid has spacing .1 and endpoints $-5, 25$ (up to the recorded binary64 representation). The 129 speed nodes are uniform on the intervals in Table 6. These ambient nominal grids include unsafe states; safety is imposed by the filter, and the physical simulated state is never projected onto a nominal-grid boundary.

Table 6: Speed intervals of the HJB nominal grids.

Model	Lower speed endpoint	Upper speed endpoint
C1	−3.887104503152740	1.960000000000000
C2	−6.964027580075817	5.035972419924184
C3	−1.923076923076923	3.423076923076923

The semi-Lagrangian Bellman recursion uses the forward Euler departure $d^+ = d - \delta t\, v$, $v^+ = v + \delta t\,[a(d, v) + u]$ and bilinear spatial interpolation of the next value array:

$$V_n(d_i, v_j) = \min_{|u|\le 5}\{\delta t\, \ell(d_i, v_j, u) + \mathcal{I}V_{n+1}(d^+, v^+)\}, \qquad V_{200} = 0.$$

On each crossed speed-interpolation cell, the objective is a convex quadratic in u. The C++ implementation evaluates its clipped stationary point on that cell and selects the least value over the intersected cells. Thus it uses a continuous bounded actuator search, rather than a finite actuator grid. The enlarged gap box includes the full all-control travel envelope and 200 interpolation-cell widths plus three extra cells on each side. A dependency outside the enlarged box or on a nonfinite value node invalidates the update; no invalid retained update is accepted.

Value updates use binary64; the minimizing action array is stored as binary32. The nominal queries use multilinear interpolation in time and both state coordinates, with the terminal action set to zero and extended by zero for $t \ge 10$. Every queried state must lie in the retained rectangle, apart from the documented roundoff tolerance. The source is `manuscript/numerics/engine.py` and `kernel.cpp`; the interpolation used by the sampled experiments is implemented in `numerics_v7/reference_v6.py`. The six files are `manuscript/numerics/policies/Cj__w5_stopped__TASK__medium.npz`, where `Cj` is C1–C3 and `TASK` is `gap` or `tracking`. Complete file and array SHA-256 hashes, exact grid endpoints, and generator checks are in `reproducibility/hjb_policy_inventory.json` and `reproducibility/hjb_regeneration_check.json`.

From the package root, `python3reproduce.py--hjb` regenerates the policies in a separate directory without replacing the supplied inputs. All six regenerated action and grid arrays agree elementwise with the distributed arrays in the recorded environment. The policy is shared between controllers; no filter-specific HJB retraining is performed.

13.2 Certificates and finite arithmetic

The reference upper enclosure is the constant-authority construction in Section 8.5. Its clock brackets are validated with interval arithmetic. A dense ODE solution provides plots and diagnostic optimal times; certification instead uses the upper enclosure and the lower optimal-clock bound.

For each held input the numerical flow is represented by piecewise cubic polynomials. Their residuals are bounded on whole cells using polynomial range bounds and bounds on the velocity law and its derivatives. The Metzler error system propagates uncertainty between cells and sampling instants. Polynomial join discrepancies and the difference between polynomial endpoints and stored state centers belong to the error budget. Matrix exponentials and transcendental coefficient bounds are enclosed independently of the ODE solver.

After the initial full-throttle rest launch, the band controller tests its sampled nominal and then uses the finite candidate rule in Section 13.5. Every applied command passes its specified whole-hold test. The search does not assume that the physical-time admissible set is convex, or compute the exact projection onto the spatial interval. The analytic short-hold witness supplies the theorem's fallback; rejection by a finite-error trajectory test is never identified with physical infeasibility.

The progress-aware baseline uses $r = \sqrt{\theta}$, $c = 5$, and $\kappa_h = \kappa_v = \kappa_M = 10$. With $\bar{v}(d) = \min\{M, (d-\epsilon)/\gamma\}$, it requests

$$L = \max\{a - U, -10v\}, \qquad R = \min\{a + U, (-v + 10h)/\gamma, 10(M - v)\},$$
$$p_{\text{req}} = -rv\bar{v}'(d) + c(r\bar{v}(d) - v), \qquad p_{\text{prog}} = \text{clip}_{[\max\{L, \min(p_{\text{req}}, R)\}, R]}(a + k),$$

and the requested actuator is $\text{clip}_{[-U,U]}(p_{\text{prog}} - a)$. The implementation takes $\bar{v}' = 1/\gamma$ for $d \le \epsilon + \gamma M$ and zero otherwise. It uses the same whole-hold safety certificate and candidate search as the band method, but omits the energy-floor constraint. This is a specified CLF/CBF comparison design; it is not an implementation of a particular published sampled-data algorithm.

For naive ZOH, set $\widehat{s} = \text{clip}_{[0,S_e]}(d_0 - d)$, $E = v^2/2$, and $p = a + k$. If $E \le \theta E^+_{\text{diag}}(\widehat{s}) + 10^{-10}$, replace p by $\max\{p, \theta(E^+_{\text{diag}})'(\widehat{s})\}$; if $E \ge \frac{1}{2}\min\{M, (d-\epsilon)/\gamma\}^2 - 10^{-10}$, then replace p by $\min\{p, 0\}$ on the speed-cap branch and by $\min\{p, -(d-\epsilon)/\gamma^2\}$ on the headway branch (the latter branch includes the equality $d = \epsilon + \gamma M$). The held actuator is $\text{clip}_{[-U,U]}(p - a)$, including on trial states outside the band. The maximum diagnostic horizon is 20; entry terminates the episode, and a negative-speed diagnostic below -10^{-6} terminates an episode that has not entered. No state is projected back to safety. The first safety escape is failure evidence regardless of later behavior.

13.3 Entry observation and common continuation

The ordinary trajectory root gives an estimated first-entry time. Confirmed sampled entry instead requires the propagated upper gap enclosure to satisfy $d \le d_\eta$. Conservative state enclosures can delay this confirmation, so the exact-observation detection theorem is not blindly assigned to an interval observer. The recorded confirmation time is separately compared with $\underline{T}^*/\sqrt{\theta} + \bar{\Delta}$, where $\underline{T}^* \le T^*$ is a verified lower bound on the optimal clock. A successful inequality is stronger than comparison with an unverified optimal-time estimate. These numerical checks use $T^*/\sqrt{\theta}$; no validated numerical enclosure of the sharper W_θ bound is claimed. The physical-entry theorem has no sampling-dependent additive term, whereas exact sampled detection adds at most $\bar{\Delta}$. Neither statement orders the sampled and continuous trajectories for a fixed nominal.

After confirmed entry, successful methods use a common sampled stopping continuation. Its exact-state version is proved in Section 8. Numerical state uncertainty is accounted for in every subsequent hold. Where needed, the command is adjusted to enforce a verified nonnegative forcing in $\dot{v} + \alpha v = \alpha\mathcal{V}(d) + u$; the headway and speed-cap certificates remain required. Correction magnitudes are recorded. No negative simulated speed is repaired by projecting the state. Only the completed finite-horizon holds of this corrected implementation are certified; asymptotic convergence of the corrected policy is not inferred from the exact-state stopping proposition.

13.4 Metrics

The common-horizon metrics use $H = 10$:

$$J_H = \int_0^H \ell(d, v, u)\, dt, \qquad I_{1,H} = \int_0^H |u - k|\, dt, \qquad A_H = \left(H^{-1}\int_0^H p^2 dt\right)^{1/2}, \quad p = a + u.$$

Squared intervention is $I_{2,H} = \int_0^H (u-k)^2\, dt$. Both use the actual time-varying nominal along the trajectory; counterparts relative to the sampled request are stored separately. For holds separated by t_n, the acceleration variation and largest jump are

$$V_{a,H} = \sum_n \int_{t_n}^{t_{n+1}} |\dot{p}(t)|\, dt + \sum_{0<t_n<H} |p(t_n^+) - p(t_n^-)|,$$
$$A_{\text{jump,H}} = \max_{0<t_n<H} |p(t_n^+) - p(t_n^-)|.$$

with the final interval truncated at H. These functionals are evaluated numerically, including both sides of each jump; they do not imply a bounded jerk. The same initial state, nominal policy, target, actuator hold, cost, and continuation are used within every matched comparison.

The case matrix is deterministic and uses no random sampling. Counts and percentages describe these prescribed cases, not independent draws from a population. Different nominal labels can produce identical baseline trajectories when the progress constraint determines the same input. Accordingly, no independence assumption or population confidence interval is attached to the paired winner counts. Comparisons are also reported by task, model, initial speed, and floor fraction in the release data.

Runtime includes candidate certification and rejected candidates. Reference-enclosure preparation is reported separately. Average runtime below the requested period does not establish a hard real-time deadline: the upper tail and maximum decision times must also be reported. The theorem assumes that the selected command is applied at its specified update; computation latency is not included in the physical plant model.

13.5 Implementation choices and what the algorithm certifies

The spatial upper-enclosure grid has maximum spacing 10^{-4} and extends to $d = \epsilon + .002$. The maximum trajectory-certificate cell width is .001. DOP853 with relative/absolute tolerances $2\,10^{-12}$ and $2\,10^{-14}$ proposes each held trajectory; these solver tolerances do not determine its accepted error radius. The polynomial residual test supplies that radius separately. The trajectory and upper-reference tests use mpmath 1.3.0 interval arithmetic at 40 decimal digits. The separate uniform .025 verifier uses 45 digits, while the analytic .01 constants checker uses 60. These are three different computations, not conflicting precision claims. All convert interval endpoints outward to binary64; the online elementary operations and nonnegative sums are enlarged as specified in their code.

The finite candidate rule is fixed as follows. At an update with state center (d, v), elapsed time t, travel $s = d_0 - d$, and trial period δ, let k_{req} be the clipped nominal actuator for the band method, or the progress-controller actuator for the baseline. Let $\widehat{q} = \theta E^{+}_{\mathrm{diag}}$, where E^{+}_{diag} is the diagnostic profile used only to propose inputs. Write $\mathcal{C}(x) = \mathrm{clip}_{[-U,U]}(x)$. Here E^{+}_{diag} integrates $E' = a(d_0 - s, \sqrt{2E}) + U$ with DOP853, relative/absolute tolerances 2×10^{-11} and 2×10^{-13}, and maximum spatial step $S_e/512$, until the first detected ceiling contact. It then follows B exactly at the profile-evaluation level. Its dense output and one-sided branch derivative propose commands and diagnostic clocks only. They are never substituted for the verified reference enclosure in acceptance tests.

1. At $t = 0$, $v = 0$, the band method tests only U for the trial period. A rejection triggers the period-reduction rule below. At every other band update, and at all progress updates, test k_{req} first and apply it immediately if certified.

2. After a rejected request, test in order u_c, U, $-U$, then the grid $\{-U + 2Uj/20 : j = 0, \ldots, 20\}$, ordered by increasing $|u - k_{\mathrm{req}}|$ (ties by increasing u). The interior seed is

$$v_c = \sqrt{\max\{0, 2[.65\widehat{q}(s) + .35B(s)]\}}, \qquad u_c = \mathcal{C}\left(\frac{v_c - v}{\max\{\delta, .02\}} - a(d, v)\right).$$

   Stop this list at its first certified candidate. Repeated input values are not removed from the list.

3. If the entire list fails, set $s_f = \min\{S_e, \max\{0, s + v\delta\}\}$ and test the five seeds below in the order $\eta = .15, .35, .5, .65, .85$:

$$\begin{aligned} v_\eta &= \sqrt{\max\{0, 2[(1-\eta)\widehat{q}(s_f) + \eta B(s_f)]\}}, \\ d_\eta^{\mathrm{mid}} &= d - \tfrac{1}{4}(v + v_\eta)\delta, \\ u_\eta &= \mathcal{C}\Big(\frac{v_\eta - v}{\delta} - a\Big(d_\eta^{\mathrm{mid}}, \tfrac{1}{2}(v + v_\eta)\Big)\Big). \end{aligned}$$

   Again stop at the first certified input.

4. For a first certified alternative u_f, initialize $u_b = k_{\mathrm{req}}$. Repeat ten times: test $u_m = (u_b + u_f)/2$; if certified replace u_f by u_m, otherwise replace u_b by u_m. Apply the final certified u_f.

The band test enforces safety, travel-domain containment, and the floor; the baseline test enforces safety. The same candidate machinery is used with these different requests and acceptance conditions. The diagnostic profile proposes candidates; only the verified test accepts them. Keeping one certified endpoint during refinement does not require convexity. It also does not establish global minimum deviation from the request, minimum integrated intervention, or minimum acceleration RMS among all certified inputs.

If no command is accepted, the period is halved up to six times, with every shortened period recorded. The analytic fallback uses a verified opposite-boundary clearance of at least $g/4$ and a duration bounded by $g/(8KM)$ and the remaining extension. The smaller constants allow for carried uncertainty. Its comparison proof certifies the true band; a conservative Q query is not required to accept its floor. If the state enclosure is too wide to verify either clearance, the implementation reports the need for refinement. The exact-state recursive-feasibility theorem is not a termination guarantee for an arbitrarily coarse finite-arithmetic enclosure.

The saved records contain accepted holds, selection modes, candidate-evaluation counts, accepted certificate margins, and initial requested-period rejection events. They do not contain the complete certificate of every rejected nominal candidate or failed halving trial. Candidate evaluations, including rejected trials, are included in the reported decision runtime.

The outward recurrence has an optional C99 implementation, compiled with floating-point contraction and fast-math transformations disabled. A representative comparison with the Python recurrence is bitwise identical. Source and the Python fallback are supplied; a compiled binary is not treated as portable evidence.

13.6 Completed matrix and feasibility evidence

All 360 scheduled runs are represented in `data/episodes.csv`. Both certified methods reach confirmed entry in all 120 of their respective runs and complete the common horizon. In the main matrix, all 96 band episodes sustain the requested .05 period without shortening or analytic fallback. Among 24 additional band sweep episodes, only C3 gap recovery, rest/braking nominal, $\theta = .5$, requested period .1, requires one shortened .05 hold. All 120 progress episodes retain their requested periods. There are no unresolved certificate cases in the final certified-controller data.

The global gap-task existence bounds range from 5.84×10^{-8} to 2.14×10^{-7}. They are useful for the existence proof but do not explain the practical .05 holds. The file `reproducibility/generic_period_constants.csv` lists S_e, g, P, K and δ_0 for all 24 model/start/fraction/target combinations. Those CSV entries are numerical displays of the theorem formula, not additional rigorous certificates. The validated whole-trajectory test is the evidence for those much longer holds. The boundary-clearance theorem (Theorem 8.9) and its OVM specialization (Theorem 8.12) provide a useful analytic period .01 over the complete specified episodes. Section 8.9 separately

certifies .025 for all band states on those episodes. Neither statement asserts a uniform .05 period or changes the candidate-search policy underlying the reported performance comparisons. No analytic fallback occurs in the completed matrix. This is stated explicitly rather than reporting the requested period for an adaptively shortened run.

All 120 band and all 120 progress runs pass the outward observed-clock inequality. Diagnostic energy-floor departures below -10^{-8} occur in 55/96 main progress runs (57.3%) and 18/24 additional sweep runs, hence 73/120 overall (60.8%). The minimum recorded margins are $-.1231241$ in the main matrix and $-.1508481$ overall. These departures are trajectory diagnostics, not interval-certified lower-margin values for the baseline: its acceptance test does not impose the floor. The floor is sufficient for the proved nominal-independent bound, but is not necessary for an individual trajectory to satisfy that bound. The minimum main-matrix certified headway is 4.12×10^{-10}, with nonnegative certified speed and floor bounds. At launch, an analytic floor lower bound is zero and is recorded as zero. Across all band runs the largest carried state-error component is approximately $3.56\,10^{-5}$, including the post-entry horizon. The largest recorded band tail-input correction is less than $7.62\,10^{-6}$. These bounds concern numerical error, not unmodeled physical disturbances.

In contrast, naive ZOH has 70 entries, 24 negative-speed terminations, and two horizon nonentries in the main matrix. There are 60 original-domain escapes: 38 headway violations and 24 negative-speed violations, with two overlaps. There are 88 energy-floor violations; this count is distinct from the original-domain count. Only 34 naive runs enter without an observed original-domain escape. All 96 naive runs violate the domain, the floor, or both. The most negative observed headway is $-.1103507$. These sampled diagnostics identify counterexamples; they are not presented as complete certificates of naive safety when no escape is observed.

Figure 2 shows a moving-start example in which naive holding crosses the headway boundary.

13.7 Matched speed, cost, and motion comparisons

For $\theta = .25$, the band median and maximum entry ratios are 1.74468 and 1.92854; the matched progress values are 1.46672 and 1.89393. For $\theta = .5$, the band values are 1.30302 and 1.39072, versus 1.13042 and 1.38324. Thus all tested progress cases also happen to satisfy the comparison time factor. No universal optimum-relative theorem for that baseline is inferred from the sample.

Table 7: Paired main-matrix comparison on the common ten-unit horizon.

Metric	Band lower	Progress lower
Entry time	18/96	78/96
J_{10}	18/96	78/96
$I_{1,10}$	65/96	31/96
$I_{2,10}$	25/96	71/96
Acceleration RMS	24/96	72/96
Acceleration TV	1/96	95/96

Table 7 summarizes the paired comparisons. The band controller has lower entry time in 18 of 96 pairs and lower cost in 18. Its median paired entry-time ratio is 1.0452. The cost ratio must be interpreted separately by objective: the median $J_{10}^{\text{band}}/J_{10}^{\text{progress}}$ is 1.12409 for gap cost and 1.09157 for tracking cost. Band cost is lower in 10/48 gap pairs and 8/48 tracking pairs. The two different running costs are not pooled as one physical utility.

Absolute nominal intervention is lower in 65/96 pairs, but the median ratios within the gap and tracking groups are .99372 and .99875, respectively. Many reductions are small. Squared intervention is lower in only 25 pairs. Acceleration RMS is lower in 24, and numerical variation in one; the median band/progress variation ratio is 4.1658. These results preclude a general

comfort advantage. The largest acceleration jump is lower in ten pairs, higher in 85, and equal in one. A hard held-input band filter can increase motion variation even while respecting its performance floor. These cost and motion comparisons characterize the specified finite candidate selection, initial launch, and common continuation. The band certificate determines which holds may be accepted; it does not uniquely select a command or optimize these metrics. A different minimum-deviation selection would require a separately specified controller and new matched experiments. Its performance cannot be substituted into the present tables.

Figure 3 displays the matched time, cost, and intervention distributions. Figure 4 shows the period sweep and its separately verified observation bounds.

13.8 Measured computation and limits of deployment claims

A separate serialized benchmark reruns the four stress cases at both values of θ with both certified methods: 16 runs. It times pre-entry decisions, including nominal evaluation, candidate generation, trajectory proposals, residual bounds, profile queries, and rejected candidates. It excludes reference preparation and the common tail. The environment uses Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0, and the compiled outward recurrence on Linux x86-64.

Table 8: Serialized pre-entry decision times in milliseconds.

Method	Decisions	Median	95th pct.	Maximum	Above 50 ms
Band	312	69.19	124.26	235.80	269
Progress	299	2.50	4.09	10.26	0

The measured latencies in Table 8 are from the original serialized benchmark. Its CPU/allocation details were not recorded and cannot be inferred from the current machine. With one model time unit mapped to one second, these overruns prevent a .05-second deadline claim for the band implementation. Parallel production elapsed times are not substituted for these serialized latencies. Certified upper-profile preparation takes 1.68–3.22 s for 7,980–9,380 spatial cells in the benchmark. The separate diagnostic profile takes approximately .15–.22 s. The mathematical clock starts with the profile available. Fewer refinement tests or optimized code may reduce latency, but that is not demonstrated by this study.

13.9 Data, reproduction, and complete paired tables

The directory `numerics_v7/` contains source, the fixed protocol, raw cases, held commands, rejected-period records, metrics, and timing. `summary.json` summarizes the 360 cases in `data/episodes.csv`. Profile-cache loading validates an exact identity for the initial data, extension, grid, model, and construction settings. The file `environment.json` records the numerical environment. Statistics use the 360 cases identified by the fixed protocol and summarized in `data/episodes.csv`. The numerical README specifies the metrics, file layout, and reproduction commands.

All commands are issued from the package root through a common wrapper:

```
python3 reproduce.py --sampled --fresh --workers 4
python3 reproduce.py --sampled-benchmark
python3 reproduce.py --render-only
```

The wrapper uses an isolated working copy and fresh simulation by default. An explicit reuse option permits validated caches. The optional C build has a Python fallback. A fresh simulation directly computes absolute and squared actual-nominal intervention. Rendering uses the supplied

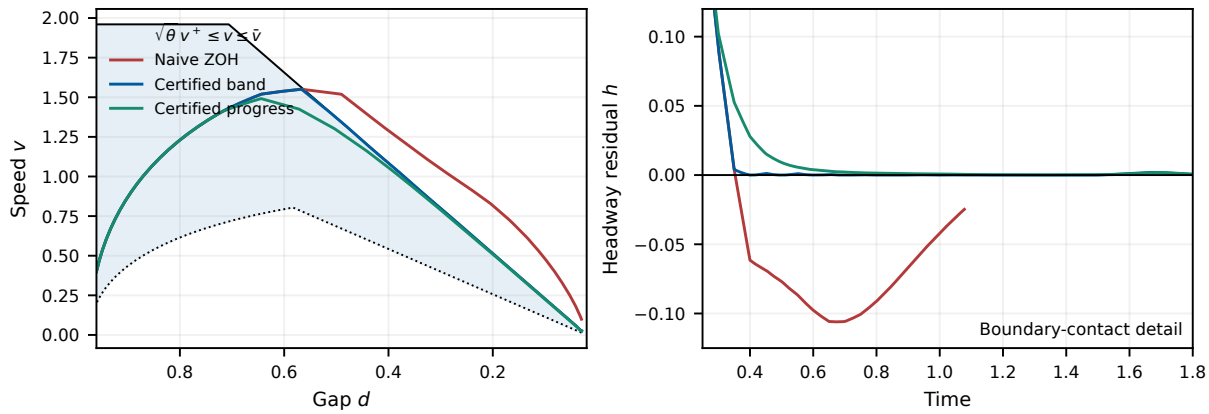


Figure 2: C3 moving-start gap task, HJB nominal, $\theta = .25$ and requested period .05. The naive held command crosses the headway boundary. Both certified controllers remain safe. The shaded speed band is the fixed initial-state reference; it is not reset per sample.

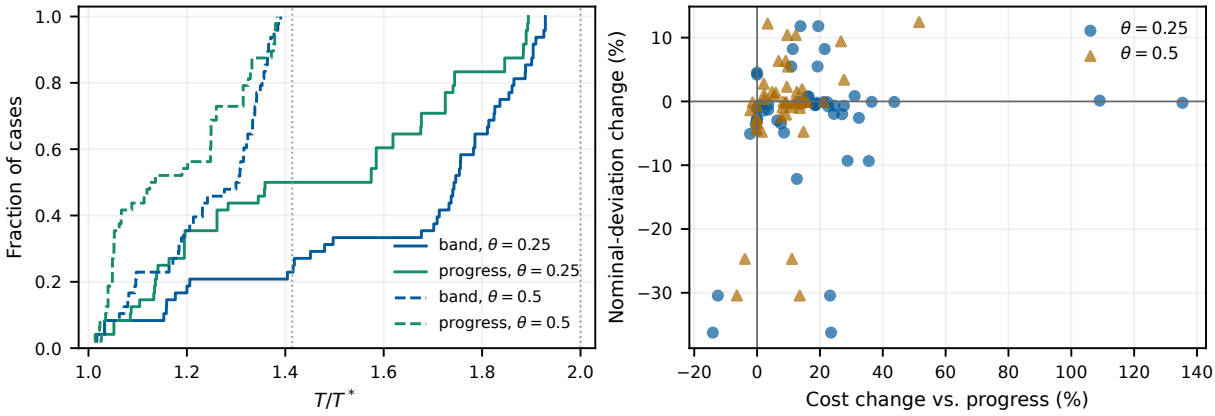


Figure 3: Main-case entry distributions and matched percentage changes of cost and absolute nominal intervention. Negative changes favor the band controller. Vertical entry-ratio references are $\sqrt{2}$ and 2.

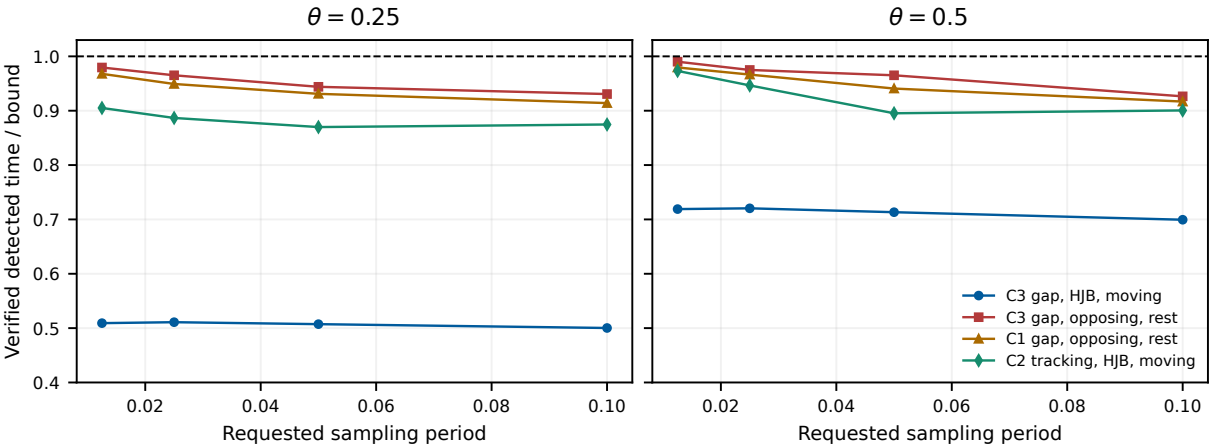


Figure 4: The four specified stress cases at both floor fractions. The ordinate compares interval-confirmed detection time with the verified sufficient bound $\underline{T}^*/\sqrt{\theta} + \bar{\Delta}$. The .05 points belong to the main matrix; the other three periods are the additional sweep. Every point is below one. Requested periods are shown; the single shortened hold is disclosed in the text and Table 11.

figures and tables and does not rerun trajectories. The top-level README maps each paper artifact to its script and data source.

The .05 comparison uses the supplied 360-case production snapshot and the cache-corrected sources identified in `numerics_v7/revision_v8_validation.json`; that complete study and its serialized timing were not rerun for this release. Its derived tables were reaggregated, and targeted episode/cache checks from the preceding release are preserved. The fixed-period study in Section 13.10 is a separate new computation.

Tables 9–11 expose all 96 main matched pairs and the 24 additional band sampling tests. In each case label, G/T denotes gap/tracking cost; R/M denotes rest/moving start; H/Z/C/O denotes HJB, zero actuator, drift cancellation, and braking nominal. The raw data additionally include squared and sampled-request intervention, safety/uncertainty margins, all naive outcomes, and every sweep baseline. Displayed costs and motion metrics are numerical integrals, not interval-certified functionals. Costs use seven and absolute intervention six significant digits so that rounding does not hide the stored paired ordering; these extra digits are not a claim of interval-certified accuracy. Independent 12/24/48-node Gauss and adaptive Gauss–Kronrod checks of the four closest selected comparison pairs preserve all winner signs and all 40 scalar metrics to three significant digits. Acceleration variation changes by up to .00127, so its table entries use three significant digits rather than claiming three-decimal-place accuracy. This is a targeted check, not an interval enclosure of every cost functional.

Table 9: All 96 matched main cases: physical entry estimates, cost and absolute intervention. Times use six decimals; cost uses seven and intervention six significant digits. B denotes band; P denotes certified progress.

Case	θ	T^*	T_B	T_P	J_B	J_P	$I_{1,B}$	$I_{1,P}$
C1-GR-H	0.25	1.317781	1.519543	1.429835	27.98554	27.99199	2.45937	2.36053
C1-GR-H	0.5	1.317781	1.441937	1.405841	27.9898	28.03218	2.36546	2.44788
C1-GR-O	0.25	1.317781	2.452122	2.075050	36.04852	28.94193	74.7583	75.2522
C1-GR-O	0.5	1.317781	1.782642	1.646124	31.35131	28.15946	75.2145	75.4407
C1-GR-C	0.25	1.317781	1.850631	1.454875	35.4114	28.73797	1.30685	1.87873
C1-GR-C	0.5	1.317781	1.543892	1.363678	31.17975	28.06697	1.75497	2.3304
C1-GR-Z	0.25	1.317781	2.443796	2.075050	36.01715	28.94193	24.7583	25.2522
C1-GR-Z	0.5	1.317781	1.787686	1.646124	31.24092	28.15946	25.2114	25.4407
C1-GM-H	0.25	1.278313	1.480607	1.389210	27.97569	27.98018	2.50124	2.39291
C1-GM-H	0.5	1.278313	1.401344	1.363209	27.98225	28.01719	2.36409	2.47364
C1-GM-O	0.25	1.278313	2.433576	2.069399	37.82044	29.63075	74.5325	75.0448
C1-GM-O	0.5	1.278313	1.751800	1.610409	32.0455	28.16259	75.0417	75.2901
C1-GM-C	0.25	1.278313	1.813227	1.452142	36.35848	29.4276	0.960201	1.50612
C1-GM-C	0.5	1.278313	1.512236	1.324414	31.86377	28.05396	1.43684	2.06541
C1-GM-Z	0.25	1.278313	2.429864	2.069399	37.65453	29.63075	24.5408	25.0448
C1-GM-Z	0.5	1.278313	1.746142	1.610409	31.99724	28.16259	25.031	25.2901
C1-TR-H	0.25	0.708877	1.211190	0.846988	12.11559	12.16982	13.9205	14.4344
C1-TR-H	0.5	0.708877	0.932723	0.737096	13.60527	13.90632	16.2163	16.4459
C1-TR-O	0.25	0.708877	1.238062	0.846988	12.25819	12.16982	72.2672	72.6662
C1-TR-O	0.5	0.708877	0.932224	0.737096	14.67914	13.90632	74.2846	73.6586
C1-TR-C	0.25	0.708877	1.206294	0.802986	13.2236	15.10942	1.30685	1.87873
C1-TR-C	0.5	0.708877	0.925996	0.716966	15.08816	15.69618	1.75497	2.3304
C1-TR-Z	0.25	0.708877	1.228237	0.846988	12.43014	12.16982	22.3409	22.6662
C1-TR-Z	0.5	0.708877	0.928083	0.737096	13.70771	13.90632	23.6232	23.6586
C1-TM-H	0.25	0.669409	1.175726	0.844280	11.68957	11.95694	13.6669	14.3993
C1-TM-H	0.5	0.669409	0.895776	0.701886	14.36479	14.06446	17.3143	16.8489
C1-TM-O	0.25	0.669409	1.221170	0.844280	12.38717	11.95694	72.161	72.4973
C1-TM-O	0.5	0.669409	0.898019	0.701886	14.38596	14.06446	73.9601	73.7048
C1-TM-C	0.25	0.669409	1.174063	0.800700	12.78921	14.88463	0.960201	1.50612
C1-TM-C	0.5	0.669409	0.895610	0.678524	14.53802	15.53412	1.43684	2.06541
C1-TM-Z	0.25	0.669409	1.214017	0.844280	12.37734	11.95694	22.2058	22.4973
C1-TM-Z	0.5	0.669409	0.898219	0.701886	14.38259	14.06446	23.9746	23.7048
C2-GR-H	0.25	1.555930	1.606894	1.636404	15.85813	15.85981	1.73366	1.78435
C2-GR-H	0.5	1.555930	1.607233	1.636404	15.85975	15.85981	1.73509	1.78435
C2-GR-O	0.25	1.555930	2.938144	2.872067	23.20766	19.15788	59.562	59.625
C2-GR-O	0.5	1.555930	2.123001	2.141418	18.9041	16.803	59.7316	59.7546
C2-GR-C	0.25	1.555930	1.867290	1.764895	21.44152	17.64657	1.88709	1.74383
C2-GR-C	0.5	1.555930	1.683309	1.637646	18.0611	16.03094	2.32351	2.28955
C2-GR-Z	0.25	1.555930	2.901177	2.872067	22.72067	19.15788	9.57164	9.62495
C2-GR-Z	0.5	1.555930	1.916051	2.061150	18.21103	16.57808	10.304	9.77133
C2-GM-H	0.25	1.549370	1.599915	1.629625	17.24675	17.25934	1.90136	1.92296
C2-GM-H	0.5	1.549370	1.600366	1.629625	17.24155	17.25934	1.83995	1.92296
C2-GM-O	0.25	1.549370	2.988020	2.919268	27.10862	22.15914	59.2331	59.3016
C2-GM-O	0.5	1.549370	2.145738	2.125261	21.25881	18.49121	59.4232	59.4637
C2-GM-C	0.25	1.549370	1.868367	1.803007	24.56184	20.55513	1.63379	1.46138
C2-GM-C	0.5	1.549370	1.699841	1.646134	20.42498	17.79727	2.04466	2.14711
C2-GM-Z	0.25	1.549370	2.958250	2.918394	26.24365	22.15086	9.2462	9.30197
C2-GM-Z	0.5	1.549370	1.924033	1.843703	20.48769	17.91919	9.97803	9.78955
C2-TR-H	0.25	0.642573	1.116161	1.018331	3.3199	3.119839	6.38785	6.58386
C2-TR-H	0.5	0.642573	0.850531	0.802487	4.45389	4.203695	8.85038	8.73006
C2-TR-O	0.25	0.642573	1.118331	1.018331	4.257622	3.119839	58.6531	58.6903
C2-TR-O	0.5	0.642573	0.835192	0.802487	4.813009	4.203695	59.1102	59.127

Case	θ	T^*	T_B	T_P	J_B	J_P	$I_{1,B}$	$I_{1,P}$
C2-TR-C	0.25	0.642573	0.950943	0.824914	5.269297	4.731746	1.88709	1.74383
C2-TR-C	0.5	0.642573	0.766863	0.699160	5.549538	5.296726	2.32351	2.28955
C2-TR-Z	0.25	0.642573	1.077349	1.018331	3.704114	3.119839	8.65591	8.69168
C2-TR-Z	0.5	0.642573	0.764026	0.772169	5.680513	4.483818	10.2386	9.35665
C2-TM-H	0.25	0.636013	1.135500	1.066309	3.163797	2.91388	6.09851	6.41209
C2-TM-H	0.5	0.636013	0.847954	0.793775	4.41761	4.086557	8.45921	8.53813
C2-TM-O	0.25	0.636013	1.167872	1.066309	4.186129	2.91388	58.3301	58.3777
C2-TM-O	0.5	0.636013	0.858087	0.793775	4.696177	4.086557	58.8453	58.801
C2-TM-C	0.25	0.636013	0.952236	0.863474	5.18728	4.557521	1.63379	1.46138
C2-TM-C	0.5	0.636013	0.783355	0.707904	5.302185	5.222452	2.04466	2.14711
C2-TM-Z	0.25	0.636013	1.133681	1.065505	3.57611	2.916055	8.31595	8.38061
C2-TM-Z	0.5	0.636013	0.771658	0.722641	5.58154	5.098553	9.91074	9.68901
C3-GR-H	0.25	1.723065	1.751966	1.768164	13.54162	13.55319	2.88479	2.9663
C3-GR-H	0.5	1.723065	1.752058	1.763467	13.54161	13.5521	2.88499	2.96202
C3-GR-O	0.25	1.723065	3.280321	3.255806	20.89257	18.42792	50.9832	51.0269
C3-GR-O	0.5	1.723065	2.372895	2.375294	16.35126	15.08767	51.2357	51.2669
C3-GR-C	0.25	1.723065	1.996852	1.966360	18.23649	16.44709	1.64111	1.55588
C3-GR-C	0.5	1.723065	1.831096	1.807128	15.3671	14.08132	2.14649	2.0184
C3-GR-Z	0.25	1.723065	3.252482	3.255806	19.82847	18.42792	0.993891	1.02947
C3-GR-Z	0.5	1.723065	1.860731	2.375294	15.59426	15.08767	1.42575	1.27066
C3-GM-H	0.25	1.749135	1.775825	1.793470	16.23208	16.21114	3.09287	3.12675
C3-GM-H	0.5	1.749135	1.775839	1.788518	16.23209	16.20941	3.09294	3.11751
C3-GM-O	0.25	1.749135	3.373152	3.312743	26.945	22.63119	50.7396	50.8404
C3-GM-O	0.5	1.749135	2.432557	2.419476	20.17787	18.46837	51.0563	51.1077
C3-GM-C	0.25	1.749135	2.057869	1.991435	23.59239	20.33282	1.41892	1.40812
C3-GM-C	0.5	1.749135	1.875420	1.839907	19.39743	17.27079	2.16233	1.95854
C3-GM-Z	0.25	1.749135	3.368000	3.310373	25.47409	22.60864	0.741993	0.844606
C3-GM-Z	0.5	1.749135	2.232201	2.413910	19.8672	18.45439	1.08481	1.11267
C3-TR-H	0.25	0.657482	1.145887	1.134292	1.86376	1.407549	4.5913	4.71286
C3-TR-H	0.5	0.657482	0.876838	0.864230	2.66302	2.440972	7.0916	7.11243
C3-TR-O	0.25	0.657482	1.154570	1.134292	2.942308	1.407549	50.3499	50.2882
C3-TR-O	0.5	0.657482	0.868642	0.864230	2.960613	2.440972	51.0745	51.096
C3-TR-C	0.25	0.657482	0.930959	0.893377	3.468436	2.908527	1.64111	1.55588
C3-TR-C	0.5	0.657482	0.765167	0.735606	3.818875	3.573716	2.14649	2.0184
C3-TR-Z	0.25	0.657482	1.126220	1.134292	1.907943	1.407549	1.31493	1.45053
C3-TR-Z	0.5	0.657482	0.792766	0.864230	3.699625	2.440972	1.45933	1.29794
C3-TM-H	0.25	0.683552	1.220434	1.192220	1.862872	1.421496	4.57495	4.5371
C3-TM-H	0.5	0.683552	0.932182	0.910601	2.652677	2.428805	6.94337	7.09044
C3-TM-O	0.25	0.683552	1.248061	1.192220	3.3461	1.421496	49.9807	50.0888
C3-TM-O	0.5	0.683552	0.929017	0.910601	2.819269	2.428805	50.8906	50.9417
C3-TM-C	0.25	0.683552	0.991916	0.919456	3.530918	3.032653	1.41892	1.40812
C3-TM-C	0.5	0.683552	0.809386	0.768747	3.94905	3.602546	2.16233	1.95854
C3-TM-Z	0.25	0.683552	1.237744	1.190022	1.835915	1.425438	1.1538	1.27225
C3-TM-Z	0.5	0.683552	0.893999	0.908459	3.132888	2.453722	1.18123	1.14244

Table 10: All main-case motion metrics on the common horizon. A is acceleration RMS, V_a numerical acceleration variation, and A_{jump} the largest acceleration jump. Metrics use three significant digits.

Case	θ	A_B	A_P	$V_{a,B}$	$V_{a,P}$	$A_{\text{jump},B}$	$A_{\text{jump},P}$
C1-GR-H	0.25	0.697	0.695	11.1	10.7	0.784	0.784
C1-GR-H	0.5	0.697	0.726	11.1	11.5	0.784	2.26
C1-GR-O	0.25	0.721	0.598	78.3	10	4.97	2.77
C1-GR-O	0.5	0.688	0.733	51.9	12.3	5.48	4.35

Case	θ	A_B	A_P	$V_{a,B}$	$V_{a,P}$	$A_{\text{jump},B}$	$A_{\text{jump},P}$
C1-GR-C	0.25	0.544	0.628	30.7	10.8	3.72	1.47
C1-GR-C	0.5	0.636	0.741	37.5	11.9	3.13	2.32
C1-GR-Z	0.25	0.75	0.598	87.1	10	4.97	2.77
C1-GR-Z	0.5	0.651	0.733	52.8	12.3	3.13	4.35
C1-GM-H	0.25	0.619	0.616	11.5	10.2	0.969	0.798
C1-GM-H	0.5	0.619	0.65	11.4	11	0.974	2.29
C1-GM-O	0.25	0.702	0.465	77.9	9.16	6.36	2.8
C1-GM-O	0.5	0.628	0.661	53.5	11.9	5.51	4.47
C1-GM-C	0.25	0.417	0.501	20	9.91	2.93	1.32
C1-GM-C	0.5	0.537	0.668	29.1	11.4	2.72	2.41
C1-GM-Z	0.25	0.68	0.465	79.8	9.16	4.99	2.8
C1-GM-Z	0.5	0.61	0.661	50.6	11.9	4.07	4.47
C1-TR-H	0.25	0.61	0.631	49.7	12.2	3.92	2.77
C1-TR-H	0.5	0.66	0.764	46.7	14.1	2.92	4.35
C1-TR-O	0.25	0.746	0.631	80.1	12.2	4.97	2.77
C1-TR-O	0.5	0.68	0.764	48.9	14.1	4.03	4.35
C1-TR-C	0.25	0.556	0.653	30.7	12.9	3.72	1.47
C1-TR-C	0.5	0.661	0.762	38.3	13.6	4.2	2.32
C1-TR-Z	0.25	0.774	0.631	88.4	12.2	4.97	2.77
C1-TR-Z	0.5	0.684	0.764	54.1	14.1	3.13	4.35
C1-TM-H	0.25	0.466	0.505	31.9	11.3	3.96	2.8
C1-TM-H	0.5	0.592	0.687	43.1	13.5	4.2	4.47
C1-TM-O	0.25	0.726	0.505	77.8	11.3	6.36	2.8
C1-TM-O	0.5	0.626	0.687	50.8	13.5	4.23	4.47
C1-TM-C	0.25	0.43	0.531	20	11.9	3.19	1.33
C1-TM-C	0.5	0.568	0.694	29.7	13.2	4.39	2.41
C1-TM-Z	0.25	0.704	0.505	82.1	11.3	4.99	2.8
C1-TM-Z	0.5	0.626	0.687	50.9	13.5	4.25	4.47
C2-GR-H	0.25	0.901	0.873	12.7	10.9	3.4	2.54
C2-GR-H	0.5	0.901	0.873	12.9	10.9	3.39	2.54
C2-GR-O	0.25	0.908	0.561	84.6	6.97	7.89	1.73
C2-GR-O	0.5	0.898	0.739	55.4	8.45	10	3.28
C2-GR-C	0.25	0.702	0.6	35.8	9	4.48	1.3
C2-GR-C	0.5	0.801	0.771	41.1	9.97	3.9	1.62
C2-GR-Z	0.25	0.728	0.561	47.8	6.97	5	1.73
C2-GR-Z	0.5	0.841	0.737	48.9	7.92	5	3.06
C2-GM-H	0.25	0.878	0.851	13.5	11.4	3.27	3.04
C2-GM-H	0.5	0.879	0.851	13.5	11.4	3.27	3.04
C2-GM-O	0.25	0.849	0.402	78.9	5.77	6.9	1.68
C2-GM-O	0.5	0.815	0.672	51.9	10.7	9.37	3.79
C2-GM-C	0.25	0.63	0.455	31.3	7.32	4.23	0.907
C2-GM-C	0.5	0.682	0.685	27.9	10.1	4.2	1.39
C2-GM-Z	0.25	0.652	0.402	42.5	5.67	5	1.63
C2-GM-Z	0.5	0.774	0.663	46.8	9.32	5	2.39
C2-TR-H	0.25	0.614	0.57	30.8	8	5.57	1.73
C2-TR-H	0.5	0.778	0.747	40.2	9.31	6.46	3.28
C2-TR-O	0.25	0.914	0.57	85.6	8	7.89	1.73
C2-TR-O	0.5	0.906	0.747	56.2	9.31	10	3.28
C2-TR-C	0.25	0.705	0.607	35.8	9.45	4.48	1.3
C2-TR-C	0.5	0.804	0.776	41.1	10.1	3.9	1.62
C2-TR-Z	0.25	0.735	0.57	47.8	8	5	1.73
C2-TR-Z	0.5	0.841	0.745	48.9	8.73	5	3.06
C2-TM-H	0.25	0.525	0.415	25.4	6.79	5.79	1.68
C2-TM-H	0.5	0.764	0.683	39.5	11.5	7.15	3.79
C2-TM-O	0.25	0.855	0.415	79.8	6.79	6.9	1.68
C2-TM-O	0.5	0.823	0.683	52.6	11.5	9.37	3.79
C2-TM-C	0.25	0.634	0.462	31.3	7.58	4.23	0.907

Case	θ	A_B	A_P	$V_{a,B}$	$V_{a,P}$	$A_{\text{jump},B}$	$A_{\text{jump},P}$
C2-TM-C	0.5	0.687	0.689	27.9	10.1	4.2	1.39
C2-TM-Z	0.25	0.66	0.415	43.6	6.69	5	1.63
C2-TM-Z	0.5	0.774	0.667	46.7	9.42	5	2.39
C3-GR-H	0.25	0.924	0.894	14	12.5	2.87	2.5
C3-GR-H	0.5	0.924	0.894	14	12.5	2.87	2.5
C3-GR-O	0.25	0.997	0.527	89.8	6.79	8.18	1.41
C3-GR-O	0.5	0.858	0.73	52.1	9.62	7.82	2.59
C3-GR-C	0.25	0.707	0.559	32.9	9.57	5.69	1.08
C3-GR-C	0.5	0.8	0.759	38.8	11.7	4.76	2.05
C3-GR-Z	0.25	0.679	0.527	32	6.79	5	1.41
C3-GR-Z	0.5	0.826	0.73	46.8	9.62	5	2.59
C3-GM-H	0.25	0.892	0.839	14.3	12.6	4.6	2.73
C3-GM-H	0.5	0.892	0.839	14.3	12.6	4.6	2.73
C3-GM-O	0.25	1.02	0.378	102	5.64	9.18	1.54
C3-GM-O	0.5	0.716	0.6	43.9	8.86	4.97	2.54
C3-GM-C	0.25	0.577	0.423	27.1	7.82	4.8	0.797
C3-GM-C	0.5	0.717	0.636	34	10.6	4.86	1.08
C3-GM-Z	0.25	0.527	0.378	26.2	5.58	5	1.51
C3-GM-Z	0.5	0.71	0.599	42.9	8.8	4.98	2.54
C3-TR-H	0.25	0.699	0.533	32.2	7.36	5.95	1.41
C3-TR-H	0.5	0.775	0.735	37.6	10.1	4.71	2.59
C3-TR-O	0.25	1	0.533	90.3	7.36	8.18	1.41
C3-TR-O	0.5	0.862	0.735	52.5	10.1	7.82	2.59
C3-TR-C	0.25	0.708	0.562	32.9	9.93	5.69	1.08
C3-TR-C	0.5	0.8	0.76	38.8	11.8	4.76	2.05
C3-TR-Z	0.25	0.683	0.533	32.4	7.36	5	1.41
C3-TR-Z	0.5	0.828	0.735	47.1	10.1	5	2.59
C3-TM-H	0.25	0.564	0.387	28	6.2	5.92	1.54
C3-TM-H	0.5	0.662	0.606	33.4	9.34	4.68	2.54
C3-TM-O	0.25	1.02	0.387	103	6.2	9.18	1.54
C3-TM-O	0.5	0.721	0.606	44.4	9.34	4.97	2.54
C3-TM-C	0.25	0.578	0.426	27.1	7.98	4.8	0.797
C3-TM-C	0.5	0.718	0.637	34	10.7	4.86	1.08
C3-TM-Z	0.25	0.534	0.386	26.6	6.15	5	1.51
C3-TM-Z	0.5	0.715	0.605	43.6	9.26	4.98	2.54

Table 11: All 24 additional band sampling tests. T_o is the interval-confirmed detection time; $B_o = \underline{T}^*/\sqrt{\theta} + \bar{\Delta}$ is the verified comparison bound. Sh is the number of shortened holds.

Case	θ	$\bar{\Delta}$	$\min \delta_n$	T	T_o	B_o	Sh
C1-GR-O	0.25	0.0125	0.0125	2.559	2.563	2.648	0
C1-GR-O	0.25	0.025	0.025	2.512	2.525	2.660	0
C1-GR-O	0.25	0.1	0.1	2.421	2.500	2.735	0
C1-GR-O	0.5	0.0125	0.0125	1.835	1.837	1.876	0
C1-GR-O	0.5	0.025	0.025	1.817	1.825	1.888	0
C1-GR-O	0.5	0.1	0.1	1.748	1.800	1.963	0
C2-TM-H	0.25	0.0125	0.0125	1.155	1.162	1.284	0
C2-TM-H	0.25	0.025	0.025	1.148	1.150	1.297	0
C2-TM-H	0.25	0.1	0.1	1.114	1.200	1.372	0
C2-TM-H	0.5	0.0125	0.0125	0.877	0.887	0.912	0
C2-TM-H	0.5	0.025	0.025	0.872	0.875	0.924	0
C2-TM-H	0.5	0.1	0.1	0.812	0.900	0.999	0
C3-GR-O	0.25	0.0125	0.0125	3.380	3.388	3.458	0
C3-GR-O	0.25	0.025	0.025	3.337	3.350	3.471	0

Case	θ	$\bar{\Delta}$	$\min \delta_n$	T	T_o	B_o	Sh
C3-GR-O	0.25	0.1	0.1	3.216	3.300	3.546	0
C3-GR-O	0.5	0.0125	0.0125	2.413	2.425	2.449	0
C3-GR-O	0.5	0.025	0.025	2.396	2.400	2.462	0
C3-GR-O	0.5	0.1	0.05	2.333	2.350	2.537	1
C3-GM-H	0.25	0.0125	0.0125	1.776	1.787	3.511	0
C3-GM-H	0.25	0.025	0.025	1.776	1.800	3.523	0
C3-GM-H	0.25	0.1	0.1	1.785	1.800	3.598	0
C3-GM-H	0.5	0.0125	0.0125	1.776	1.787	2.486	0
C3-GM-H	0.5	0.025	0.025	1.776	1.800	2.499	0
C3-GM-H	0.5	0.1	0.1	1.785	1.800	2.574	0

13.10 Fixed-period nominal-first experiments

This separate experiment exercises the certified periods directly. For each of the 96 combinations of plant, task, initial speed, specified nominal, and $\theta \in \{0.25, 0.5\}$, the controller uses the sampled nominal whenever a sound physical-time test certifies the complete hold, and otherwise selects a stored clearance witness. The two prescribed periods are $\Delta = 0.01$ and 0.025 in normalized model time. All four nominals, including the six supplied HJB policy tables, are retained. The .05 candidate-search controller and its supplied results are reported separately in Sections 13.2–13.9. The nominal gate uses the physical-hold admissibility condition of Theorem 8.3; it can accept inputs outside the more restrictive spatial-prefix interval $\mathcal{I}_\Delta$. Lemma 8.8 supplies the feasible stored fallback, so the two gates are not identified.

The implementation carries an outward state enclosure across holds, starting from the exact decimal initial coordinates; it never resets the numerical error radius. The nominal test uses the continuous residual-tube construction, full-cell energy-profile bounds, actuator and rest-launch guards, and the fixed episode's profile extension. The stored floor comparison can certify a floor contact that a rectangular residual tube cannot separate, including rest launch. For a sample enclosure intersecting more than one spatial cell, the fallback takes the union of the stored ceiling-clock prefixes and recomputes the boundary inputs and their opposite-face budgets for that union. It chooses a witness only when the corresponding clearance lower bound passes. The exact-decimal offline model and binary64 reference curve are reconciled by an explicit outward model-discrepancy residual; the initial-coordinate rounding is also enclosed. The printed-period/binary64 distinction is checked by verifying every stored prefix against the actual binary64 hold duration. These numerical enclosures do not introduce a general sensing-error or delay claim. The complete derivation, source files, and checks are in `fixed_period_experiments/METHOD.txt` in the release archive.

All 192 runs enter successfully, with no shortened pre-entry hold and no unresolved witness choice. The maximum numerical physical-entry ratios T/W_θ are 0.990619 and 0.980558 for $\Delta = 0.01$ and 0.025, respectively (rounded upward). Here W_θ is computed from the lower-reflected profile initialized at E_0, including moving starts; $T^*/\sqrt{\theta}$ is retained separately as the simpler factor bound. A separate interval calculation brackets physical entry and compares its upper endpoint with a rigorous lower bound on W_θ; all 192 comparisons pass, with maximum ratios 0.998107 and 0.987971 (rounded upward). For an independent lower bound on the reflected clock, use monotonicity of $\mathcal{V}$ and $d \le d_0$, $v \ge 0$ to compute an outward upper bound $m \ge \alpha\mathcal{V}(d_0) - U$; all three models give $m < 0$. Before its first floor contact E_θ^- satisfies $(E_\theta^-)' \le m$, while full floor admissibility makes it follow q after contact. Hence

$$E_\theta^-(s) \le \max\{q(s), E_0 + ms\}. \tag{116}$$

Let $\delta_s = 10^{-4}$, $s_j = j\delta_s$, and $N_s = S/\delta_s \in \mathbb{N}$ for the specified benchmark targets. Let Q_j bound E^+ on the entire cell $[s_j, s_{j+1}]$ and put $\overline{E}_j^- = \max\{\overline{\theta Q_j}, \overline{E_0 + m s_j}\}$. Outward arithmetic gives

$$\underline{W_\theta} = \underline{\sum_{j=0}^{N_s-1} \frac{\delta_s}{\sqrt{2\overline{E}_j^-}}} \le W_\theta. \tag{117}$$

The overbar and underline denote outward upper and lower evaluations. The negative slope makes the free-line term largest at the cell's left endpoint. The initial energy is enclosed from the exact decimal initial speed; this construction includes moving starts and is distinct from merely dividing a lower bound for T^* by $\sqrt{\theta}$.

The numerical entry root and sampled detection are stored separately. The largest sampled-detection delays are 0.0098585 and 0.0240219. No diagnostic floor or speed violation is observed; their minimum values are zero because rest is included. Minimum diagnostic headways are 4.6471×10^{-7} and 4.7038×10^{-7}. These finite-sample minima are diagnostics; the residual/comparison arguments supply the full-hold certificate. Maximum carried pre-entry position radii are 1.7830×10^{-7} and 1.8825×10^{-7}.

The nominal is accepted on 3047/15185 (20.1%) and 1247/5913 (21.1%) of recovery holds before confirmed entry; the remaining 12138 and 4666 holds use the stored witness. The fixed-period rule enters earlier than the matched historical 0.05 candidate-search band controller in 21/96 and 28/96 configurations. On the 72 non-HJB configurations the counts are 13/72 and 24/72. These comparisons match the initial state, task, nominal, and θ, but change both the period and selection rule; they are descriptive comparisons, not a same-period causal comparison or evidence of performance dominance. The new rule has lower numerical J_{10} in 55/96 configurations at either period, lower actual $I_{1,10}$ in 73/96 and 58/96, and lower acceleration RMS in 96/96 and 95/96. All cost and motion values use the historical 10-unit task and common sampled safety-tail convention, at the new period. Full-precision paired values and the numerical equality tolerances are saved. A 32-point quadrature replay of the six closest paired cases, with tighter ODE tolerances, changes J_{10} by at most 7.2×10^{-15}, $I_{1,10}$ by 1.18×10^{-5}, and $I_{2,10}$ by 5.57×10^{-5}; all checked comparison signs remain unchanged. These are sensitivity diagnostics, not rigorous quadrature error bounds.

Runtime evidence is separate from parallel production timings. A serial process replays all 2102 recovery decisions in a specified 16-case stress subset and reproduces every stored command and branch exactly. For $\Delta = 0.01$ the median, 95th percentile, and maximum are 5.94, 9.12, and 45.25 ms across 1520 decisions; for 0.025 they are 8.09, 11.50, and 44.33 ms across 582 decisions. The measured interval includes nominal evaluation, held-flow proposal, residual certification, and witness selection, after policy/certificate loading and warm-up. The host reports an AMD EPYC 9V74 processor, Linux 6.18.57, Python 3.12.14, and a C99 outward kernel built with `-O2 -fno-fast-math -ffp-contract=off`. The residual transcendental calculations use 40 interval decimal digits; the offline fixed-period certificates use 45. Full environment, compiler, policy and source hashes are recorded. Runtimes are wall-clock milliseconds, whereas Δ is normalized model time; no hard real-time deadline claim follows. Related configurations are not independent statistical samples.

Figure 5 and Table 12 summarize the fixed-period experiments. Table 13 compares them descriptively with the supplied .05 candidate-search runs.

Table 12: Fixed-period summary; each period has 96 cases.

Δ	Entered	Max. T/W_θ	Nominal (%)	Witness holds	p50 (ms)	p95 (ms)
0.010	96/96	0.9906	20.1	12,138	5.94	9.12
0.025	96/96	0.9806	21.1	4,666	8.09	11.50

All commands use the package root:

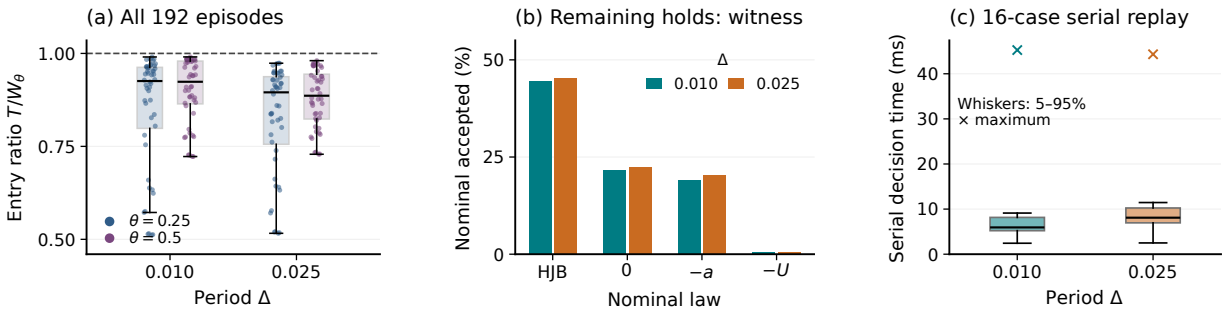


Figure 5: Fixed-period witness policy: recovery ratios, accepted nominal holds and serialized decision times.

Table 13: Number of fixed-period cases with a lower diagnostic metric than the matched .05 candidate search. Both the period and selection law change.

Δ	T lower	J_{10} lower	$I_{1,10}$ lower	Acc. RMS lower	Acc. TV lower	Input TV lower
0.010	21/96	55/96	73/96	96/96	84/96	84/96
0.025	28/96	55/96	58/96	95/96	83/96	83/96

```
python3 reproduce.py --fixed-periods --fresh --workers 3
python3 reproduce.py --fixed-benchmark
```

The complete per-hold records, source and input hashes, protocol, full-precision paired comparisons and independent entry-bracket checks are in `fixed_period_experiments/`.

A Continuous-time comparison: quantitative recovery and nominal fidelity

This study evaluates finite physical-gap-entry episodes for the reflected rate controller. Its continuous-time policies and post-entry continuation differ from the sampled-actuation study in Section 13; the two datasets are analyzed separately. The experiment specification is recorded in `numerics_v6/protocol.json`. All 13 specified controller settings are run on all 72 scenarios; no replacement gain is chosen from the outcomes. The base study therefore contains 936 episodes. All rate cases and all baseline cases at $(h_0, v_0) = (.8, 0)$ are repeated with tighter integration and event resolution, giving 504 refinement episodes.

A.1 Common plants, objectives, and nominal commands

The model constants and the three plants are specified in Section 13.1 and Table 5. Here $a = \alpha(V(d) - v)$ and $b = 1$. C1/C2 allow signed optimal-velocity drift and deliberately challenge restart; C3 uses a nonnegative bounded desired-velocity law. The original face and global CBF feasibility certificates are satisfied. The all-θ floor certificate is proved analytically in Proposition 7.3; sampled residuals are additional implementation diagnostics, not substitutes for that proof.

The initial (h_0, v_0) pairs are $(.4, 0)$, $(.8, 0)$, and $(.8, .4)$; $d_0 = \epsilon + h_0 + \gamma v_0$. The gap objective uses $h_g = 0$, $\lambda = 2$, and $\ell = .2u^2 + 50d^2$. The tracking-cost objective uses $h_g = .2$, $\lambda = .1$, and $\ell = .2u^2 + v^2 + 5(h - .2)^2$. In both cases first entry means $d \le \epsilon + h_g + .01$, with no terminal-speed

constraint. The tracking-cost designation does not claim convergence to exactly $h = .2$; the theoretical rest target is the entire accepted interval.

Each case uses one of four nominal commands:

1. The HJB nominal specified in Section 13.1.1, using continuous interpolation and its zero terminal action extended by zero after time ten. Its quadratic objective differs from the minimum-entry objective. The same policy tables are used for every controller setting.
2. Zero actuator input, $k = 0$. Its net acceleration is the plant drift $a(d, v)$, which can be positive or negative.
3. Clipped drift cancellation, $k = \mathrm{clip}_{[-U,U]}(-a/b)$, giving zero nominal net acceleration where cancellation is attainable. It cancels the rest drift at the specified initial states.
4. Maximum braking actuator input, $k = -U$, which attains the exact worst clock of every filter in the theorem's strict-fidelity class. It is an adversarial command for recovery, not a claim about the behavior of a practical nominal driver.

All commands belong to the joint local Lipschitz class used by the rate theorem on the queried state/time region. Common nominal tables are reused across methods without controller-dependent changes.

A.2 Controller definitions and comparison priorities

The controller settings are $\kappa_h = \kappa_v = \kappa_M = 10$, $N = 50$, $s_h = 1$, $s_v = M$, $h_s = .2$, and the objective-specific λ above. Ordinary CBF projects $p_N = a + bk$ onto

$$[L, R] = [\max\{A_-, -10v\}, \min\{A_+, (-v + 10h)/\gamma, 10(M - v)\}] .$$

The recovery blend and recovery CBF are the regular laws defined in Section 4.

The progress-aware CBF baseline uses $v_{\mathrm{ref}} = r\bar{v}(d)$ and the inequality

$$\frac{d}{dt}(v - r\bar{v}(d)) \geq -c(v - r\bar{v}(d)).$$

Since $\dot{d} = -v$, its requested acceleration is

$$p_{\mathrm{req}} = -rv\bar{v}'(d) + c(r\bar{v}(d) - v).$$

Safety remains hard. Minimum nonnegative slack is $\zeta = (p_{\mathrm{req}} - R)_+$; among those controls, the chosen acceleration is closest to the nominal:

$$p_{\mathrm{prog}} = \mathrm{clip}_{[\max\{L,\min(p_{\mathrm{req}},R)\},R]}\, p_N.$$

The derivative of $\bar{v}$ is zero on the speed cap and $1/\gamma$ on the headway portion, with either value at the isolated switch. This one-sided progress constraint follows the CLF/CBF design principle [33, 34]. It is a fully specified comparison instantiation, not a literal reproduction of a published adaptive cruise-control tuning. A positive slack prevents claiming its unrelaxed exponential inequality. Its interior intervention also means it need not satisfy the strict fidelity definition in Theorem 7.2.

Four baseline speed fractions $r = \sqrt{.1}, .5, \sqrt{.5}, 1$ use $c = 5$. Two additional settings use $r = .5$ with $c = 2$ and 10. All six appear in the results; the grid is not searched to select a preferred winner. Four reflected methods use $\theta = .1, .25, .5, 1$. Each uses a fixed episode maximal profile; it is never reset along the trajectory. At $\theta = 1$ the method directly follows the optimal profile, separately from the separated-boundary construction for $0 < \theta < 1$.

A.3 Common event horizon and post-entry continuation

First entries are observed through time 600. An unobserved entry remains right censored and is retained in every denominator. Some ordinary-CBF nonentries can also be proved to be permanent rest equilibria from the nominal-acceleration sign criterion; the horizon alone is not used to prove that statement.

Every controller hands over at its first entry to the same recovery-CBF law with that case's nominal and objective parameters. This defines a comparable ten-unit cost after different entry times. A method that has not entered by ten continues its episode law throughout the cost window. The handover is part of the comparison protocol. The entry-time theorem ends at the entry event; no rate or jerk guarantee is assigned to the post-entry law or its switch.

A.4 Metrics and interpretation

Write $p = \dot{v}$ for net acceleration and $H = 10$. The common-horizon metrics are

$$J_H = \int_0^H \ell(d, v, u)\, dt, \qquad I_{1,H} = \int_0^H |u - k|\, dt,$$
$$I_{2,H} = \int_0^H (u - k)^2\, dt, \qquad A_H = \left(\frac{1}{H} \int_0^H p^2\, dt \right)^{1/2}.$$

Effort $\int_0^H u^2 dt$ is stored separately. The entry-only absolute intervention $I_{1,\mathrm{ent}}$ uses $[0, T]$ for an observed entry and $[0, 600]$ for a censored case. The integration durations therefore vary by case. Persistent braking input can make a faster episode have a smaller integrated intervention even when its instantaneous intervention is larger. Neither the strict-fidelity definition nor the scalar projection asserts global minimization of $I_{1,H}$ or of an entry-only integral.

The numerical acceleration-variation estimate is

$$\widehat{\mathrm{TV}}_H(p) = \sum_i |p(t_{i+1}) - p(t_i)|.$$

Its base grid spacing is .01 and its refined spacing .005. Both grids additionally contain all known profile knots, progress-CBF ceiling-kink crossings, arc joins, and handovers, together with samples on both sides. The separate largest-jump metric uses one-sided acceleration differences at those events, with time offsets at most 10^{-8}. Differences above 10^{-6} are counted as recorded jumps. The complete event ledger stores time, type, left and right acceleration, and magnitude. A smooth saturation transition can change a derivative without an acceleration jump.

The progress-slack integral and maximum use $[0, 10]$. All progress episodes enter before ten in this matrix; the common continuation has no progress requirement and its recorded slack is zero. The field `progress_slack_fraction10` is the fraction of sampled points whose slack exceeds 10^{-8} on the event-augmented grid. It is not an exact duration fraction; the integral and maximum are the primary slack measures. The slack-free counts below likewise mean that no positive slack was observed in the stated numerical check.

Acceleration RMS, variation, and jumps describe the computed signals. The models are normalized, and no human-comfort threshold is calibrated. The reflected law and the progress baseline can have acceleration jumps. Thus no intrinsic finite jerk norm follows from these data. Actuator smoothing or sample-and-hold implementation would change the controller and requires an additional analysis.

A.5 Numerical construction and tolerances

For each of the 18 plant/task/initial-state combinations, the maximal energy profile is integrated with DOP853 at relative/absolute tolerances $2\,10^{-9}$ and $2\,10^{-11}$, using maximum spatial step

$S/256$. The clock uses $s = x^2$, maximum coordinate step $\sqrt{S}/256$, and the same tolerances. Refined runs use $2\,10^{-11}$ and $2\,10^{-13}$ with denominators 512. The fixed initial-state profile is shared across all four nominals and every rate setting.

The fixture-specific maximal-profile solver uses full throttle until first ceiling contact and then follows the admissible ceiling. On the headway branch, the outward full-throttle difference is $C + (1/\gamma - \alpha)v > 0$ for these models. C1 cannot reach its unfavorable speed cap from the specified initial states; cap contact in C2/C3 has nonnegative outward full-throttle slope. This numerical shortcut is not advertised as a generic solver for every plant in the abstract theorem.

Interior reflected arcs integrate the nominal energy and its coupled physical clock. Boundary arcs follow the fixed floor or ceiling exactly at the profile-evaluation level. Contact is event detected. Release is located by a scan over 1,024 spatial intervals (2,048 under refinement), including profile knots, followed by bracketed scalar root finding. If the initial nominal points strictly above the floor at a rest start, a physical-time nominal launch over 10^{-5} (refined 10^{-6}) is integrated first, avoiding division by zero. The analytic no-waiting branch supplies its interpretation. This event-resolution procedure is checked on the specified matrix; it is not a proof that a finite scan detects every possible crossing for arbitrary rapidly oscillating nominal commands.

The ordinary, recovery, and progress-aware baselines use Radau with relative/absolute tolerances $2\,10^{-8}$ and $2\,10^{-10}$ and maximum time step .5. Refinement uses $2\,10^{-10}$ and $2\,10^{-12}$ with maximum step .25. Post-entry continuation uses maximum steps .05 and .025. A bounded actuator-saturated extension defines vector fields on implicit-solver trial states outside D; it does not project accepted physical states. Exact ordinary-CBF rest equilibria with autonomous commands are represented by stationary arcs. Accepted-state safety is checked independently of that extension.

Costs use adaptive vector quadrature, with profile and HJB knots supplied where available. Full-episode constraint checks include integration nodes, their midpoints, profile knots, and dense samples through the entry or censoring time, rather than ending at time ten. The additional floor residual is sampled along the reflected arcs. The analytic hypotheses establish safety and feasibility; floating-point sampling is evidence about the implementation, not an interval proof.

A.6 Full comparison and nominal ablation

All 936 base episodes and 504 refinements completed. Ordinary CBF enters in 21/72 cases, with 51 right-censored outcomes at time 600. Every other specified setting enters in 72/72. Ordinary-CBF successes are 12/18 for HJB, 3/18 for zero input, 6/18 for clipped drift cancellation, and 0/18 for braking input. Its time medians are conditional on those successes and are not directly comparable as paired 72-case rankings. Table 14 summarizes all settings. Table 15 separates outcomes by nominal, and Figure 6 displays entry ratios for the same scenario matrix.

The regular recovery methods retain eventual recovery but incur large delays for unfavorable nominals: blend median/maximal ratios are 10.060/567.246, and recovery-CBF ratios are 10.351/561.026. Those are finite values for the fixed certified data. They do not show that the uniform nominal bound for those recovery laws is infinite. The specified progress baseline substantially improves entry time: at $r = .5, c = 5$, its median/maximal ratios are 1.607/2.023. At $r = 1, c = 5$ they are 1.067/1.309. The reflected $\theta = .25$ ratios are 1.976/2.000. The methods enforce different interior-intervention requirements, so faster progress-CBF cases do not contradict the exact strict-fidelity theorem.

Table 16 reports the progress-baseline slack. No positive slack is observed for $r = \sqrt{.1}, c = 5$ in any case. For $r = .5, c = 5$, 48/72 cases are slack-free and maximum slack is 1.3745. Larger references or gains can request accelerations incompatible with the simultaneous safety/actuator interval. This does not invalidate the baseline: safety remains hard and progress is relaxed explicitly. Conversely, a slack-free progress design can have its own quantitative analysis. The distinction

Table 14: All specified controller settings. Time medians and maxima condition on observed entry; all other medians use all 72 cases. The numerical variation includes known event jumps.

Method	Entry	Median T/T^*	Max. T/T^*	Median $I_{1,10}$	Median A_{10}	Median $\widehat{\mathrm{TV}}_{10}$
Ordinary CBF	21/72	1.79	5.96	1.80	0.106	0.87
Recovery blend	72/72	10.06	567.25	2.22	0.133	3.42
Recovery CBF	72/72	10.35	561.03	2.22	0.130	3.42
Progress $r = \sqrt{.1}$	72/72	2.37	3.08	2.55	0.295	4.90
Progress $r = .5$	72/72	1.61	2.02	3.28	0.458	7.46
Progress $r = \sqrt{.5}$	72/72	1.25	1.61	3.86	0.633	9.05
Progress $r = 1$	72/72	1.07	1.31	5.10	0.738	9.84
Progress $r = .5, c = 2$	72/72	1.96	2.89	3.00	0.325	4.82
Progress $r = .5, c = 10$	72/72	1.51	1.93	3.43	0.528	8.50
Reflected $\theta = .1$	72/72	3.02	3.16	2.41	0.209	3.73
Reflected $\theta = .25$	72/72	1.98	2.00	3.30	0.353	6.56
Reflected $\theta = .5$	72/72	1.40	1.41	4.21	0.552	9.63
Optimal profile ($\theta = 1$)	72/72	1.00	1.00	6.28	0.834	15.23

Table 15: Nominal ablation: median T/T^* followed by entries/18. The selected progress setting has $r = .5, c = 5$; Table 14 retains all six progress settings.

Method	HJB	Zero input	Net zero	Braking
Ordinary CBF	1.15 (12/18)	2.98 (3/18)	2.21 (6/18)	– (0/18)
Recovery blend	2.41 (18/18)	12.22 (18/18)	3.42 (18/18)	207.26 (18/18)
Recovery CBF	2.53 (18/18)	12.71 (18/18)	3.46 (18/18)	207.23 (18/18)
Progress $r = .5$	1.23 (18/18)	1.77 (18/18)	1.21 (18/18)	1.77 (18/18)
Reflected $\theta = .25$	1.60 (18/18)	2.00 (18/18)	1.58 (18/18)	2.00 (18/18)
Optimal profile	1.00 (18/18)	1.00 (18/18)	1.00 (18/18)	1.00 (18/18)

Table 16: Progress-baseline slack diagnostics. "Slack-free" means no positive slack observed; the fraction is a sampled-point share, not an exact time-duration fraction.

Progress setting	Slack-free episodes	Maximum slack	Median sampled fraction
Progress $r = \sqrt{.1}$	72/72	0.000	0.000
Progress $r = .5$	48/72	1.374	0.000
Progress $r = \sqrt{.5}$	16/72	3.404	0.004
Progress $r = 1$	0/72	6.274	0.028
Progress $r = .5, c = 2$	54/72	0.367	0.000
Progress $r = .5, c = 10$	0/72	6.274	0.008

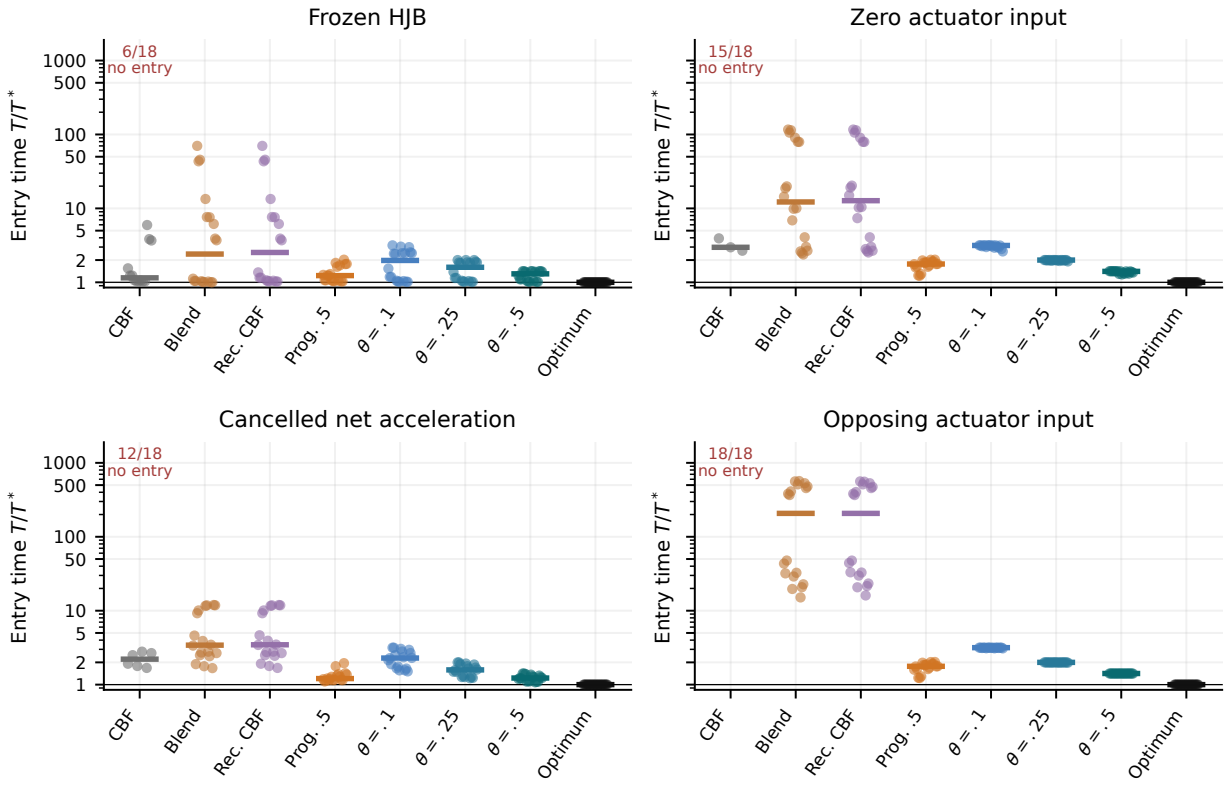


Figure 6: All specified plant/task/initial-state cases separated by nominal. Nonentries remain in the denominators. Horizontal bars are successful-case medians and are not computed using an imputed censoring time.

claimed here is the exact clock for a specified fidelity class, not an inherent inability of CBF-based performance designs to certify rates.

A.7 Exact clock, braking nominal, and moving starts

All 288 rate/optimal-profile episodes enter. For the 216 cases with $\theta < 1$, the observed T/W_θ never exceeds one. All braking-input cases attain one at returned numerical precision. Rest-start braking trajectories follow the fully admissible floor, so the equality $W_\theta = T^*/\sqrt{\theta}$ is partly built into the implementation. It is not an independent numerical proof of sharpness. Independent quadrature and physical-time integrations, reported below, check the clock calculation.

The six moving-start profiles have $v_0 = .4$ and a positive initial energy buffer. Table 17 lists their exact clocks, and Figure 7 compares nominal-dependent entry times with the clock bounds. The relative conservatism $1 - W_\theta\sqrt{\theta}/T^*$ ranges from approximately 0.1185% to 2.1848% over the specified $\theta < 1$ cases. For C2 tracking cost, for example, $T^* = .63601349$ and $W_{.25} = 1.25943085$, whereas $2T^* = 1.27202697$. Both bounds are useful; only W_θ is the exact all-faithful-filter worst value for a moving start.

A.8 Paired cost, intervention, and motion comparison

The $\theta = .25$ versus $r = .5, c = 5$ comparison is paired by plant, task, state, and nominal. The reflected method is slower in 58/72 cases. It has lower ten-unit intervention in 62/72, lower acceleration RMS in 56/72, lower numerical variation in 44/72, and lower applied cost in 31/72. These are descriptive counts on the fixed scenario matrix, not a statistical population estimate or a claim of minimum integrated intervention. Table 18 reports these paired comparisons.

Table 17: Exact comparison-clock evaluations for moving starts $(h_0, v_0) = (.8, .4)$. Values are numerical quadratures, not interval-enclosed exact constants.

Plant	Task	T^*	$W_{.1}$	$W_{.25}$	$W_{.5}$
C1	gap	1.2783	4.0045	2.5446	1.8051
C1	tracking	0.6694	2.0790	1.3268	0.9440
C2	gap	1.5494	4.8591	3.0861	2.1884
C2	tracking	0.6360	1.9708	1.2594	0.8967
C3	gap	1.7491	5.4840	3.4843	2.4707
C3	tracking	0.6836	2.1144	1.3531	0.9638

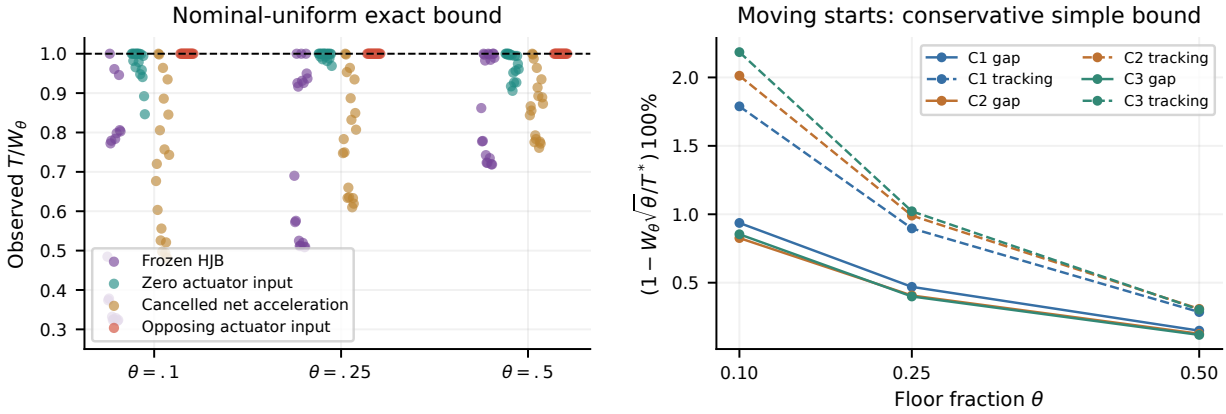


Figure 7: Nominal-dependent entry time relative to the exact worst clock, and the moving-start improvement over the simple scaled bound.

Table 18: Paired reflected $\theta = .25$ versus progress $r = .5, c = 5$. "Lower" counts smaller reflected values; each ratio is the median of the paired reflected/progress ratios. Costs are separated by objective so distinct running costs are not pooled into a utility ranking.

Quantity	Gap: lower/36	Gap ratio	Tracking: lower/36	Tracking ratio
Entry T	10/36	1.067	4/36	1.169
Cost J_{10}	9/36	1.134	22/36	0.993
Intervention $I_{1,10}$	34/36	0.977	28/36	0.989
Entry intervention $I_{1,\mathrm{ent}}$	20/36	0.992	17/36	1.005
Acceleration RMS	24/36	0.880	32/36	0.766
Grid variation $\widehat{\mathrm{TV}}_{10}$	19/36	0.957	25/36	0.818

For the gap objective, the median paired cost ratio is 1.134 and only 9/36 reflected costs are lower. For the tracking-cost objective, the median ratio is .993 and 22/36 are lower. The median paired intervention ratios are .977 and .989; acceleration-RMS ratios are .880 and .766. This supports a quantitative time–intervention tradeoff in the tested cases while preventing a blanket cost claim. Figure 8 compares entry-only intervention on the C2 rest cases.

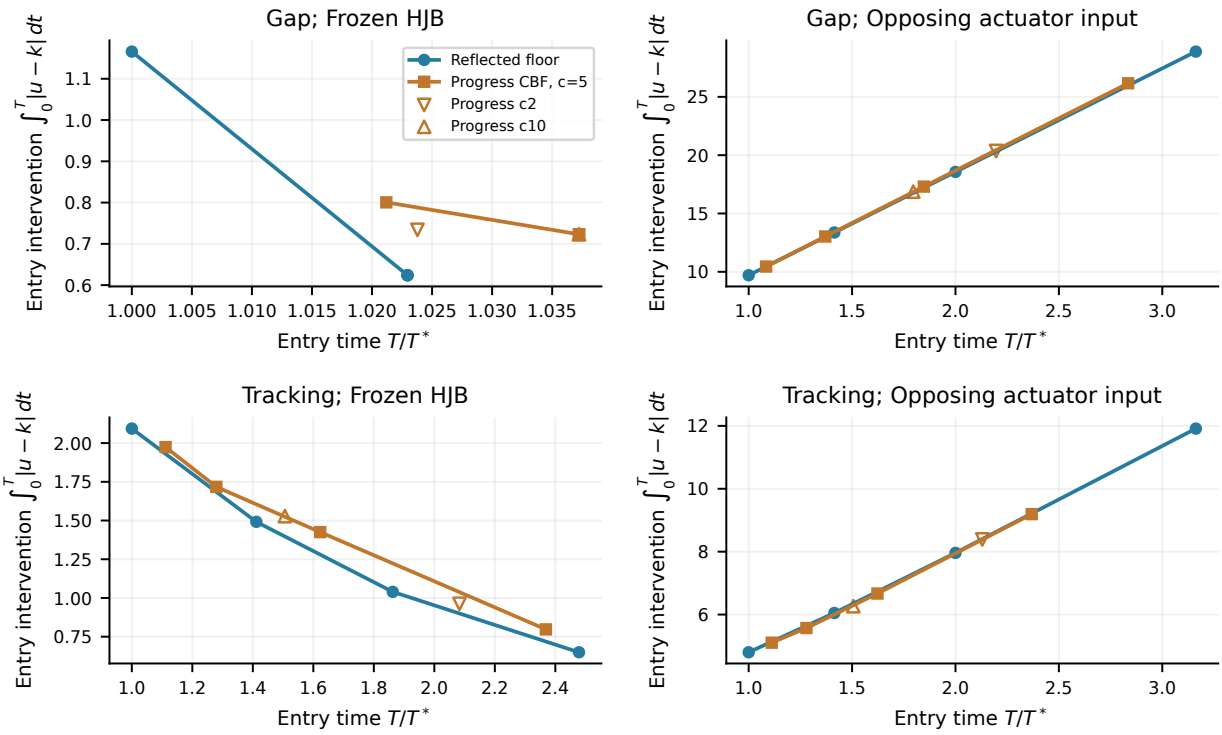


Figure 8: C2 rest case $(h_0, v_0) = (.8, 0)$, both objectives and favorable/braking commands. All specified reflected fractions and progress fractions are shown; progress sensitivity points use $r = .5, c = 2, 10$. Coordinates use entry-only intervention. Coincident points are retained. Under persistent braking input, a longer entry duration increases the accumulated intervention, even when the instantaneous correction is gentler.

Table 19 gives the complete grid for the C2 tracking-cost case with the braking nominal. In this case, ordinary CBF is a permanent rest equilibrium. Recovery CBF enters at 329.11865; progress $r = .5, c = 5$ enters at 1.04265; the reflected $\theta = .25$ filter enters at 1.28515. The last two have acceleration RMS .57616 and .41885 and numerical variation 12.2421 and 9.5835, respectively. Their costs are 2.57100 and 2.53891. Their largest recorded acceleration jumps are 2.8989 and 2.9167: lower RMS and total variation do not imply a smaller largest jump. The optimum-entry endpoint has cost 4.74075, RMS 1.07553, variation 23.2098, and a largest jump 8.0859. It optimizes entry time with free terminal speed, not the tracking cost or comfort. Figure 9 shows the trajectories for this case. Figure 10 compares entry time, cost, and motion measures across the fraction grids, while Figure 11 compares common-horizon intervention.

A.9 Refinement and signal diagnostics

The 504 refined outcomes have no entry-classification differences. Maximum absolute entry-time change is $2.0443\,10^{-7}$ and maximum relative change is $1.1846\,10^{-7}$. The largest absolute optimal-clock change is $3.3327\,10^{-8}$; the largest exact-worst-clock change is $1.0539\,10^{-7}$. Maximum relative changes in J_{10}, $I_{1,10}$, and acceleration RMS are respectively $1.8385\,10^{-7}$, $3.4153\,10^{-6}$, and $3.0651\,10^{-7}$. Entry-only intervention has a larger maximum relative change, $4.5901\,10^{-5}$, over potentially much longer integration intervals.

Table 19: Complete specified grid for C2 tracking cost, rest $(h_0, v_0) = (.8, 0)$, braking nominal. $T^* = .642573$. $I_{1,\mathrm{ent}}$ integrates to entry; the dagger marks the censored integral through 600. Cost, RMS, variation, and maximum progress slack use $[0, 10]$.

Method	T/T^*	J_{10}	$I_{1,\mathrm{ent}}$	$\mathrm{RMS}_{10}(p)$	$\widehat{\mathrm{TV}}_{10}(p)$	$\zeta_{\max}$
Ordinary CBF	–	18.17	3173.37†	0.00	0.00	0.00
Recovery blend	512.19	17.17	1833.34	0.01	0.49	0.00
Recovery CBF	512.19	17.17	1833.34	0.01	0.49	0.00
Progress $r = \sqrt{.1}$	2.37	2.18	9.19	0.36	7.93	0.00
Progress $r = .5$	1.62	2.57	6.67	0.58	12.24	0.19
Progress $r = \sqrt{.5}$	1.28	3.21	5.57	0.76	14.94	2.22
Progress $r = 1$	1.11	3.84	5.11	0.90	17.52	5.09
Progress $r = .5, c = 2$	2.13	2.29	8.39	0.38	7.98	0.00
Progress $r = .5, c = 10$	1.51	2.80	6.27	0.67	12.92	5.09
Reflected $\theta = .1$	3.16	2.62	11.91	0.23	5.46	0.00
Reflected $\theta = .25$	2.00	2.54	7.96	0.42	9.58	0.00
Reflected $\theta = .5$	1.41	3.08	6.05	0.67	15.07	0.00
Optimal profile	1.00	4.74	4.80	1.08	23.21	0.00

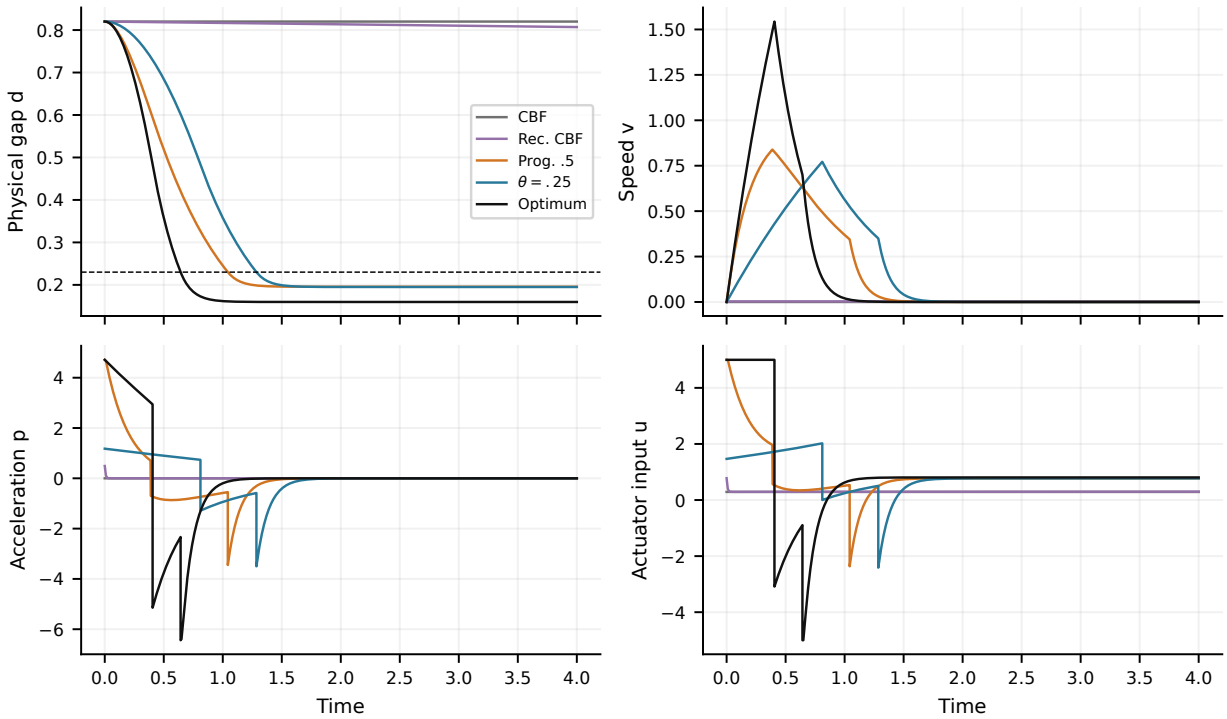


Figure 9: The same braking-nominal example. The dashed gap is the entry target .23. All methods use the common post-entry recovery-CBF rule, whose braking contributes to the common-horizon metrics. The traces show genuine acceleration jumps rather than a smoothed actuator realization.

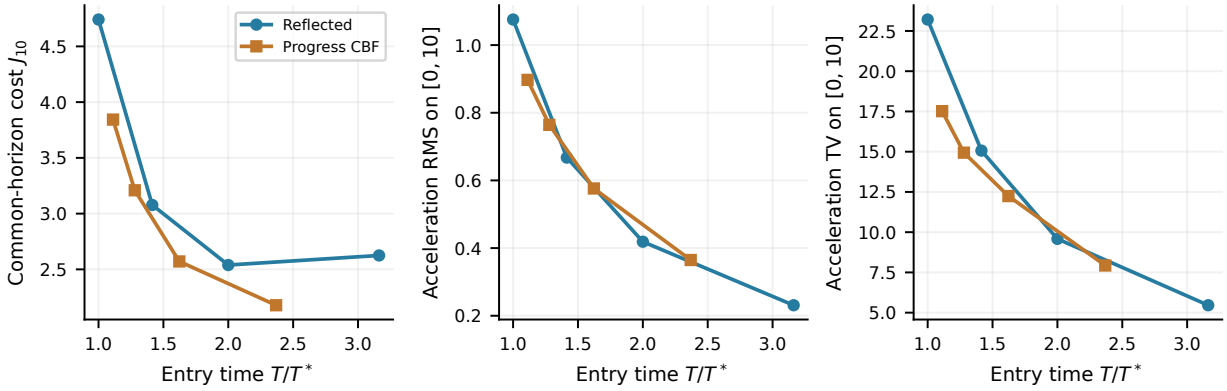


Figure 10: The C2 braking-command tracking-cost example across the specified main fraction grids. Entry speed, common-horizon cost, acceleration RMS, and numerical variation have different optima. All gain-sensitivity settings remain in the tables and source data.

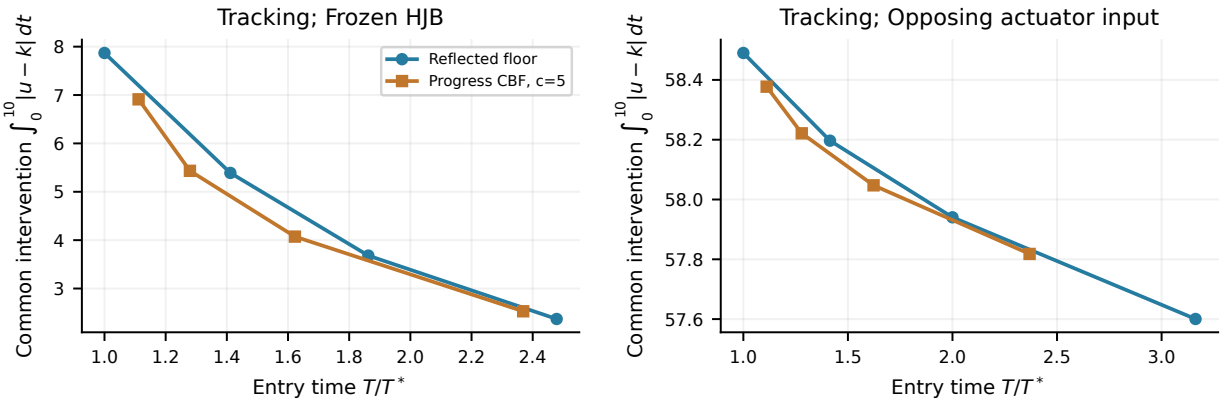


Figure 11: The same C2 fraction sweeps using common-horizon intervention $I_{1,10}$ rather than entry-only intervention. The fixed post-entry handover is part of this comparison. These integrals assess empirical performance; they are not the optimization criterion of the strict interior-fidelity theorem.

Event extraction includes both sides of all arc joins, profile knots, and progress-ceiling kinks. The ledger contains 3,274 event records across the two resolutions. The maximum variation change under refinement is .04286 absolute and .46448% relative. The largest recorded jump changes by at most $4.27\,10^{-7}$.

Across full accepted base trajectories the smallest headway residual is $-7.2550\,10^{-14}$, smallest speed is $-4.8297\,10^{-14}$, smallest speed-cap margin is .24593, and maximum actuator excess is $1.7764\,10^{-15}$. The minimum sampled CBF-interval width is .20408. The smallest sampled reflected-floor residual is $-1.3579\,10^{-13}$. These are floating-point residuals from the stated sample checks; they are not exact lower bounds on continuous trajectories.

A.10 Computation and reproducible archive

The production/refinement matrix took about 186 seconds with four workers in the recorded Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0 environment. These times include offline simulation, quadrature, and diagnostics; they are not controller latency measurements.

A separate sequential Python benchmark measured profile preparation over the 18 profiles: median 91.36 ms, range 76.67–118.50 ms. Supplied-state scalar-kernel medians were approximately 3.16 μs for ordinary CBF, 3.60 μs for recovery blend, 3.31 μs for recovery CBF, 4.21 μs for progress $r = .5$, and 17.97 μs for reflected $\theta = .25$. Every kernel used the same supplied states; the reflected kernel includes fixed-profile interpolation, with half the states on its floor and half strictly inside the tube. This excludes event location, a sampled-data realization, and any worst-case real-time claim. The scalar CBF baseline is already inexpensive. No computational dominance over it is claimed.

The mathematical entry clock starts with the episode profile already available. Profile preparation is excluded from T and T^*. A deployment computation delay would need separate accounting, particularly because the theorem specifies an immediate rest launch.

From the reproduction package root, the following commands rerun the continuous-time matrix, regenerate figures and tables from saved rows, and measure scalar-kernel timing, respectively:

```
python3 reproduce.py --continuous --fresh --workers 4
python3 reproduce.py --render-only
python3 reproduce.py --continuous-benchmark
```

The input policy hashes, locked protocol, software versions, source hashes, raw outcomes, refinement pairs, and event ledger identify the computations. The separate LaTeX archive needs no numerical execution to compile the paper and technical notes.

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